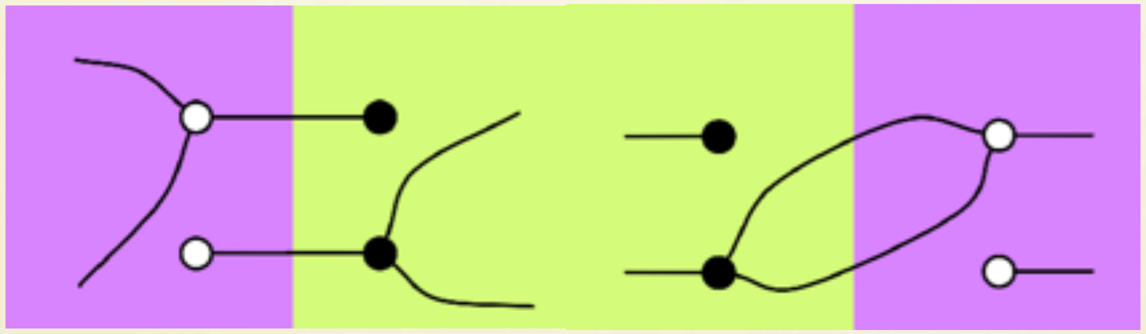
Graphical Linear Algebra



Fabio Zanasi

Introduction

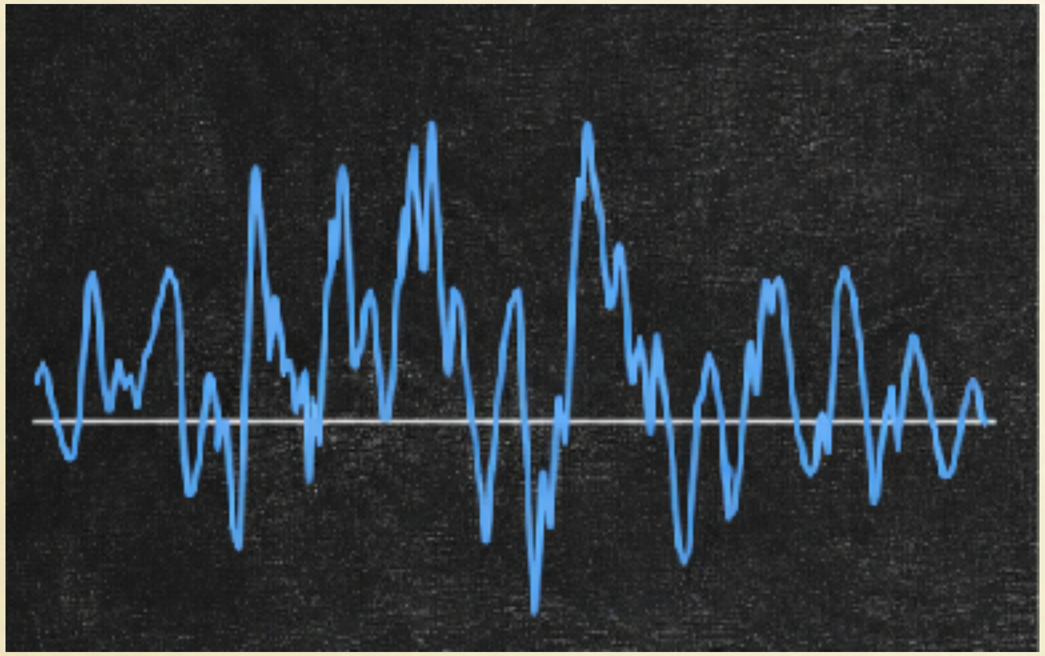
About me

- **Professor** of **Computer Science.**
- At UCL since **2016.**
- Background in **Philosophy** and **Mathematical Logic.**
- I develop **formal methods** for **artificial intelligence** and **programming** using **category theory** as unifying mathematical language.

LA: the Workhorse of Science

Abstraction Matters

Music: physical description vs score



Abstraction Matters

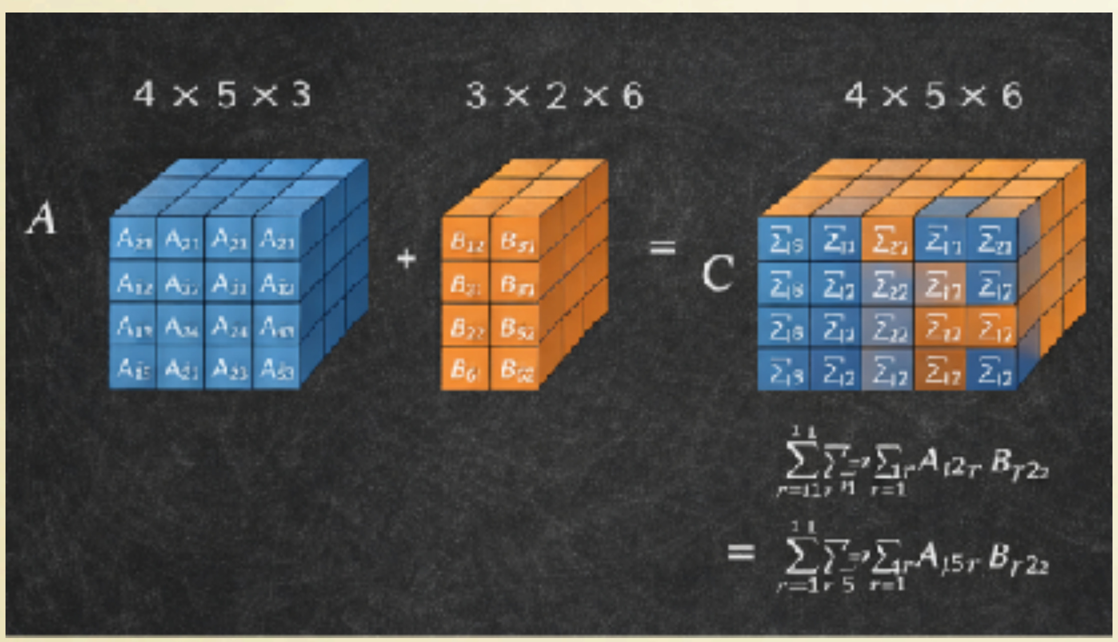
Assembly code vs Python

```
mov eax, [esp+8]
mov ebx, [esp++12]
cmp eax, ebx
jge .L1
.L2:
    call _find_max
    add esp  4
.L1: push [esp+8]
    push eax
   call _find_max
pop ebx
cmp eax, ebx
jle .L3
mov eax, ebx
L3: ret
```

```python
def find_max(a, b):
    return max(a, b)
```

Abstraction Matters

Are we doing linear algebra at the right level of abstraction?



Abstraction Matters

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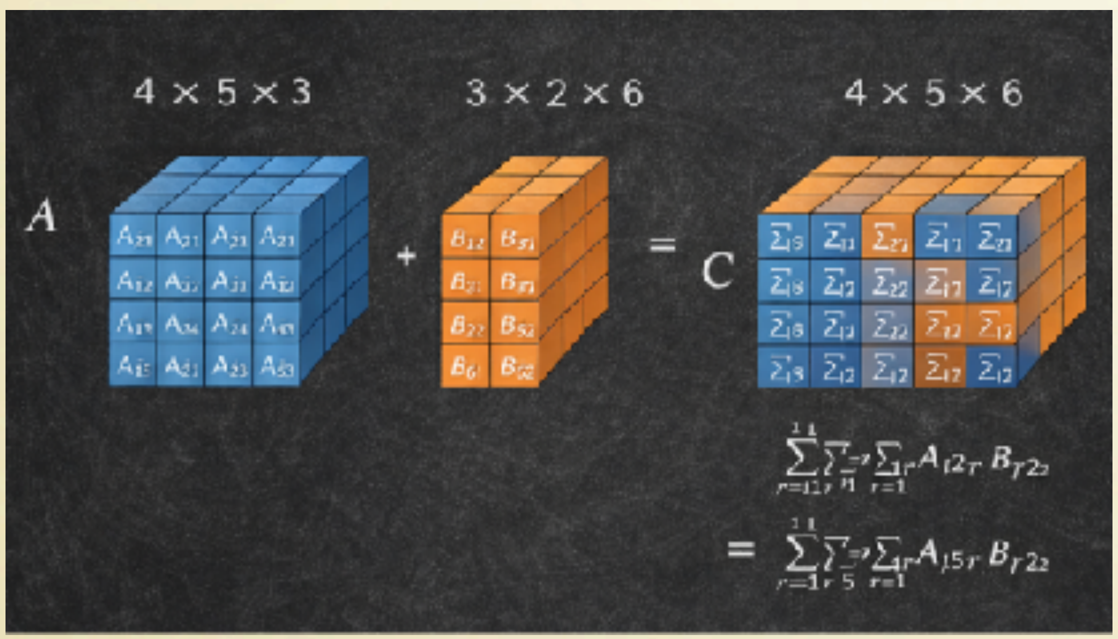
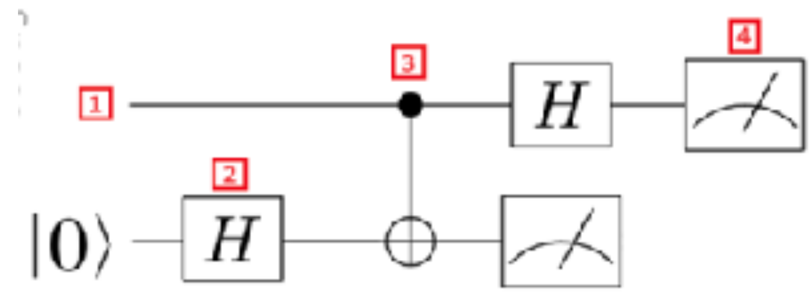
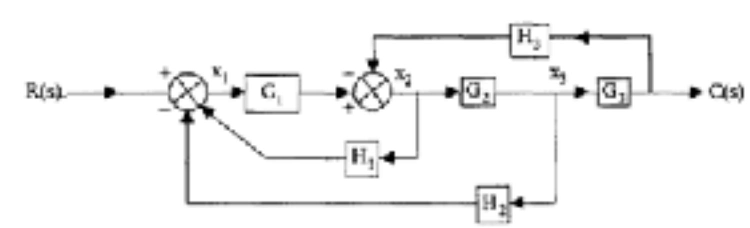


quantum circuits



signal flow graphs



…

In this Talk

Graphical Linear Algebra

An high-level language for linear algebraic calculations

Exposes the underlying structure and
symmetries of linear algebra

Graphical but completely formal

Rooted in category theory

In this Talk

Outline

GLA for linear maps

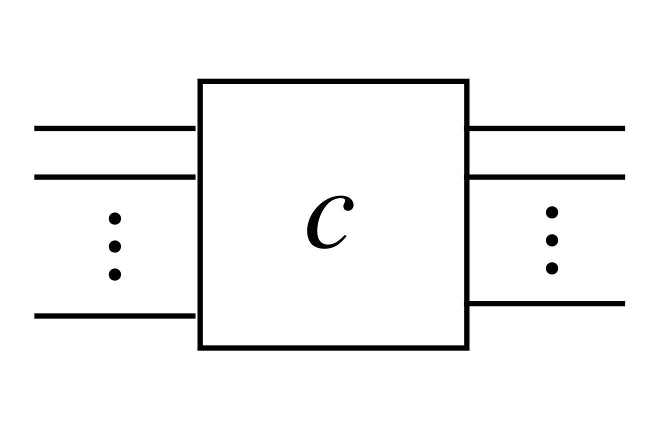
Full GLA (subspaces)

Applications: GLA in action

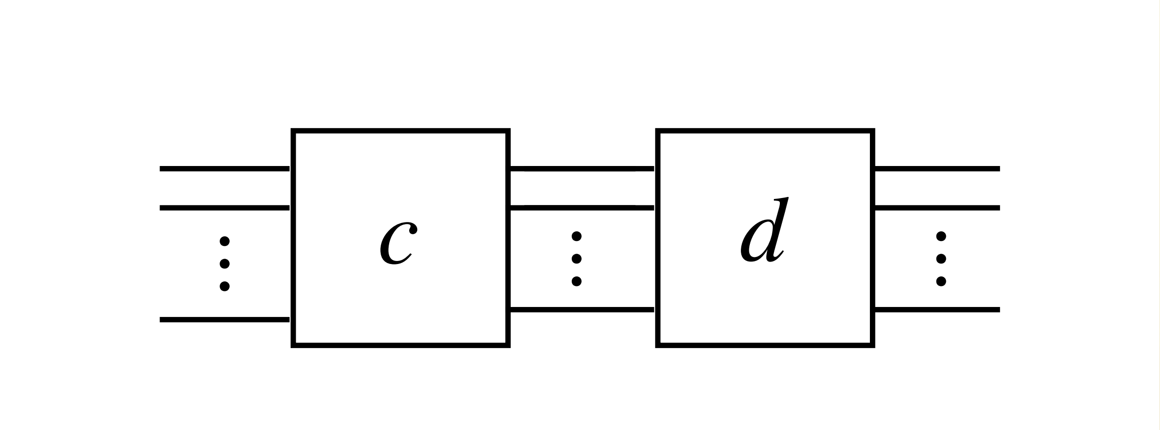
“Causal" Linear Algebra: the matrix calculus

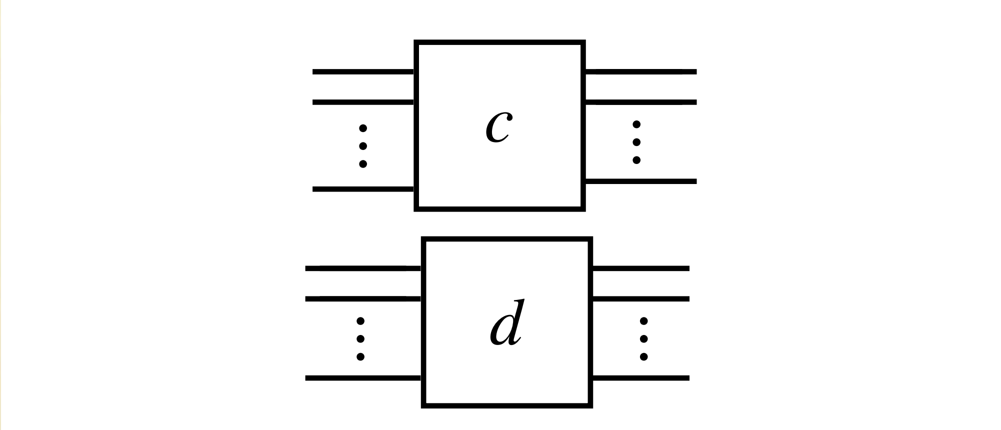
String Diagrams

A generic string diagram

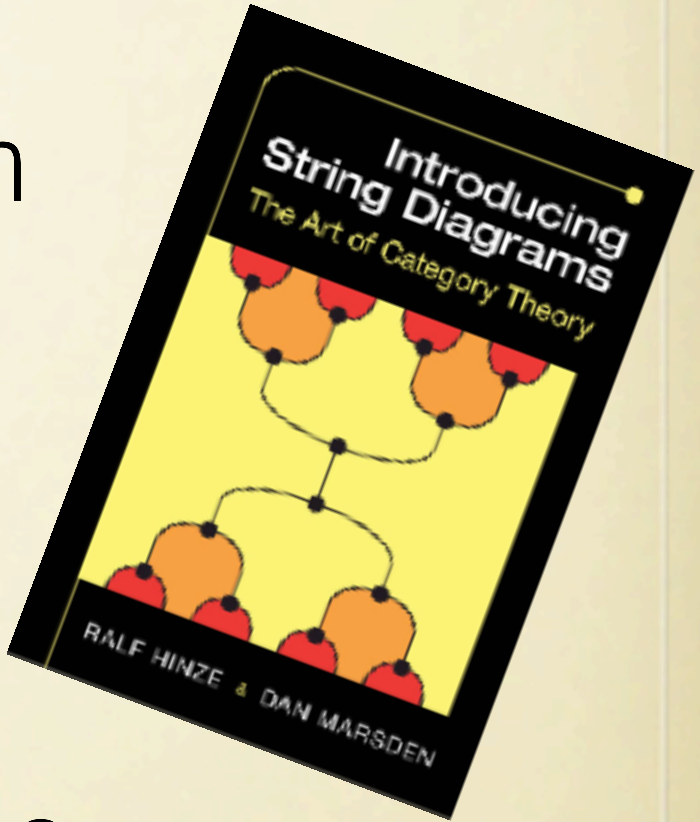


Composing string diagrams

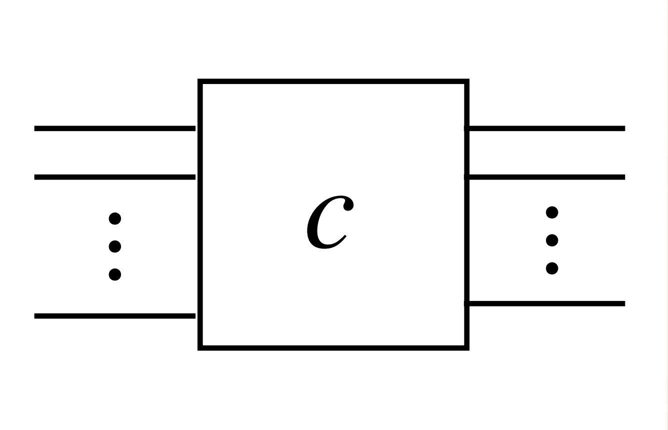




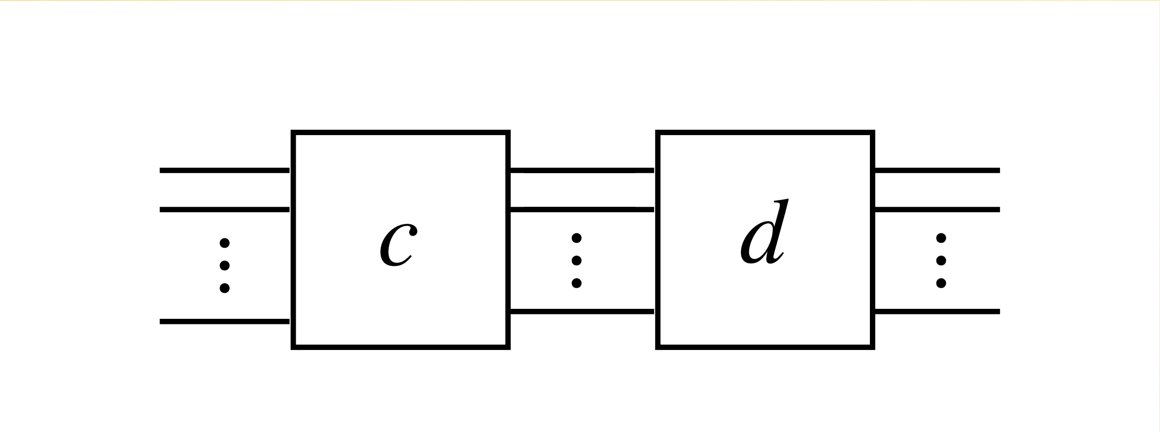
String Diagrams

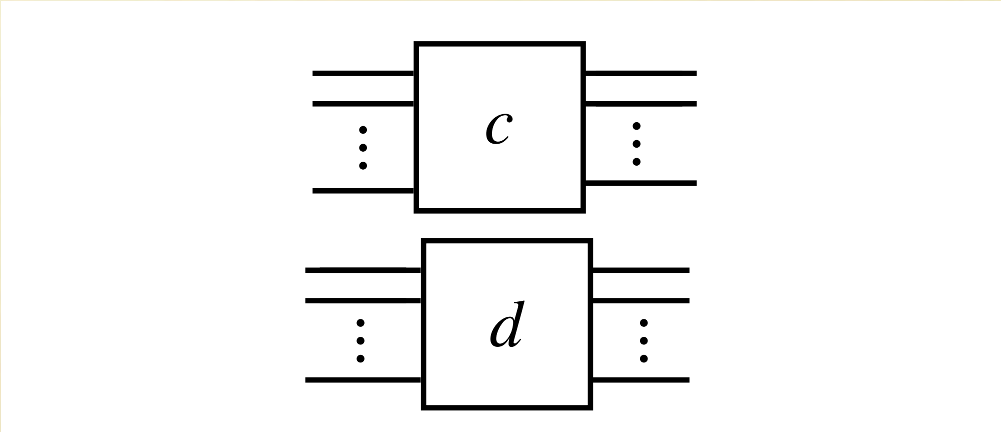


A generic string diagram



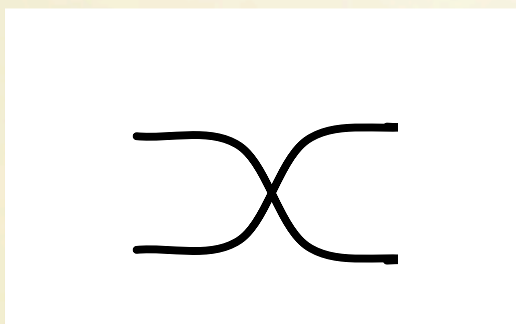
Composing string diagrams

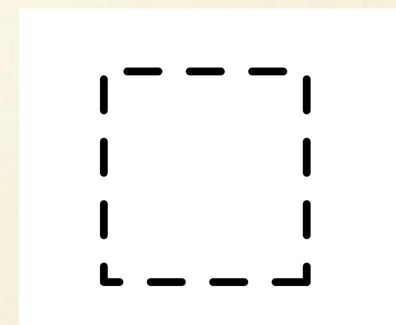




Basic wiring

Operations





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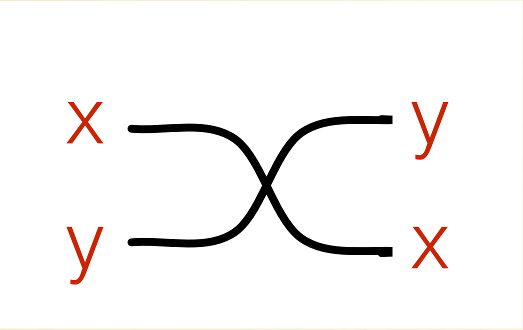

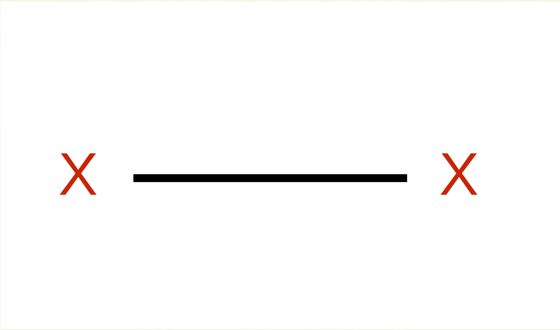

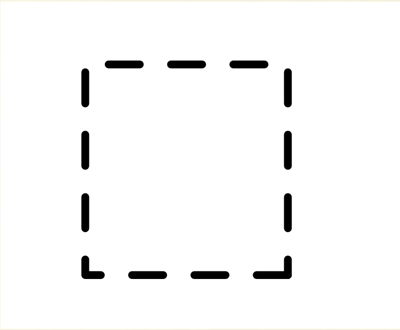

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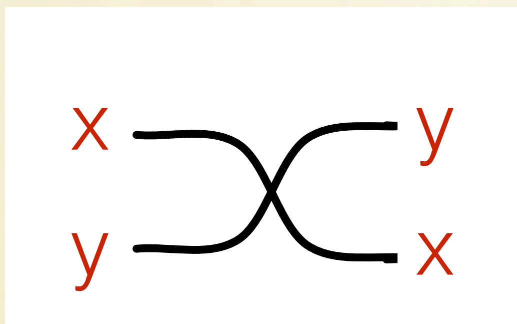
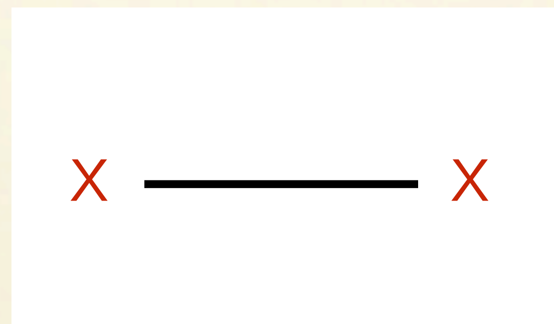
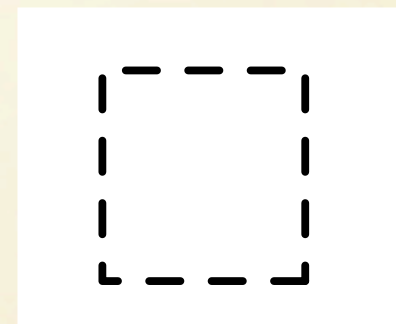
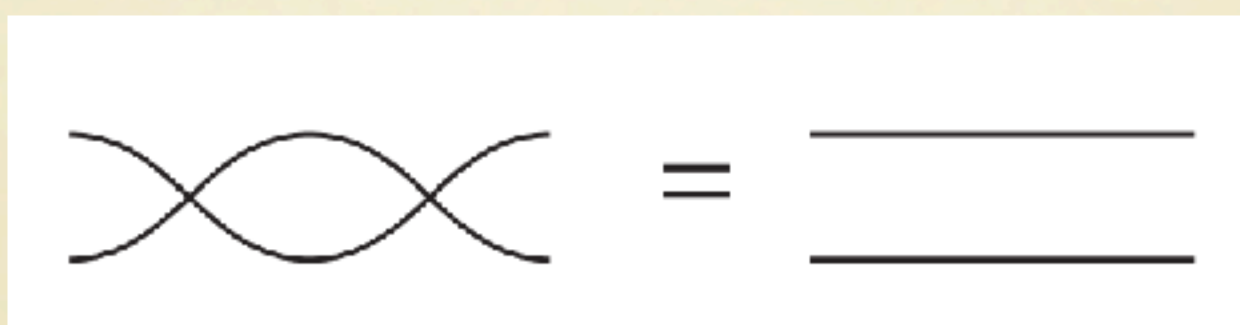






Equations



Basic wiring

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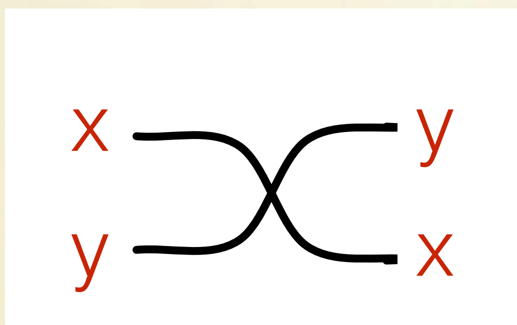

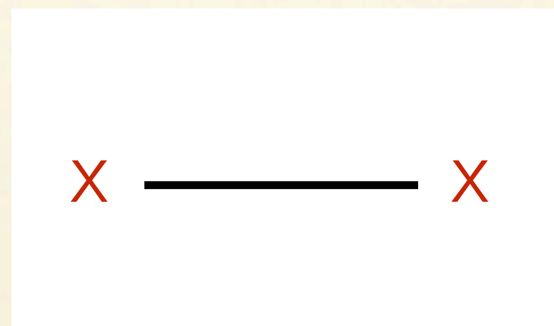

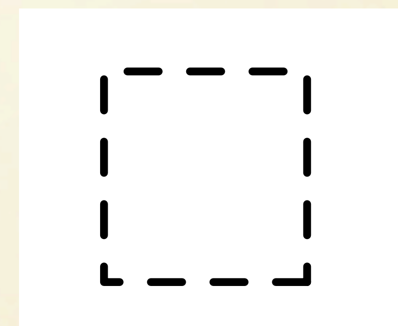

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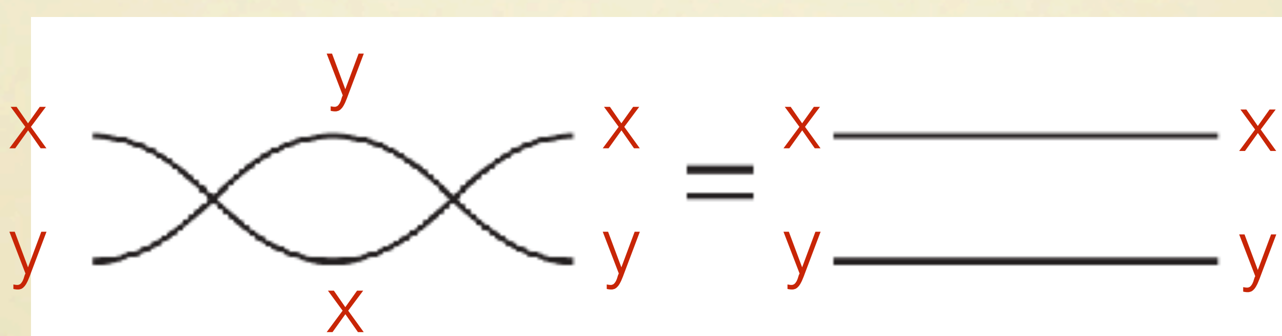

Basic wiring

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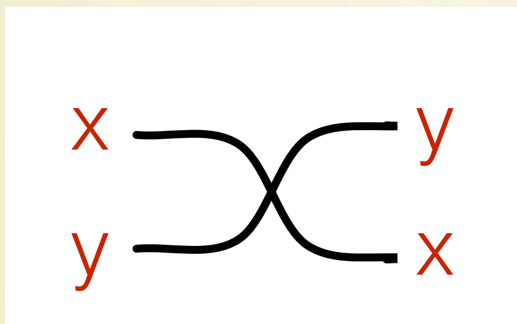

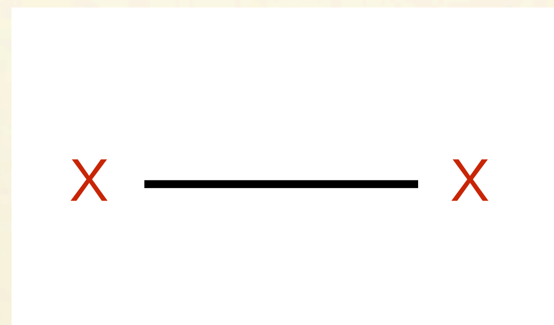

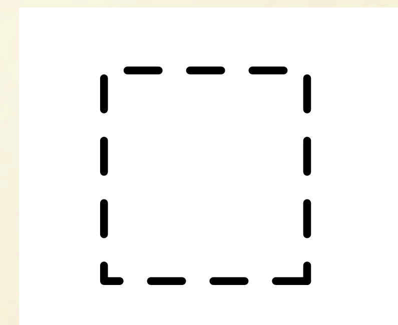

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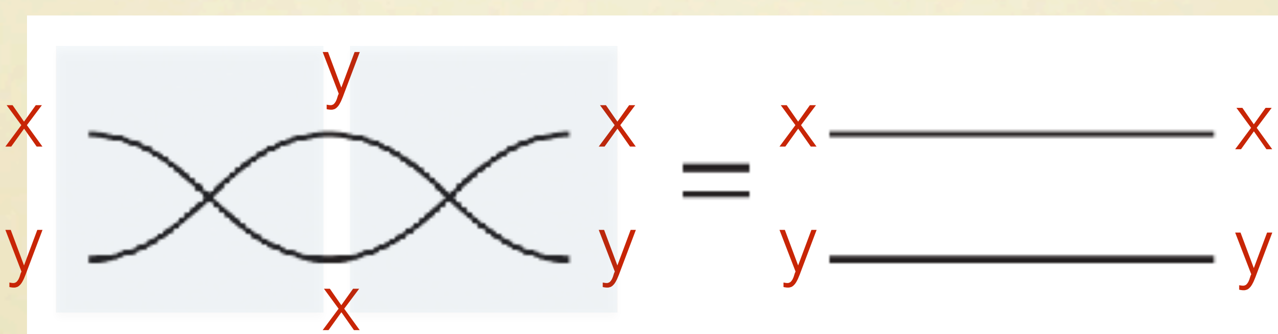

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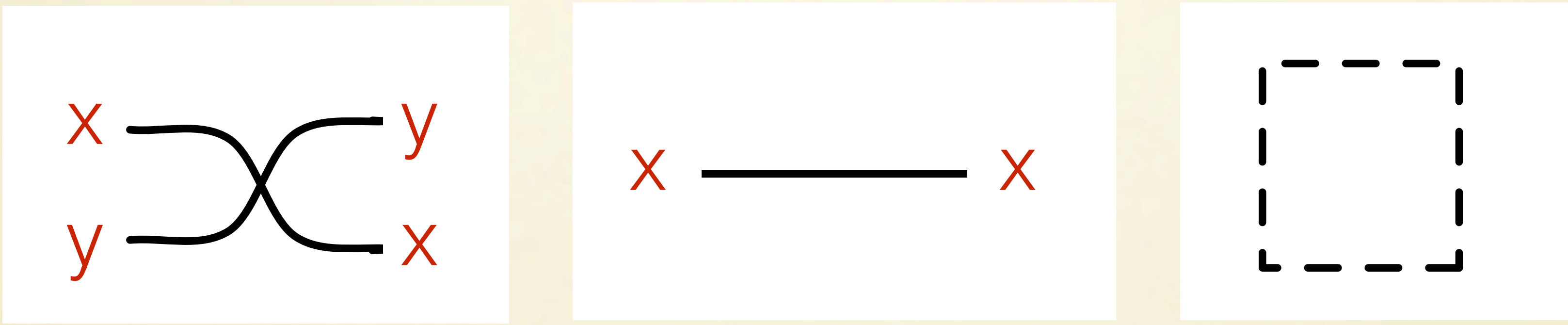

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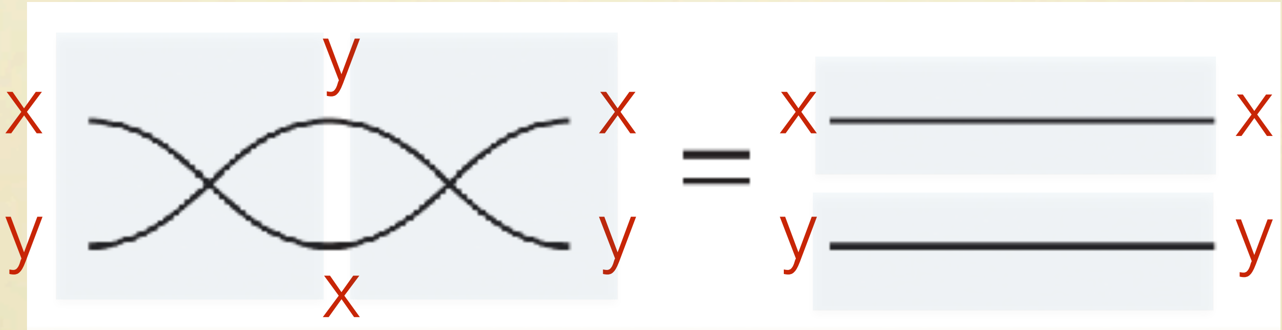
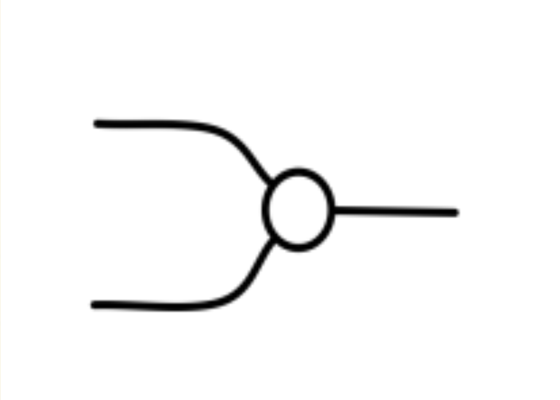
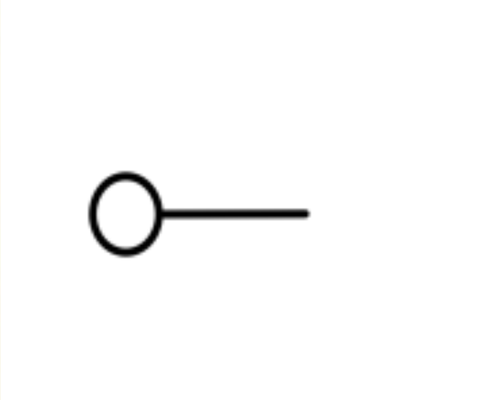


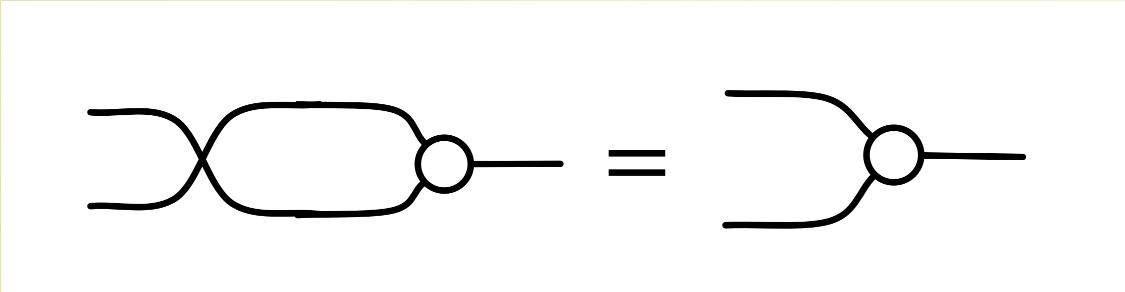
Addition

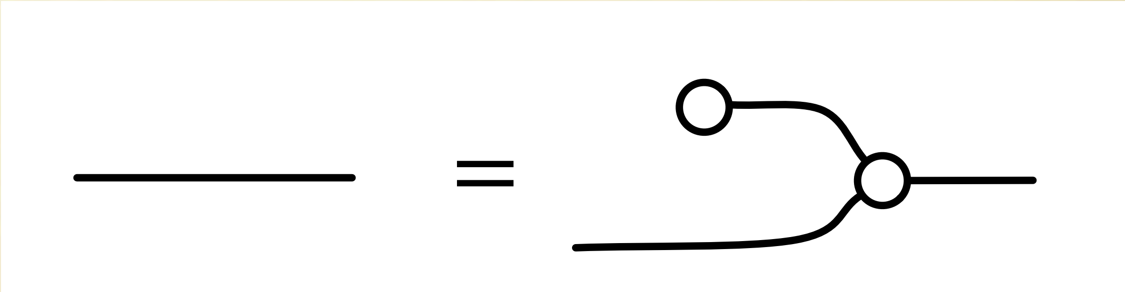
Operations

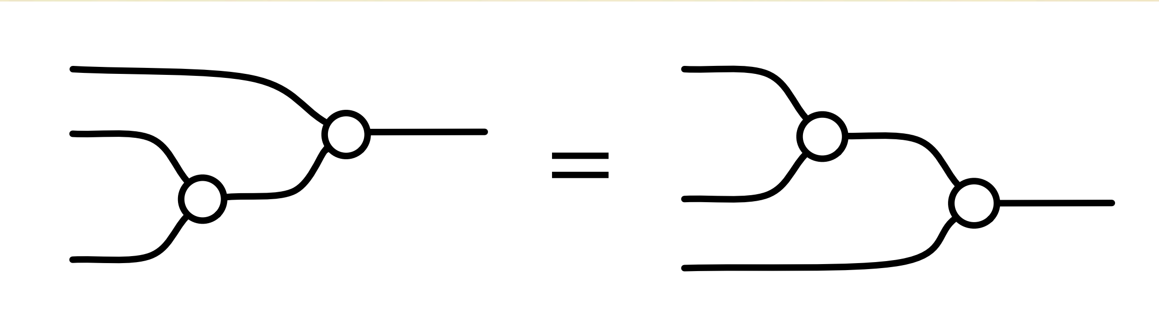




Equations







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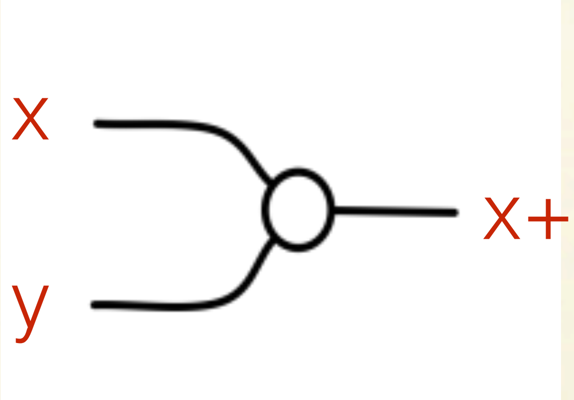

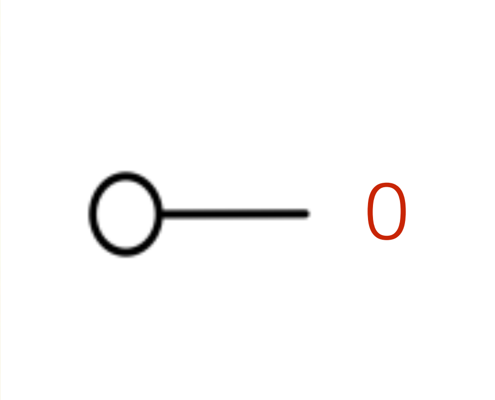

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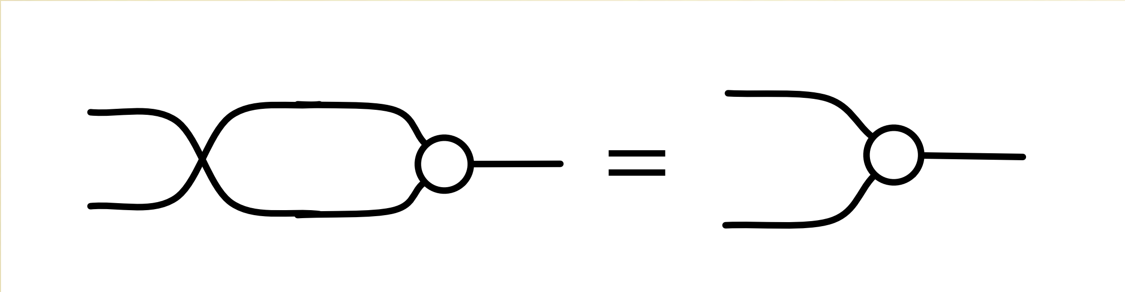

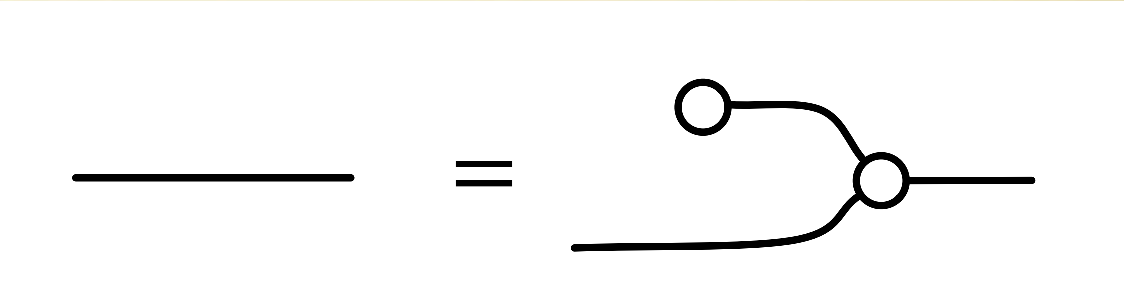

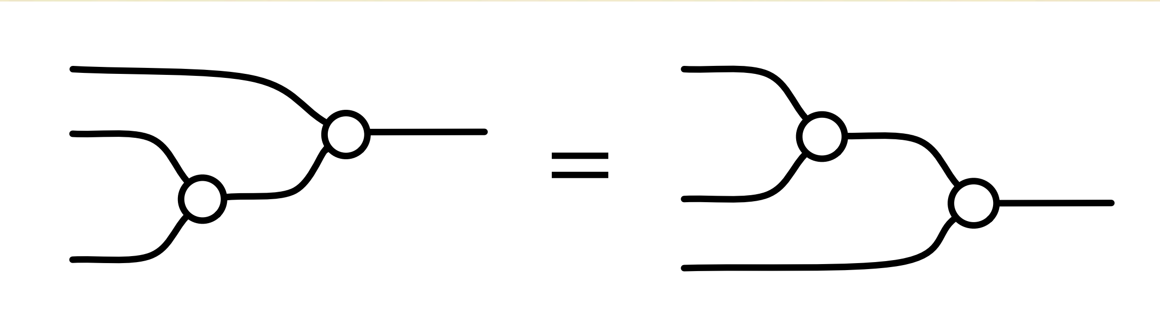

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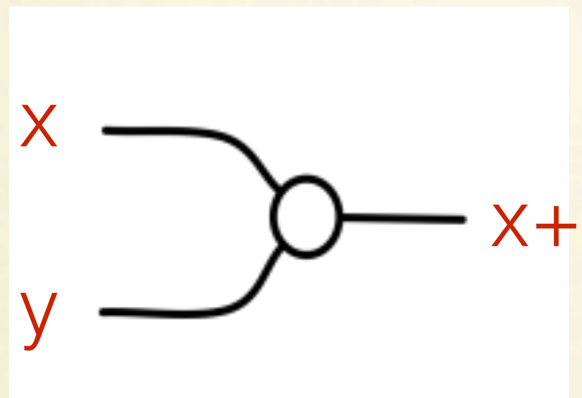
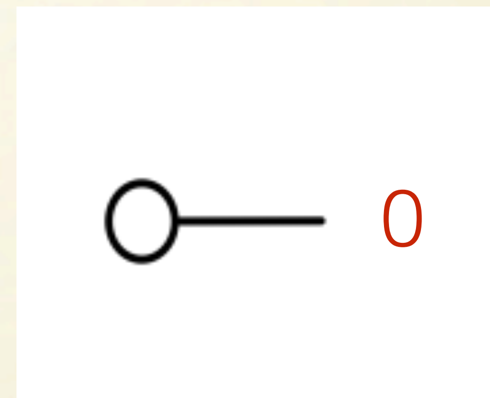
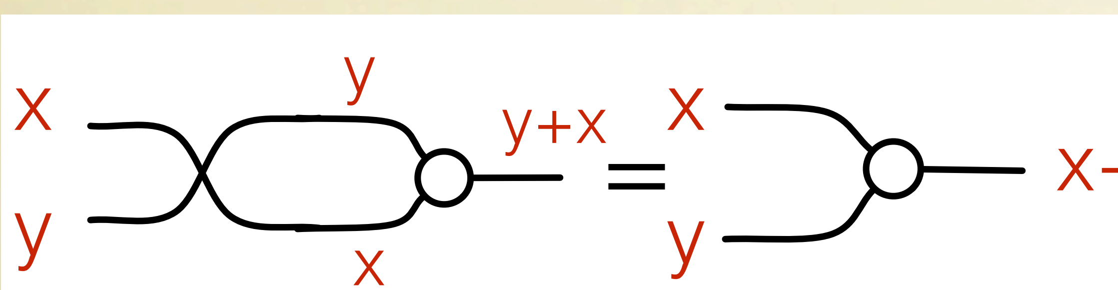
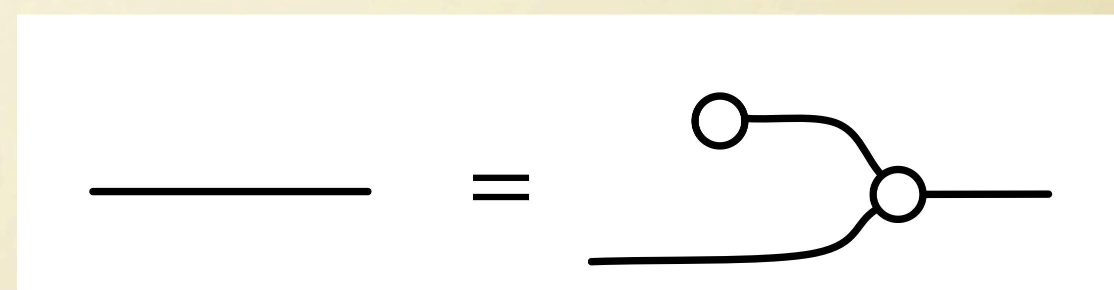
Operations

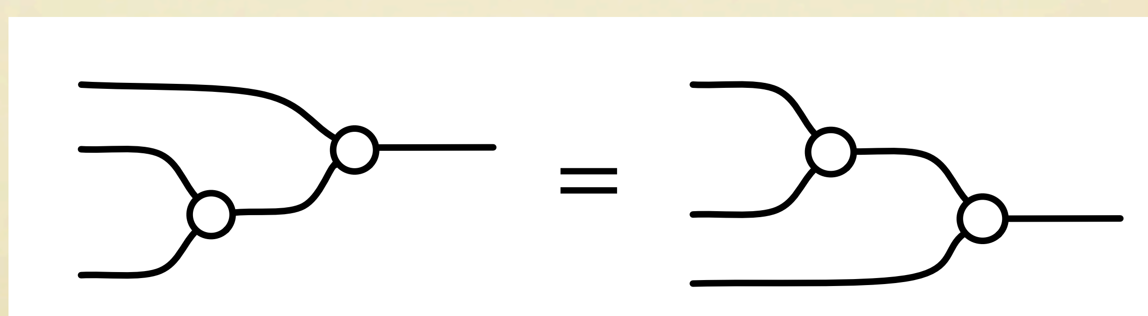




Equations







Addition

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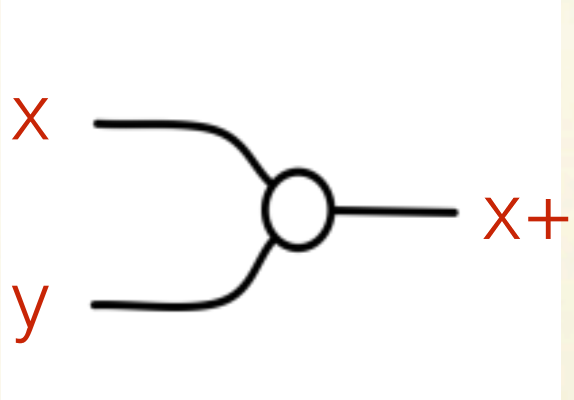

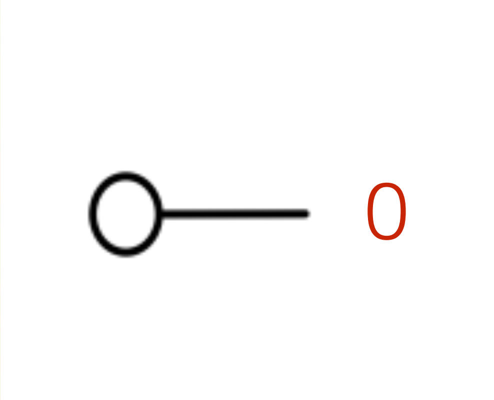

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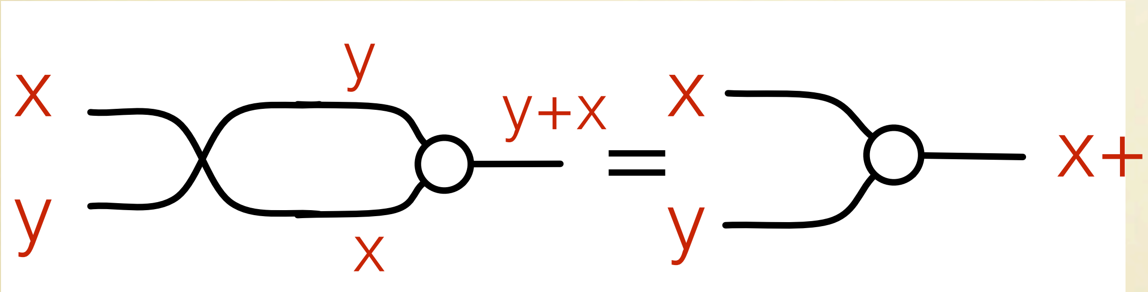

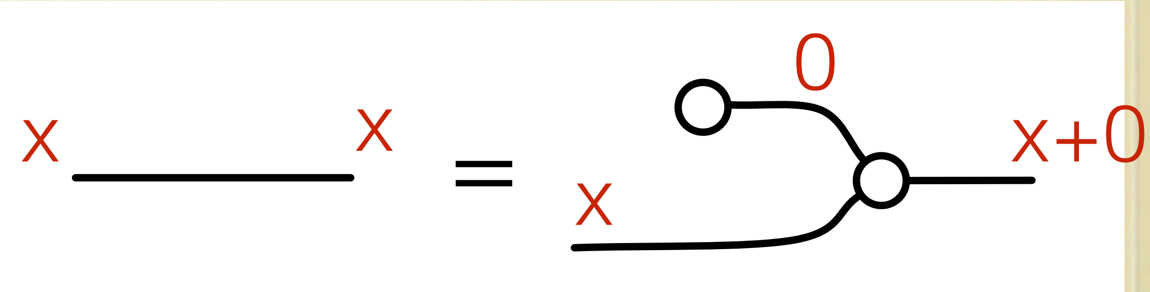

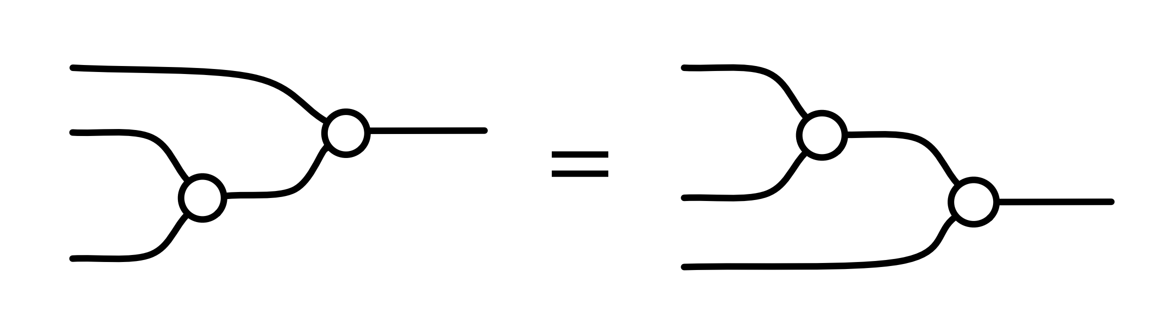

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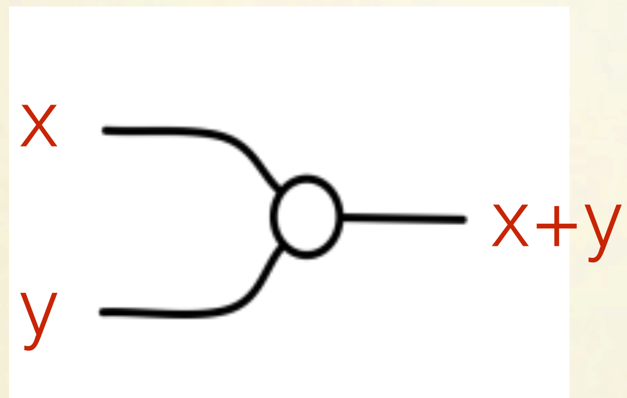
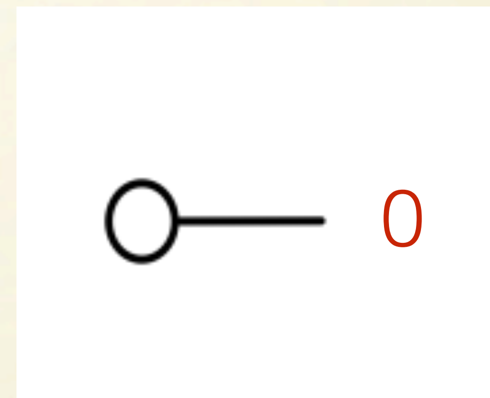
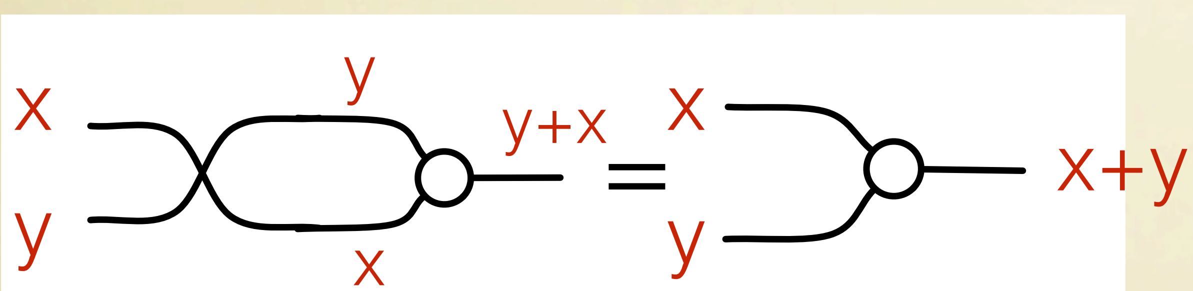
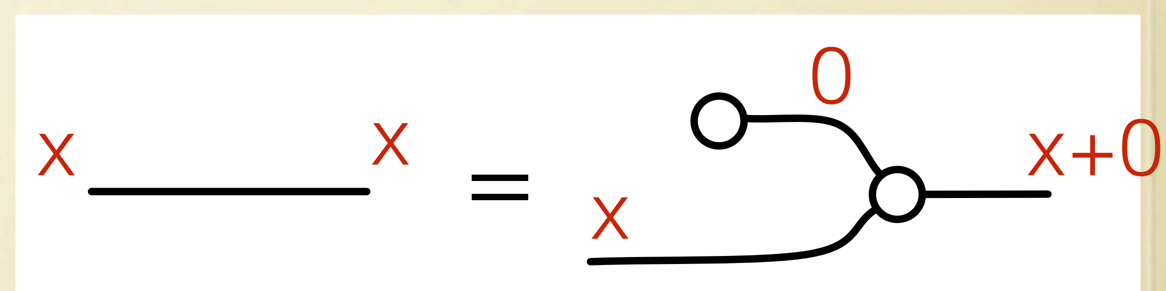
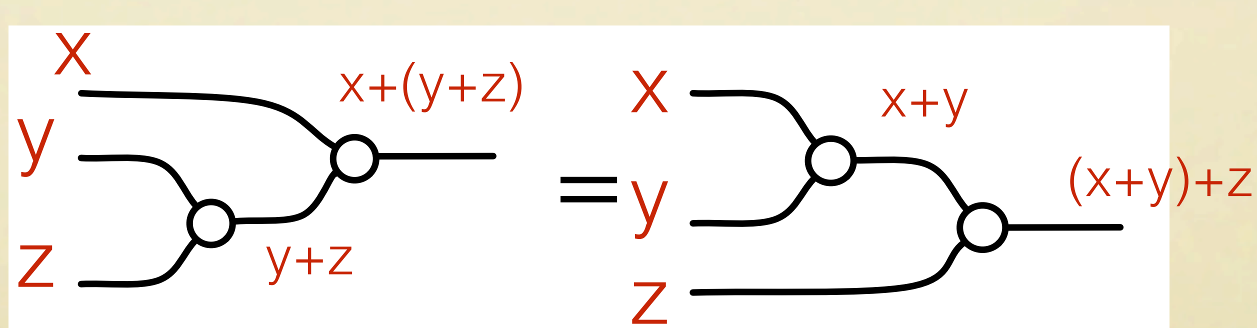




Equations

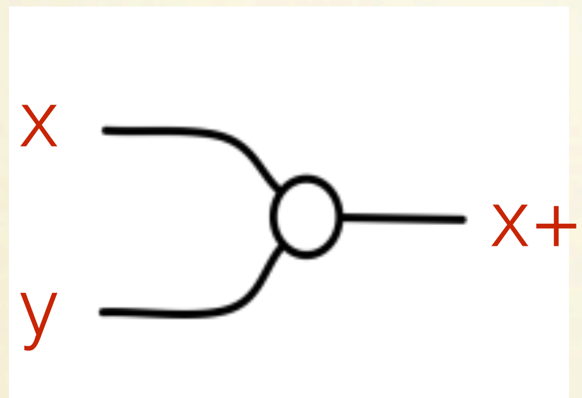


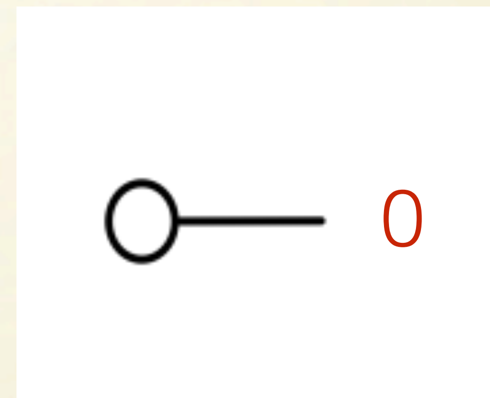




Addition

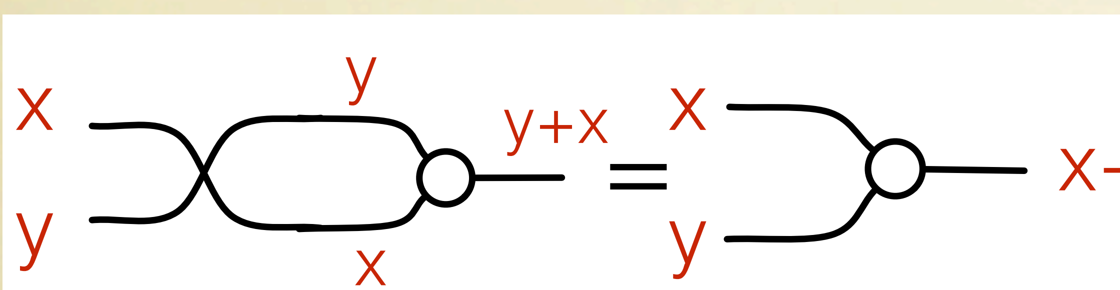
Operations

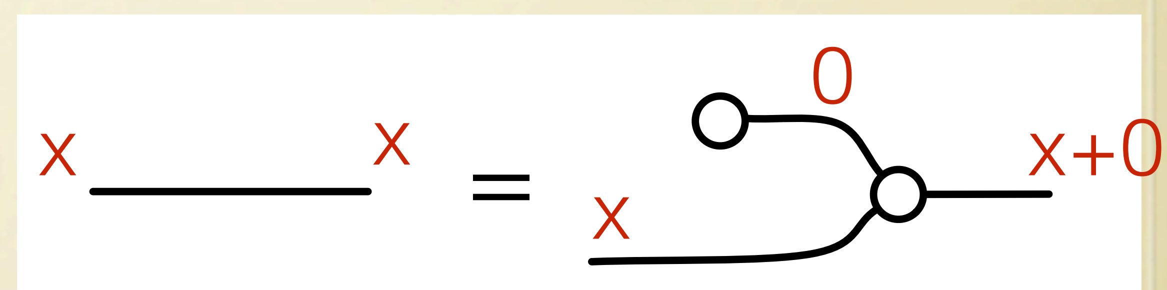


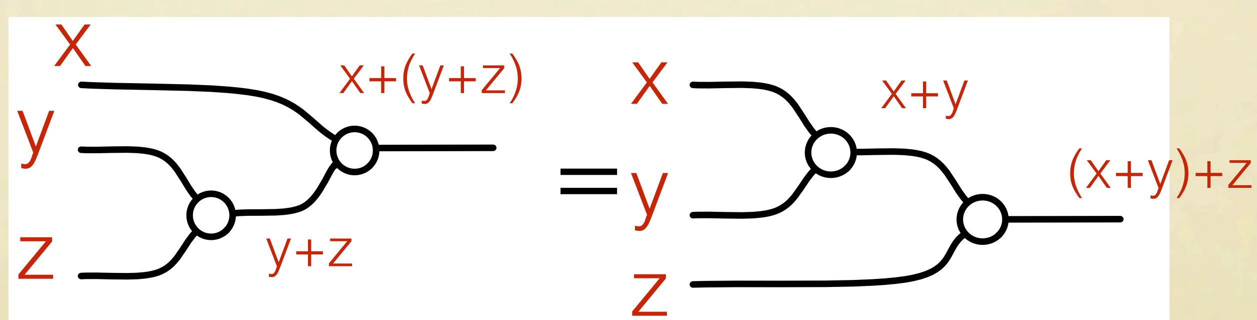


Addition forms a **commutative monoid**

Equations







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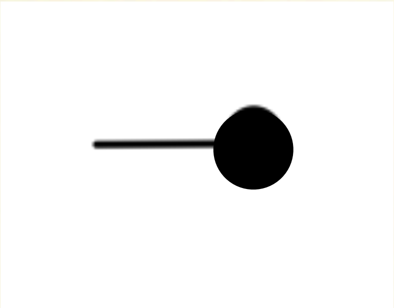
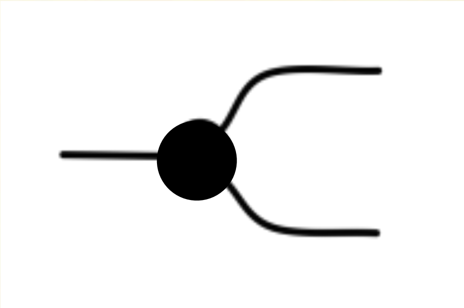
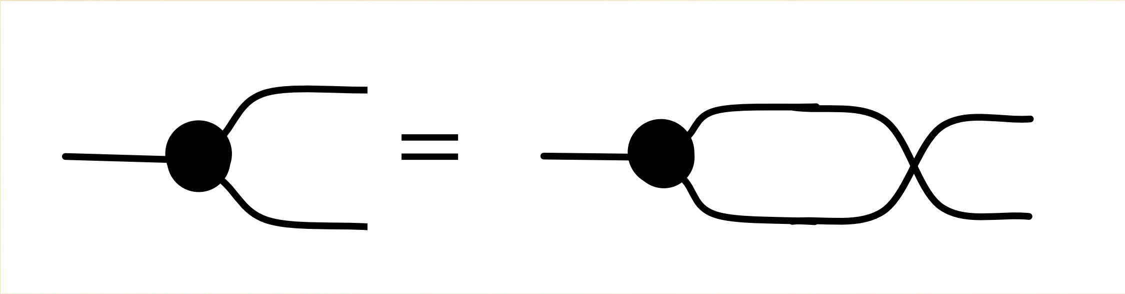
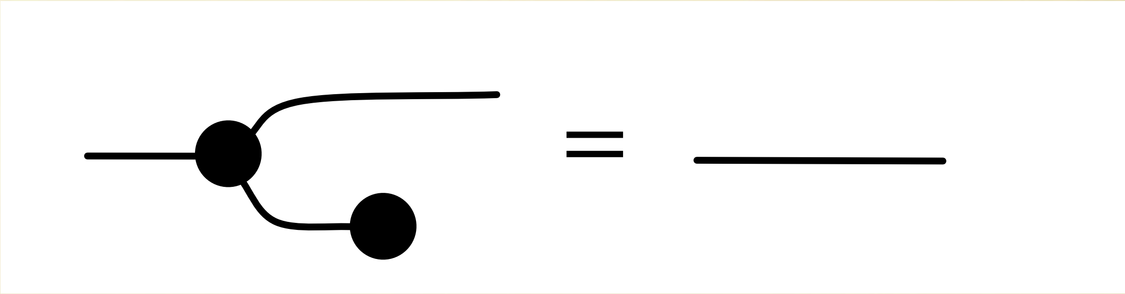
Operations

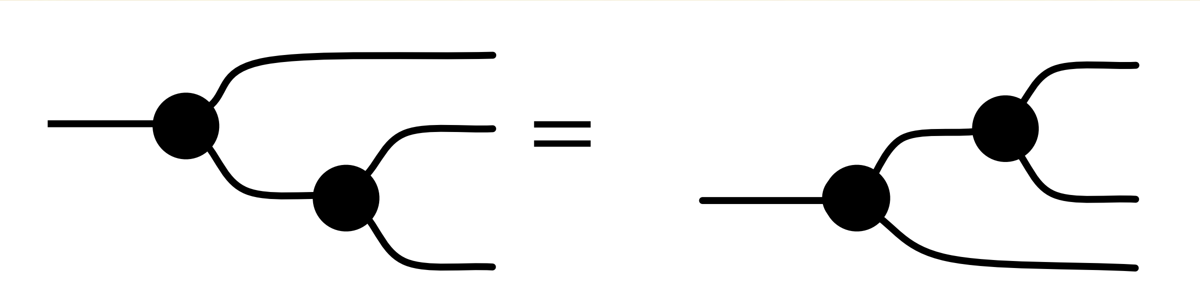




Equations







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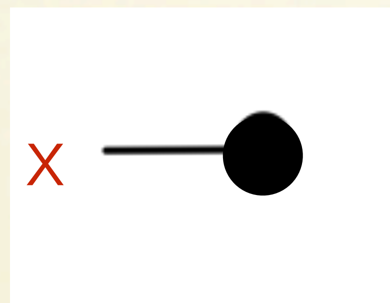
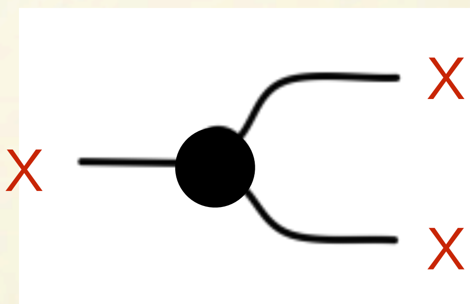
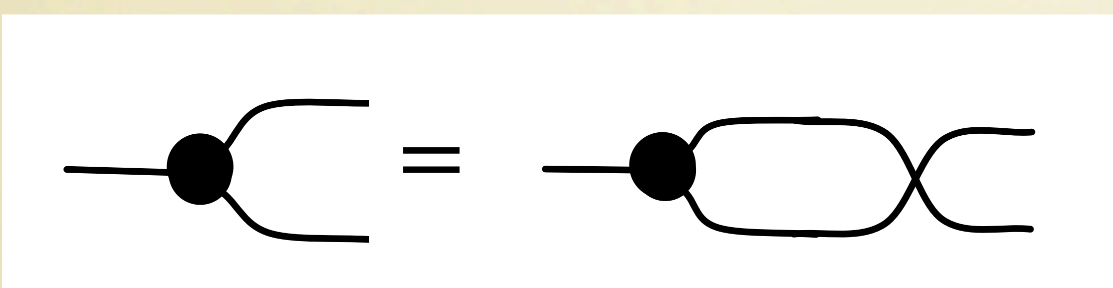
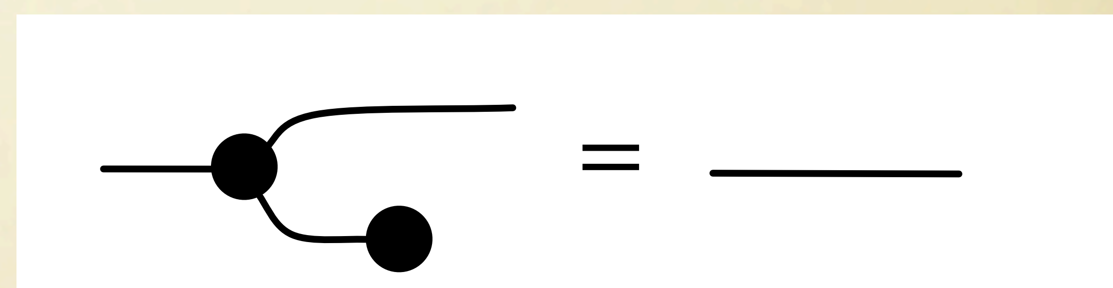
Operations

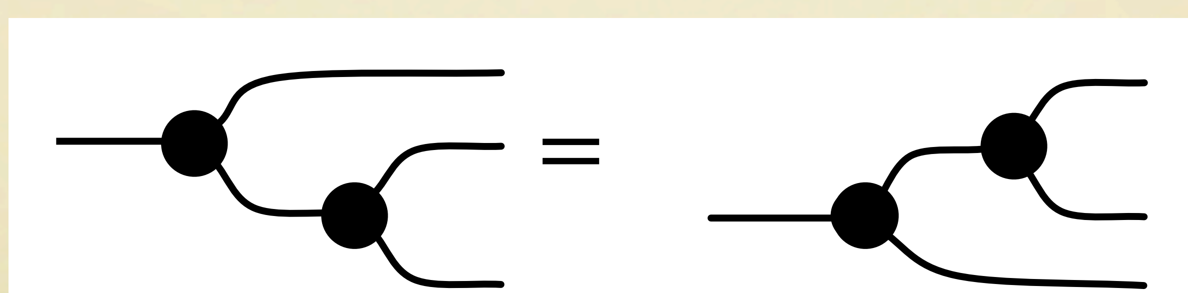




Equations







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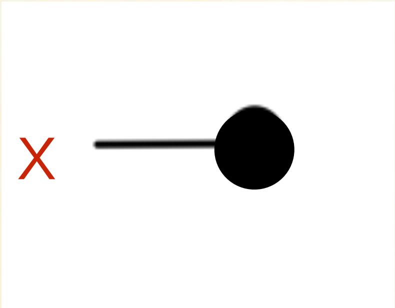
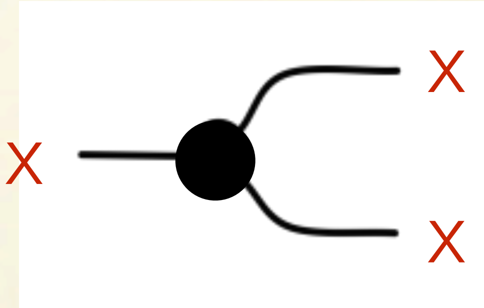
Operations





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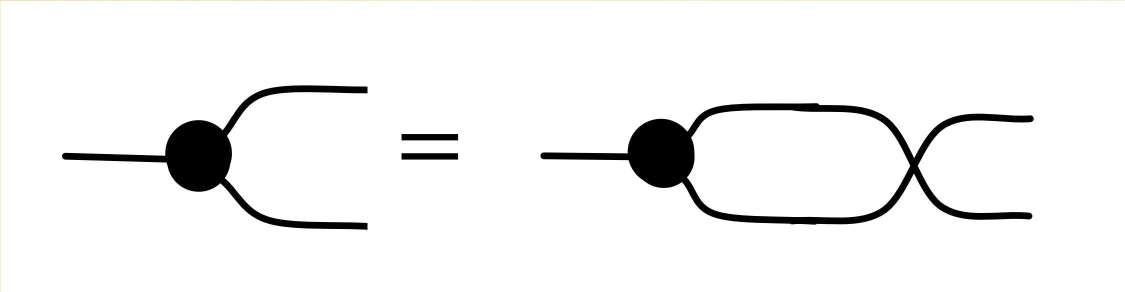
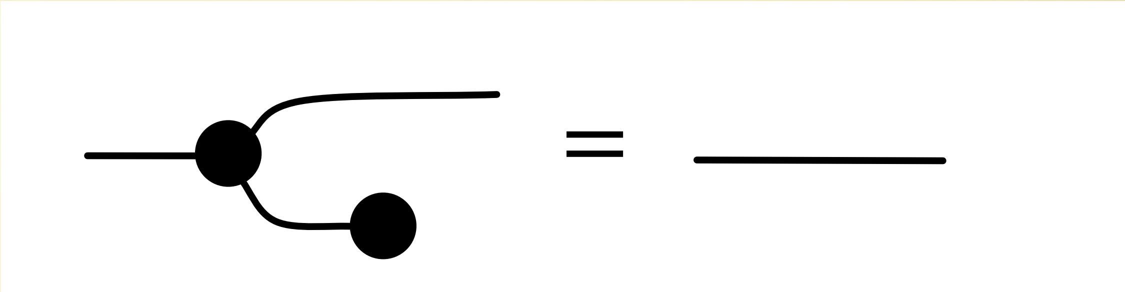
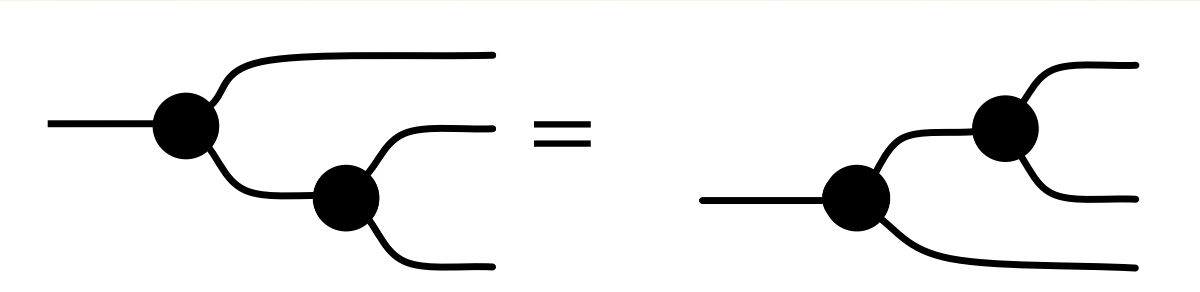
Equations







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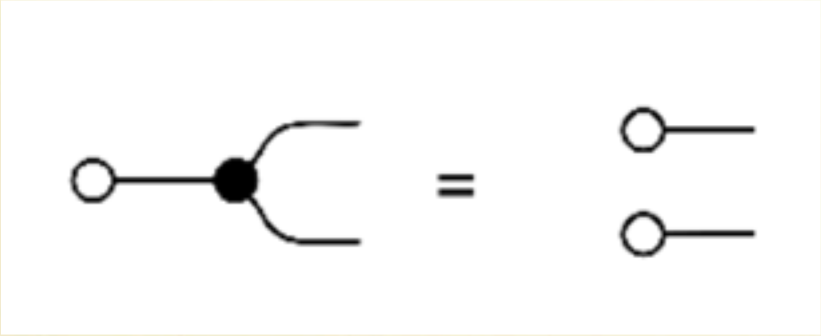

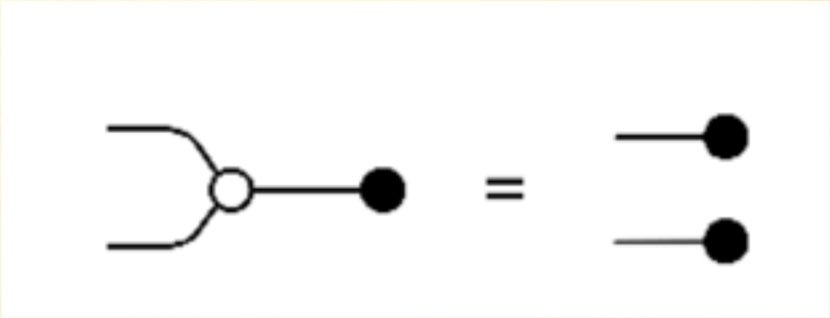

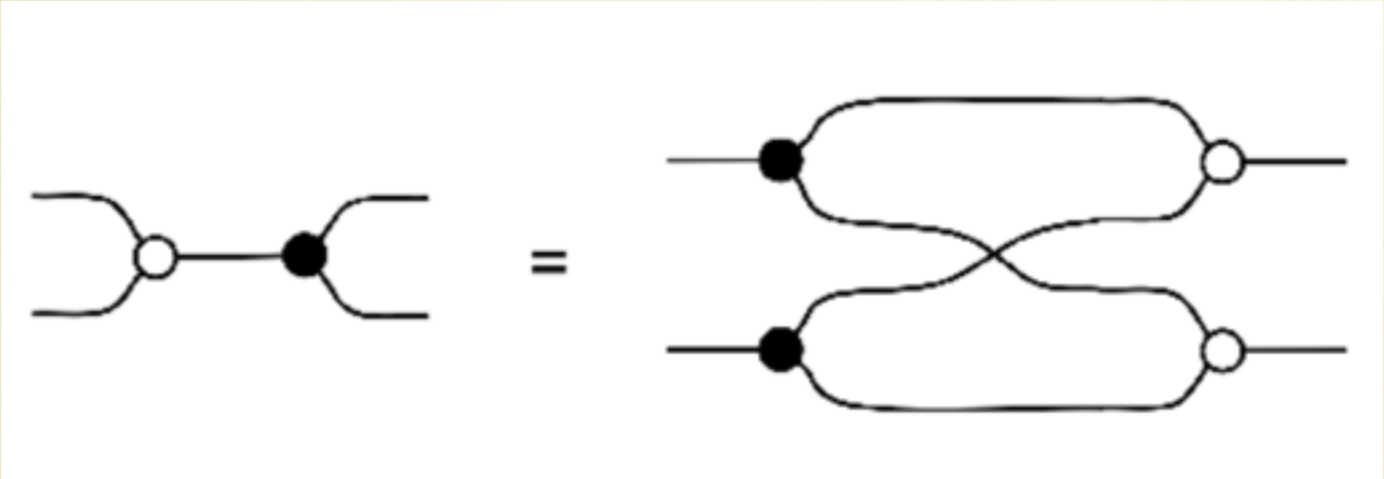

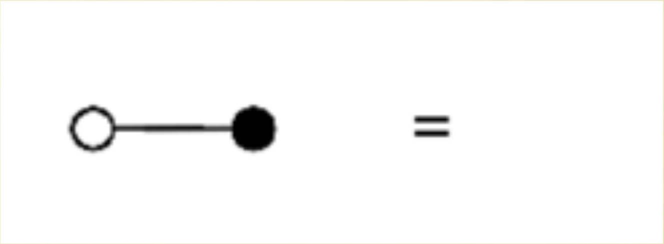

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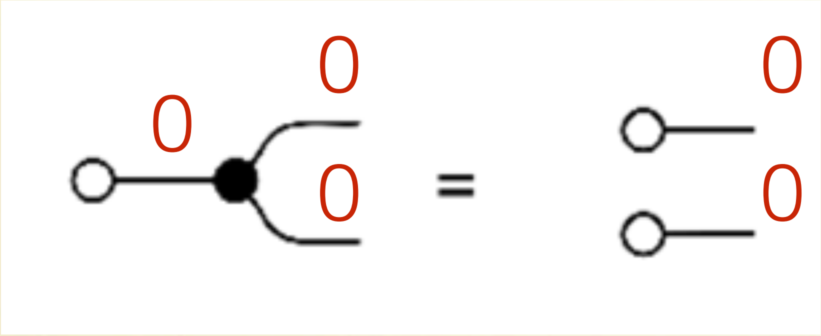

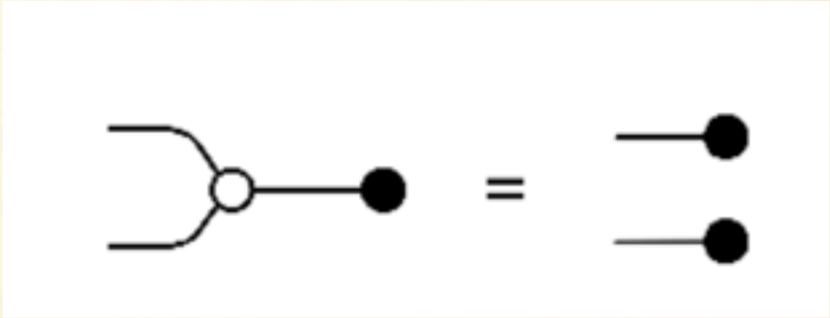

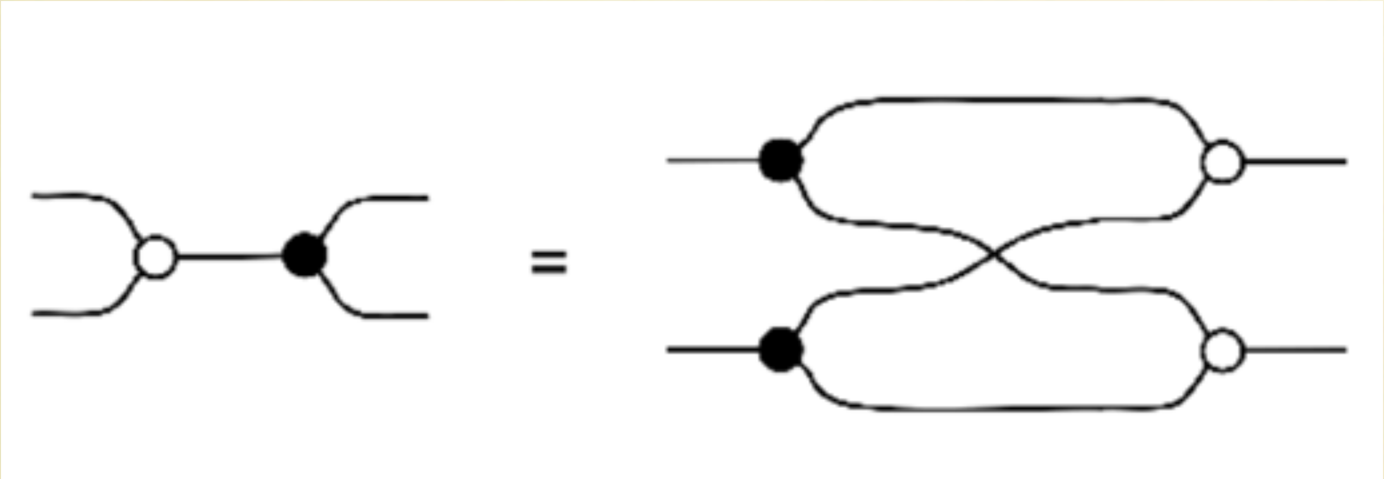

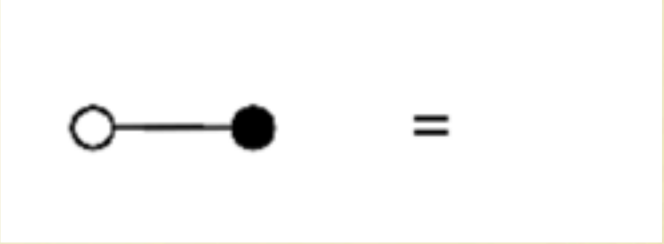

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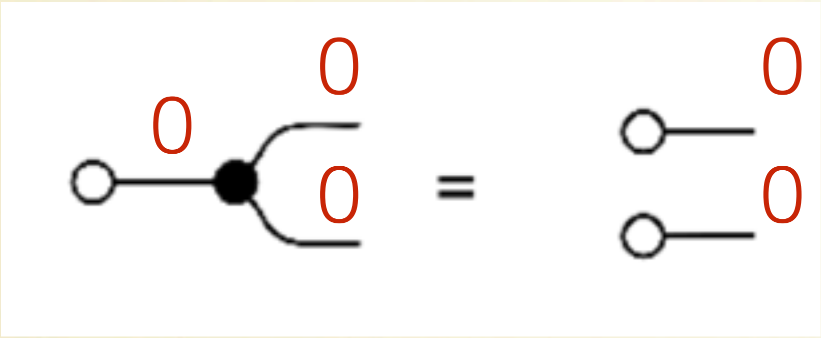

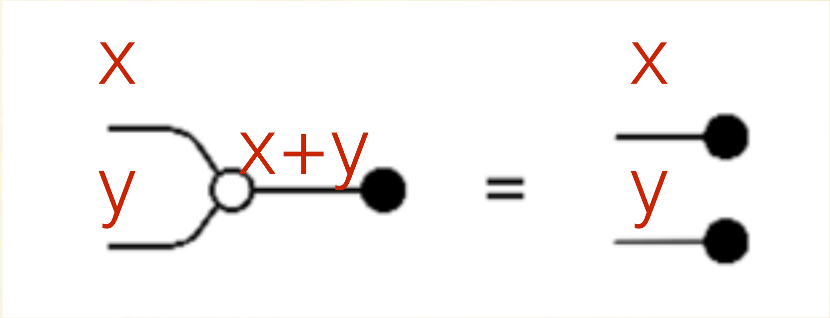

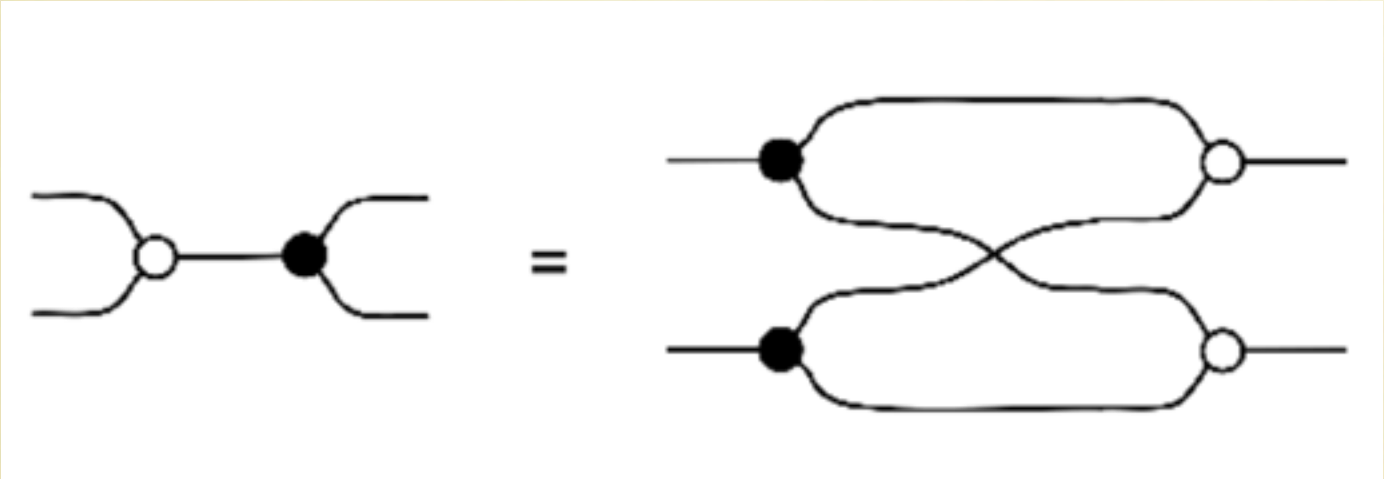

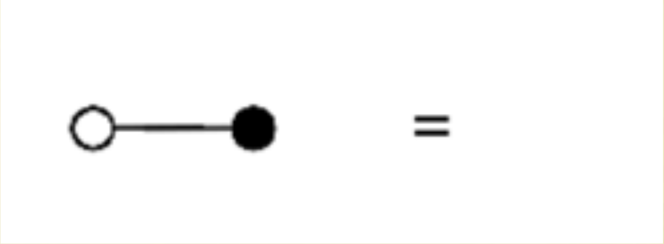

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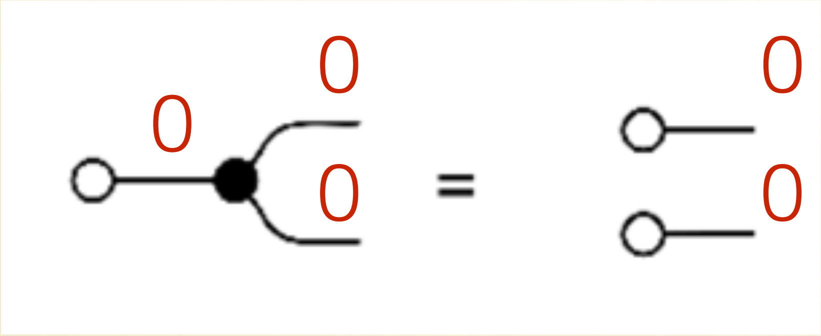

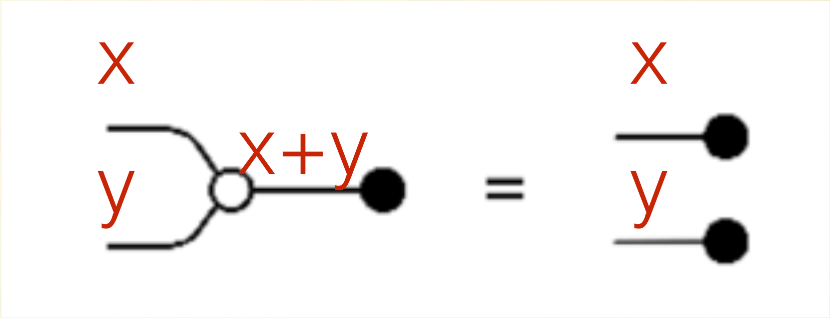

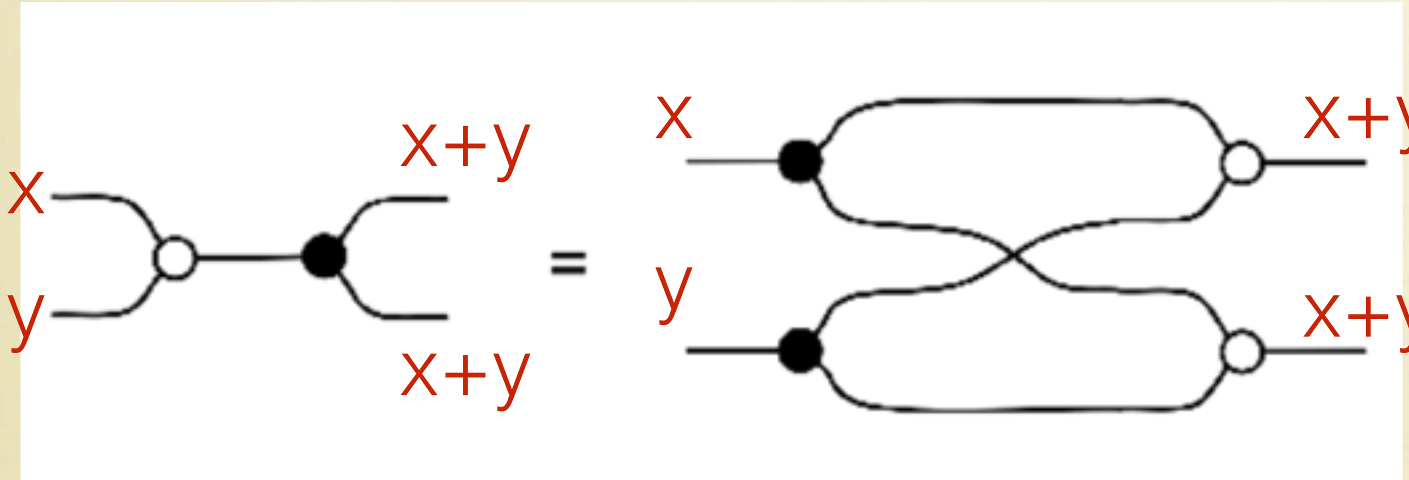

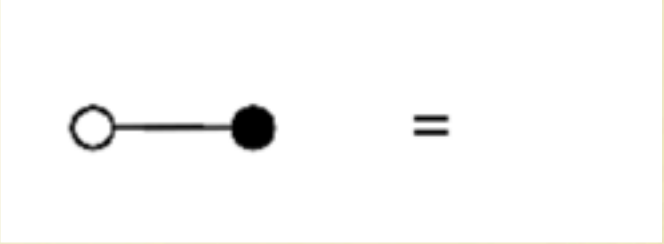

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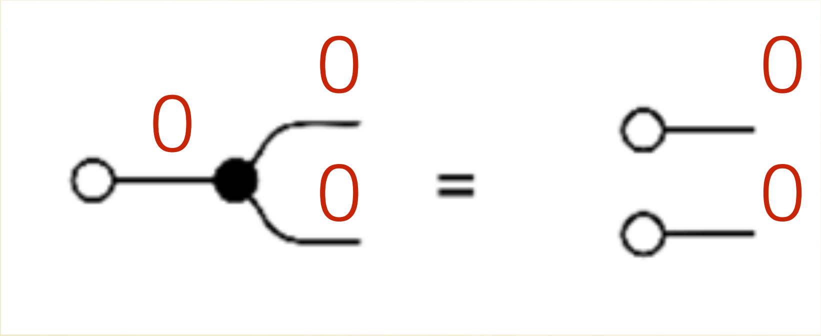
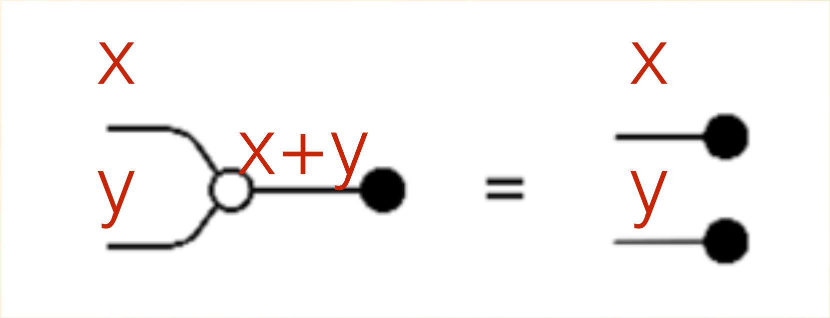
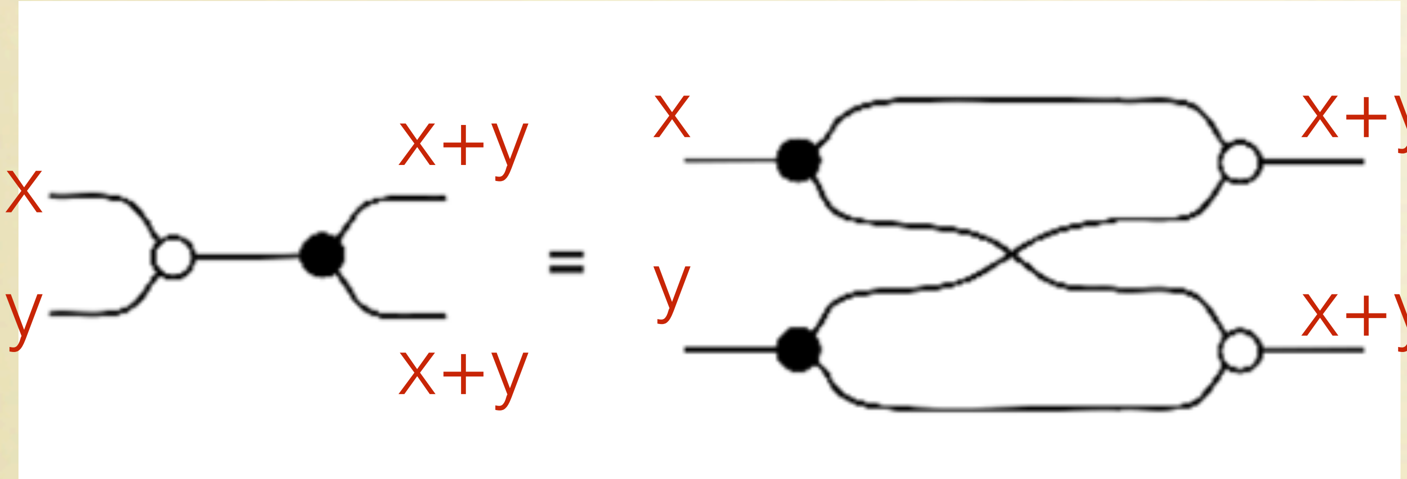
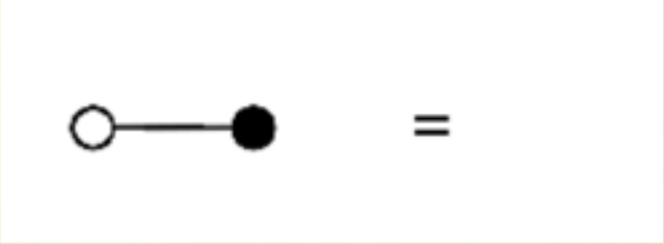
Bialgebra Equations

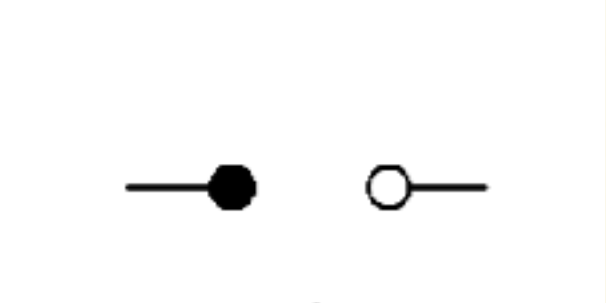


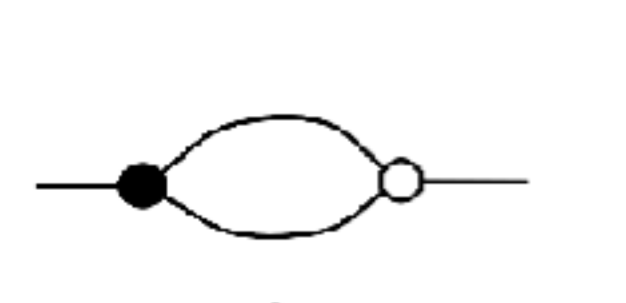


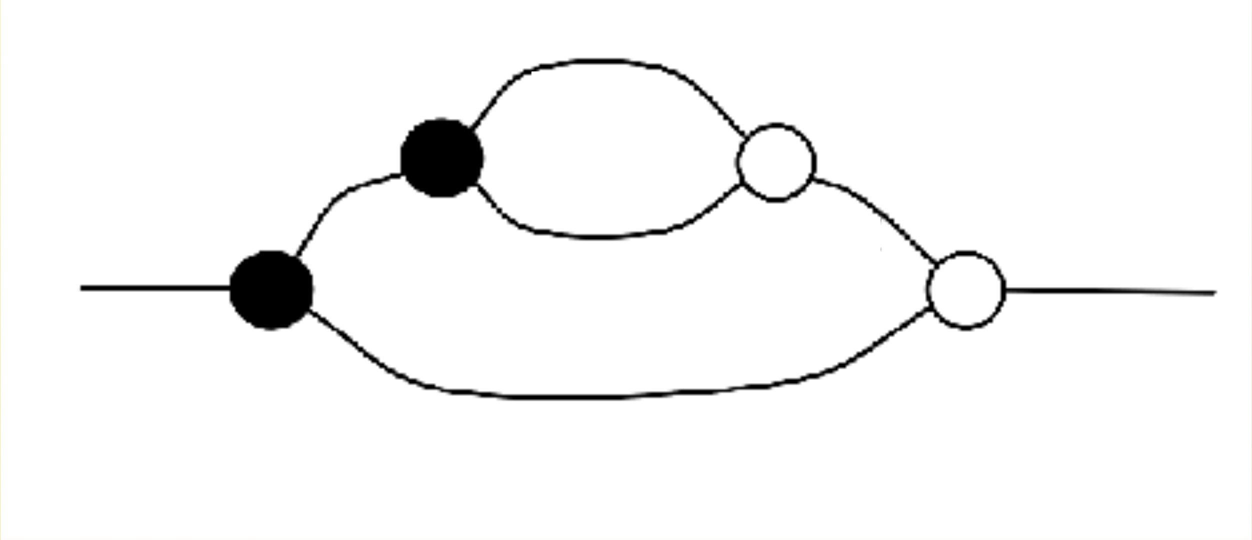




Natural Numbers







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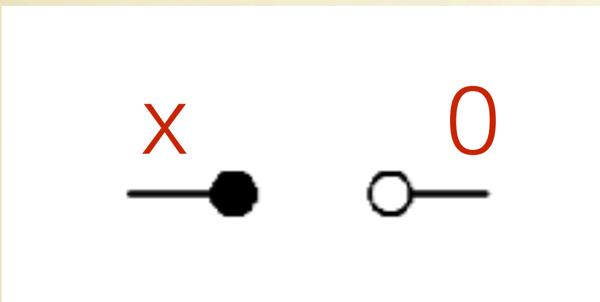

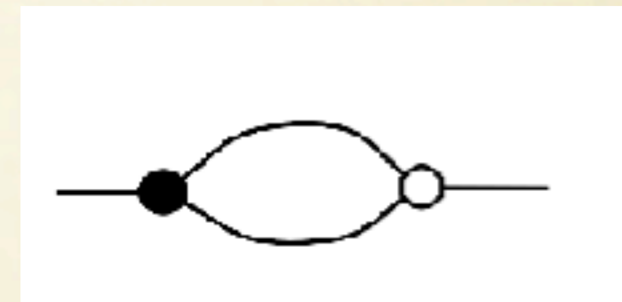

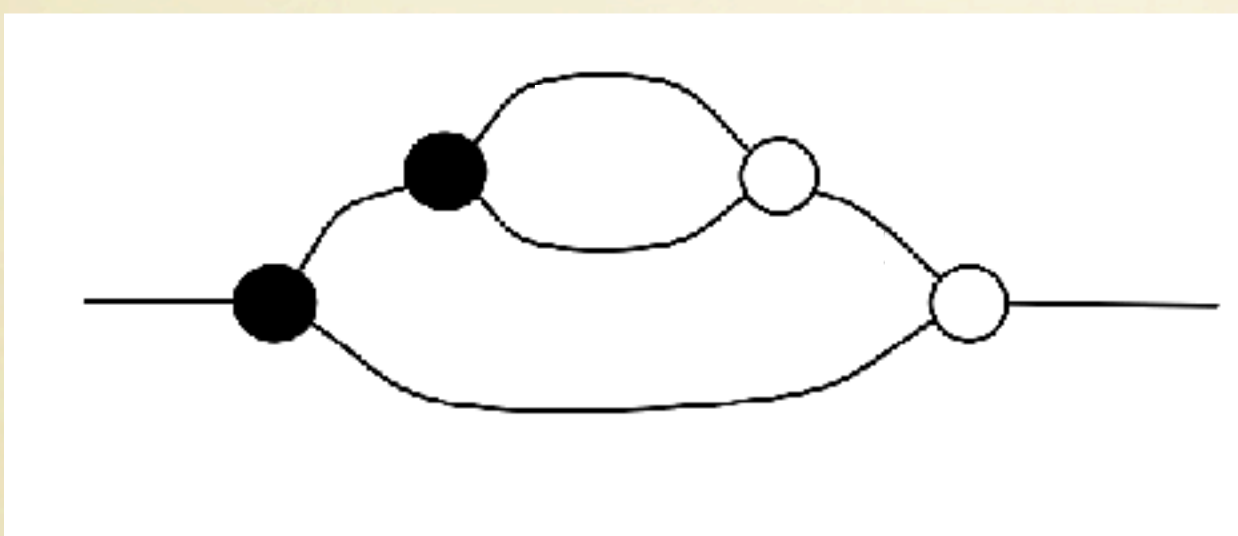

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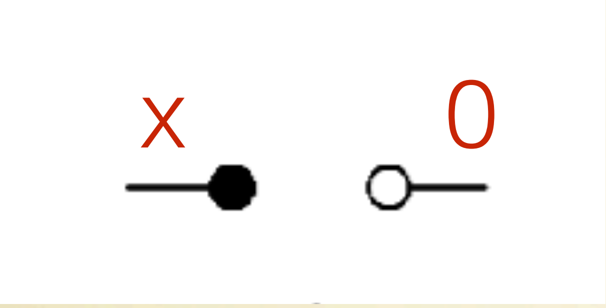

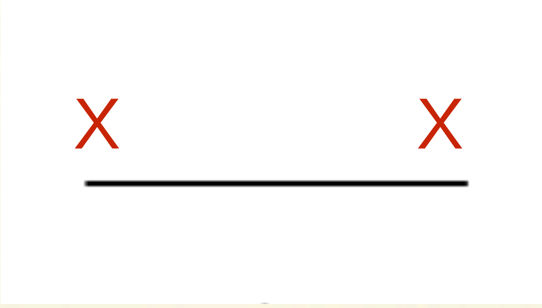

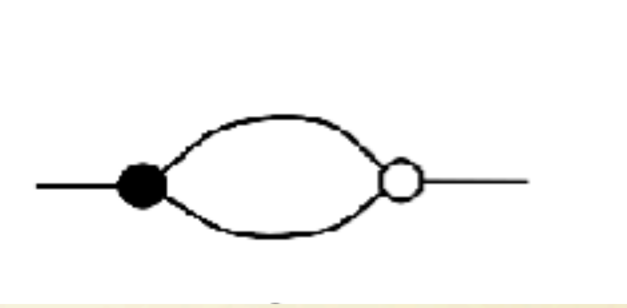

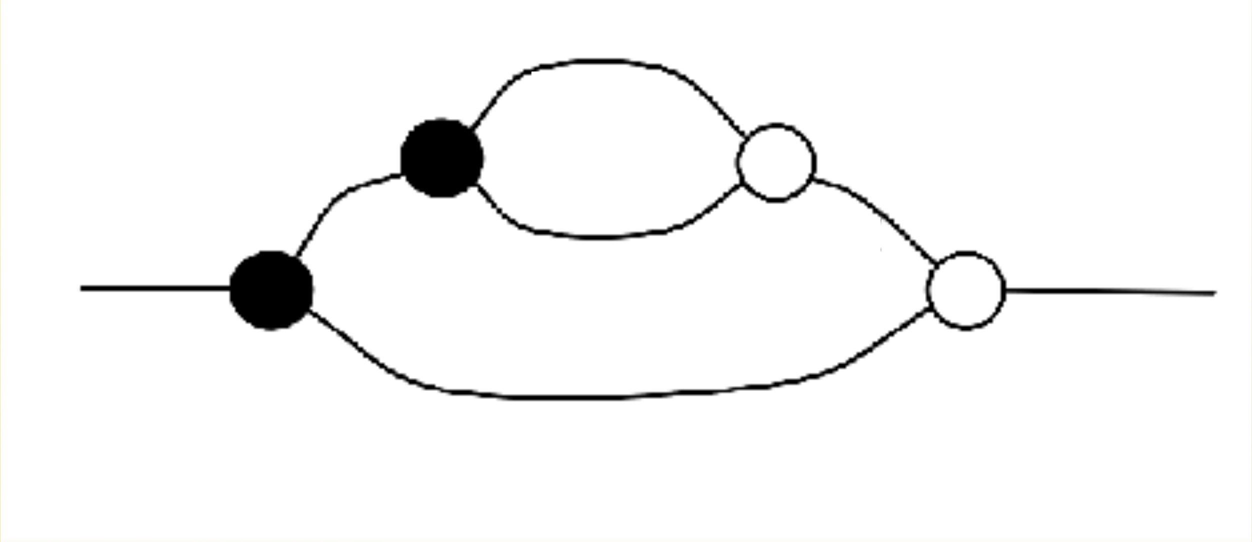

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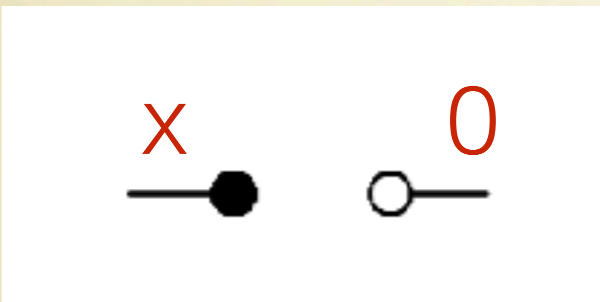
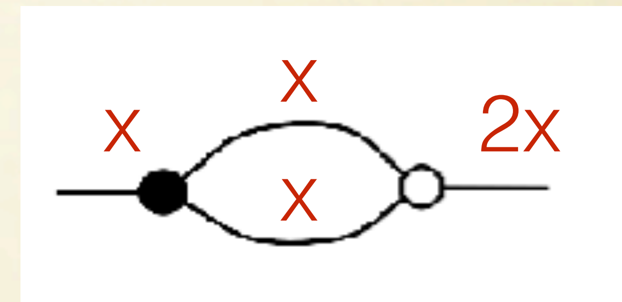
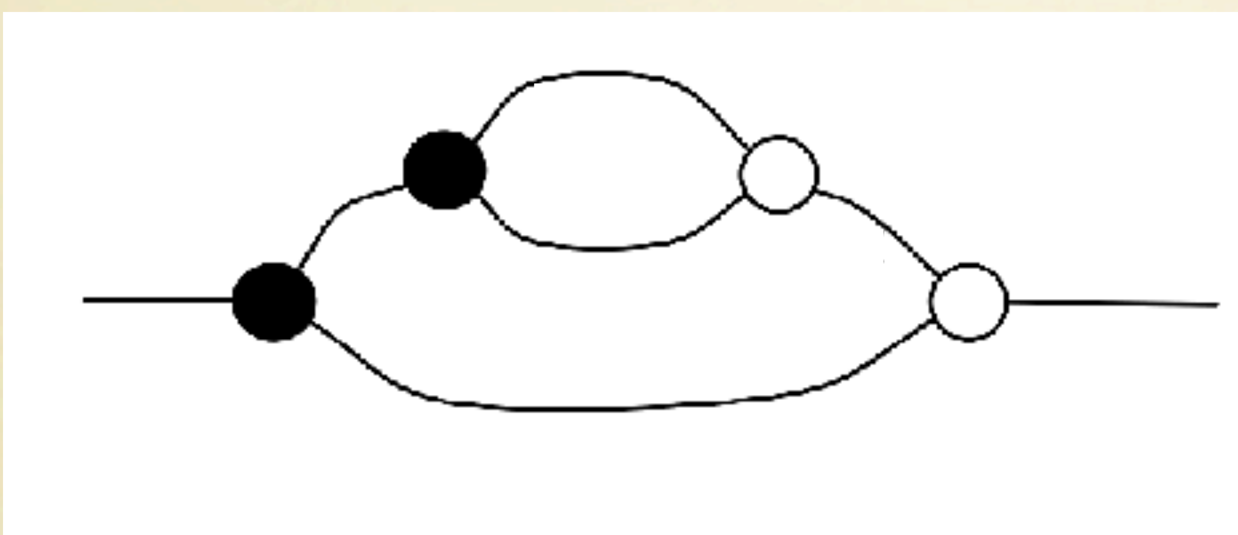
Natural Numbers







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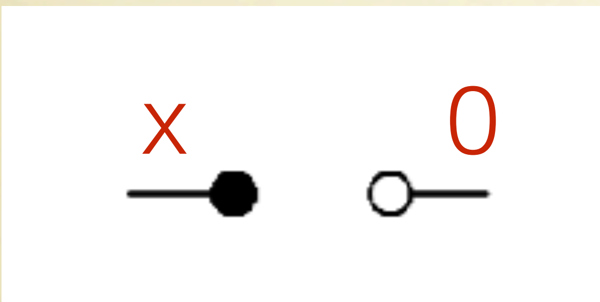
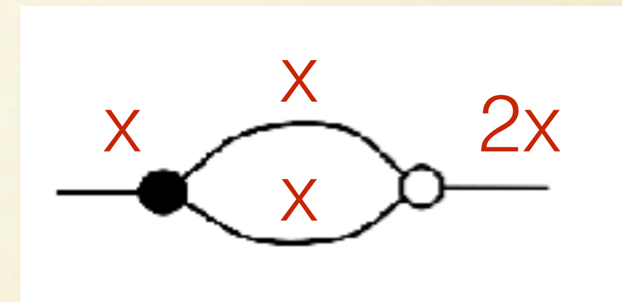
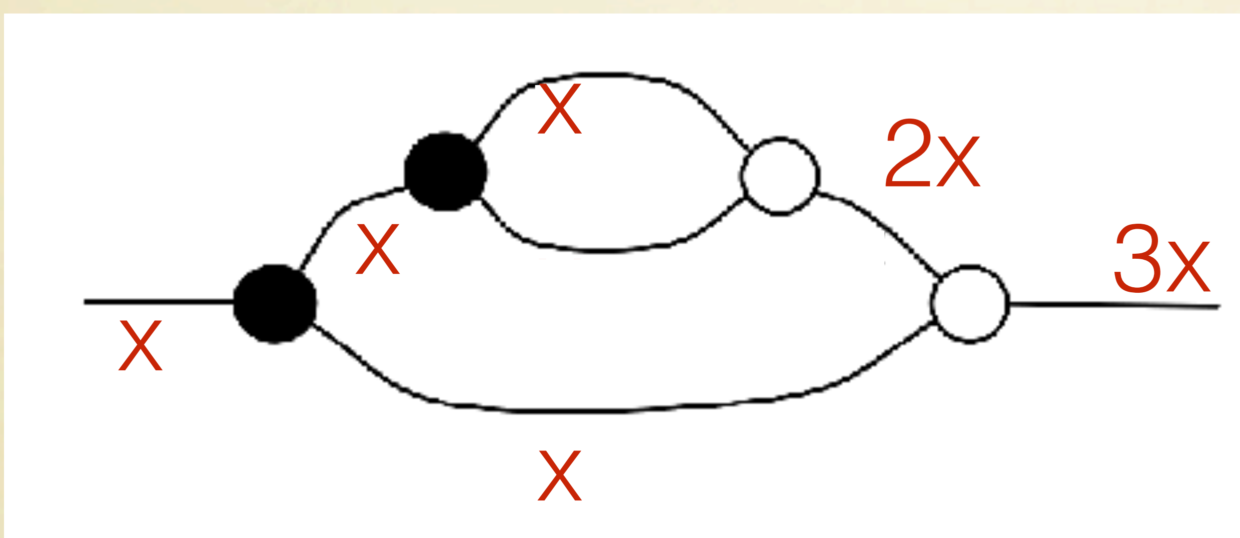
Natural Numbers







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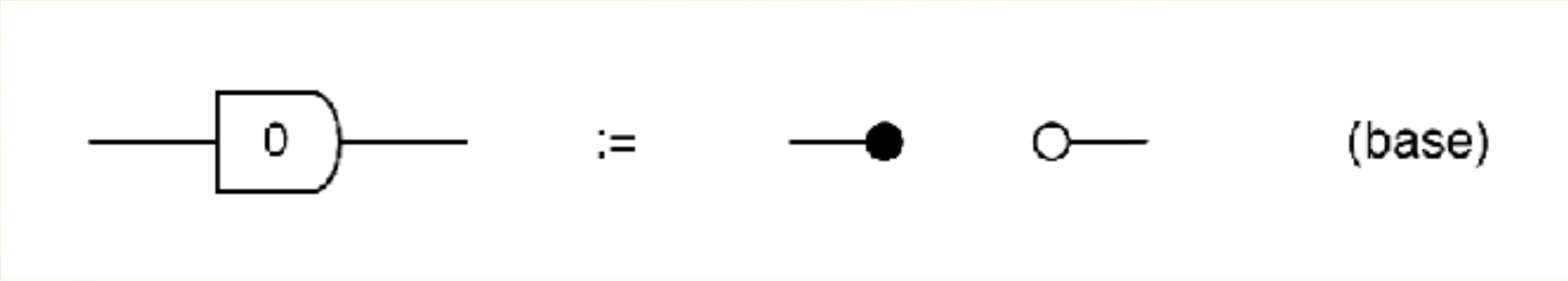
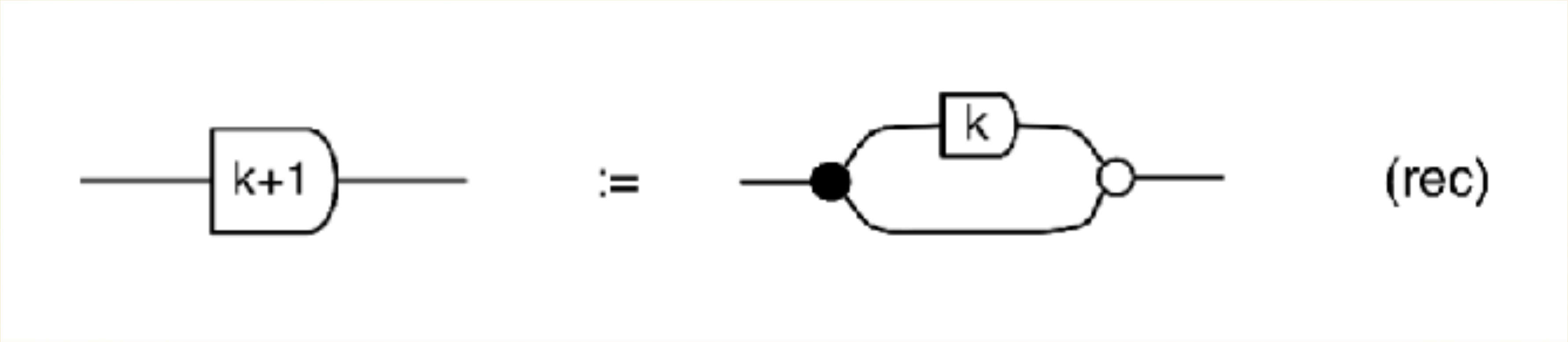
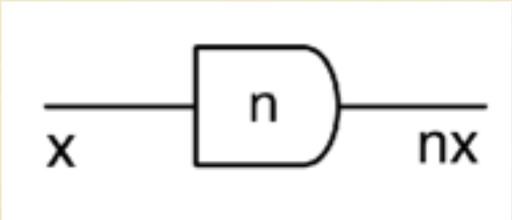
Natural Numbers

Syntactic sugar:





The semantics of n is to **multiply** by n its input.



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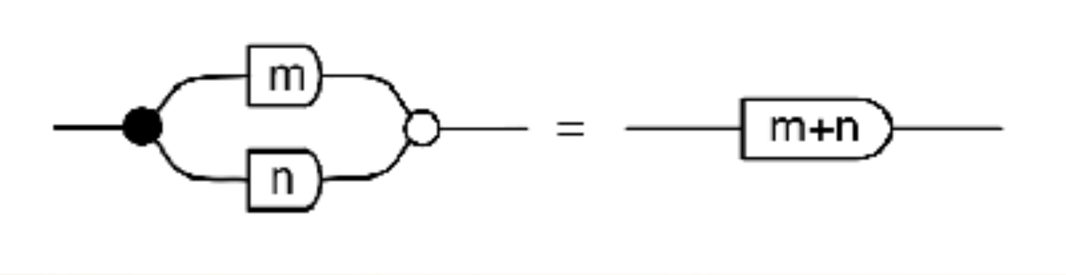
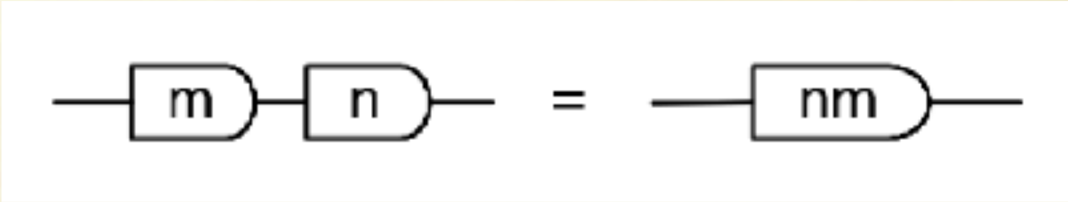
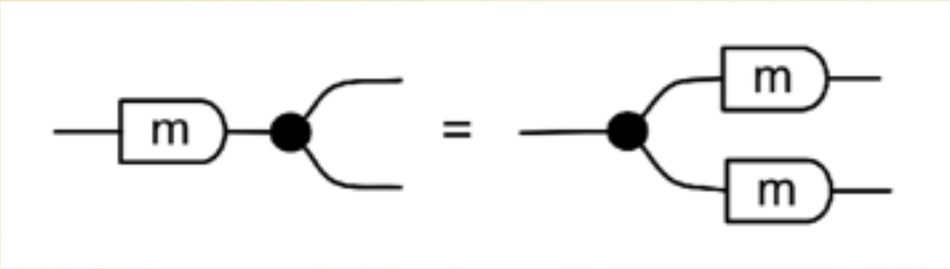
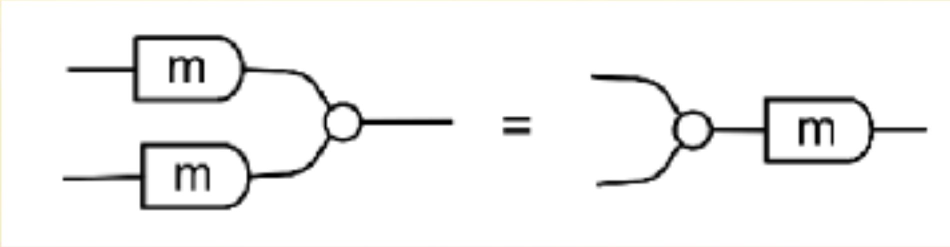
Using the available equations we can prove:

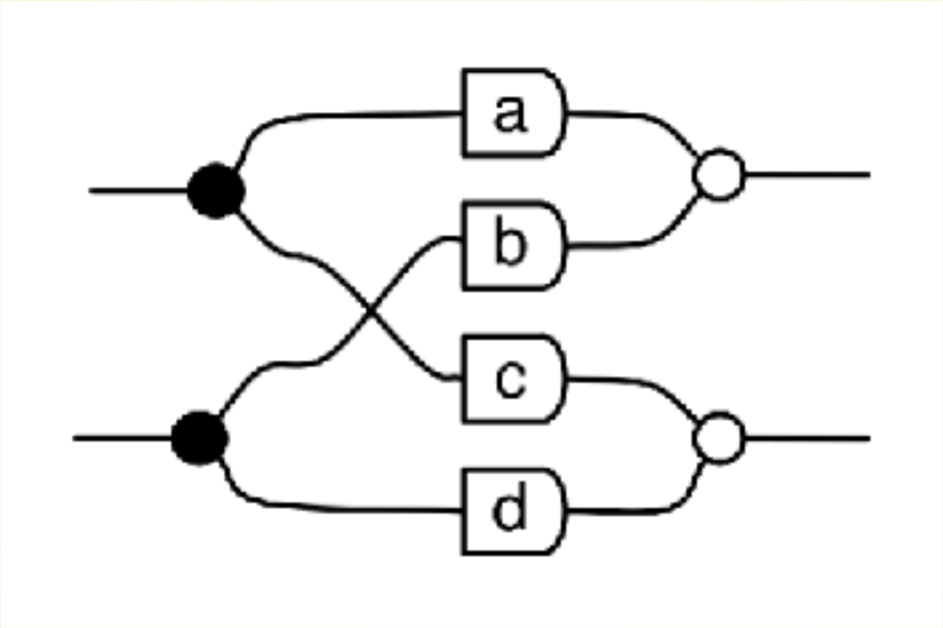




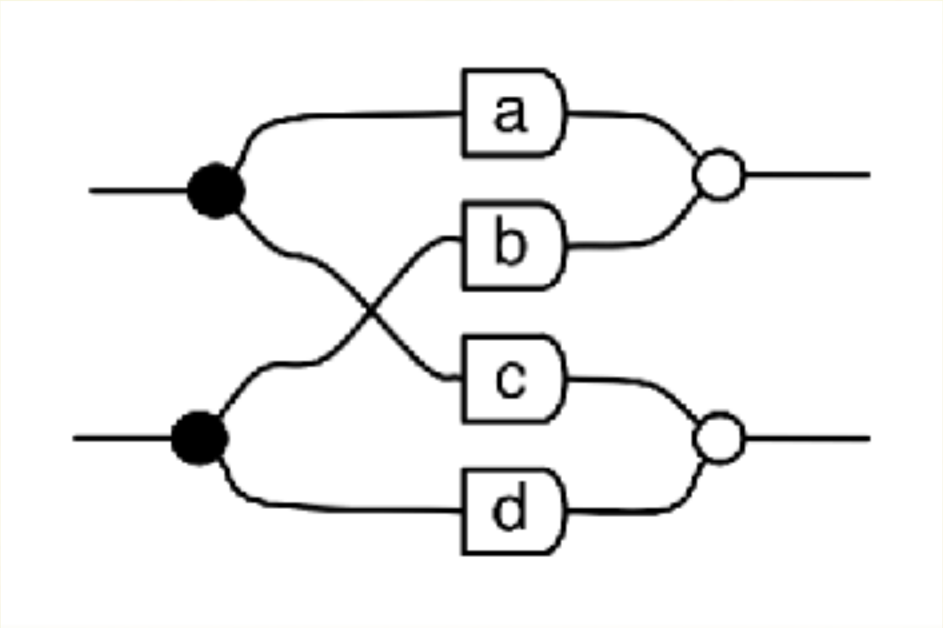




Matrices of Natural Numbers



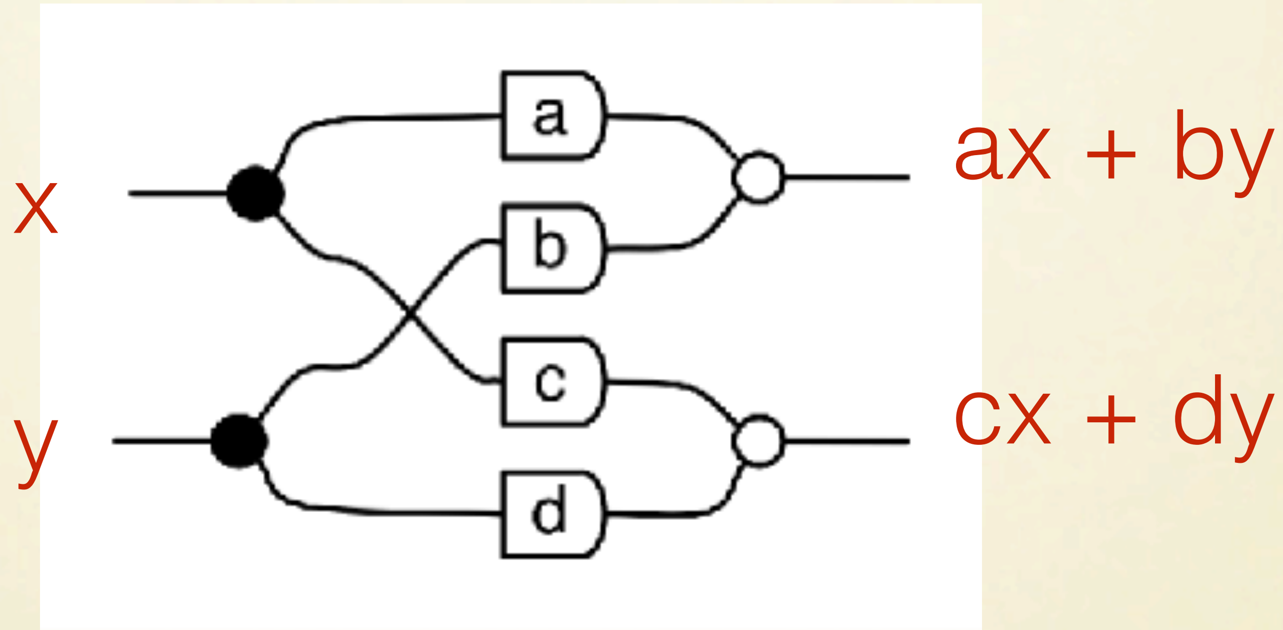
Matrices of Natural Numbers



The semantics of this string diagram is the matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Matrices of Natural Numbers



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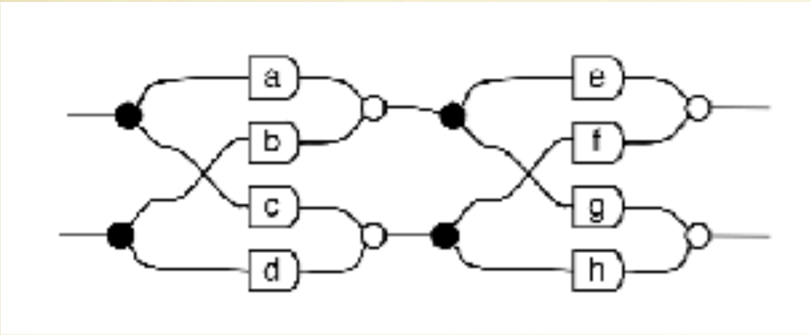
Matrix Multiplication

Sequential composition of string diagrams corresponds to **matrix multiplication**



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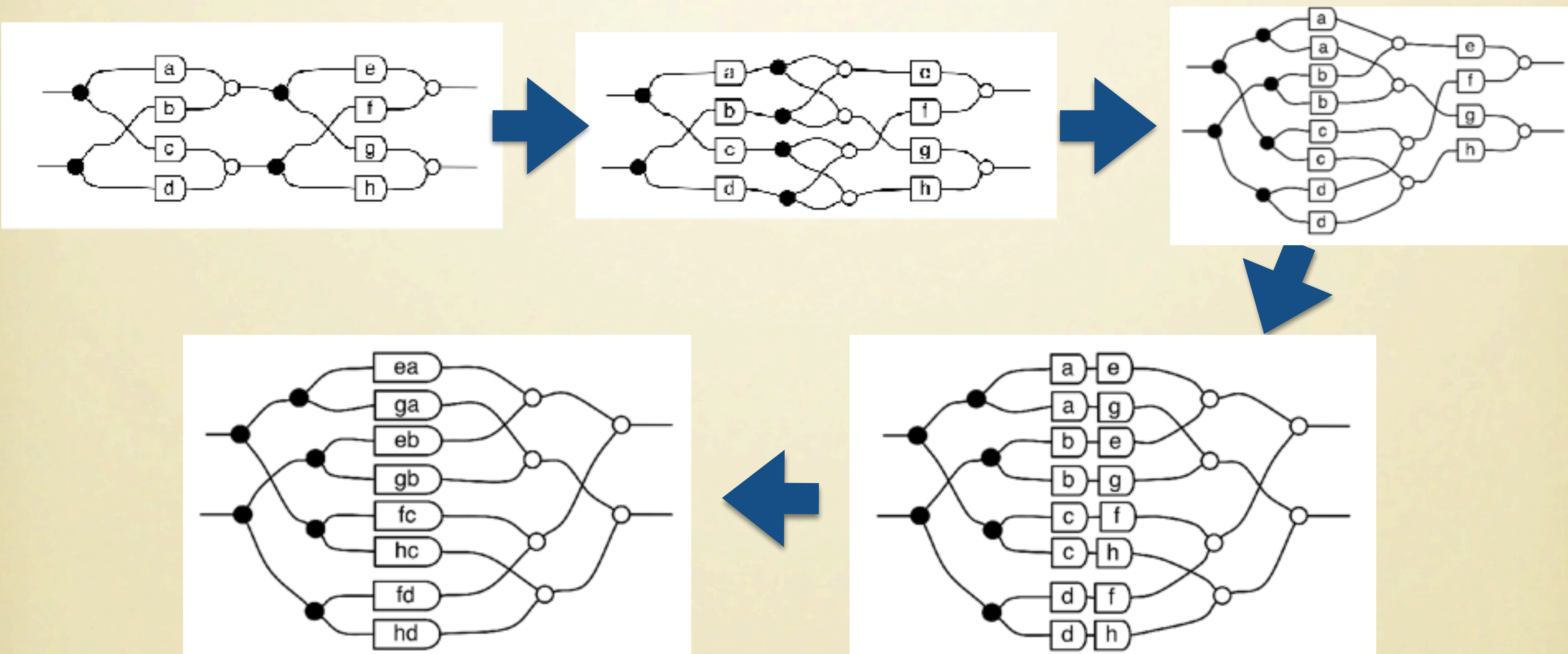

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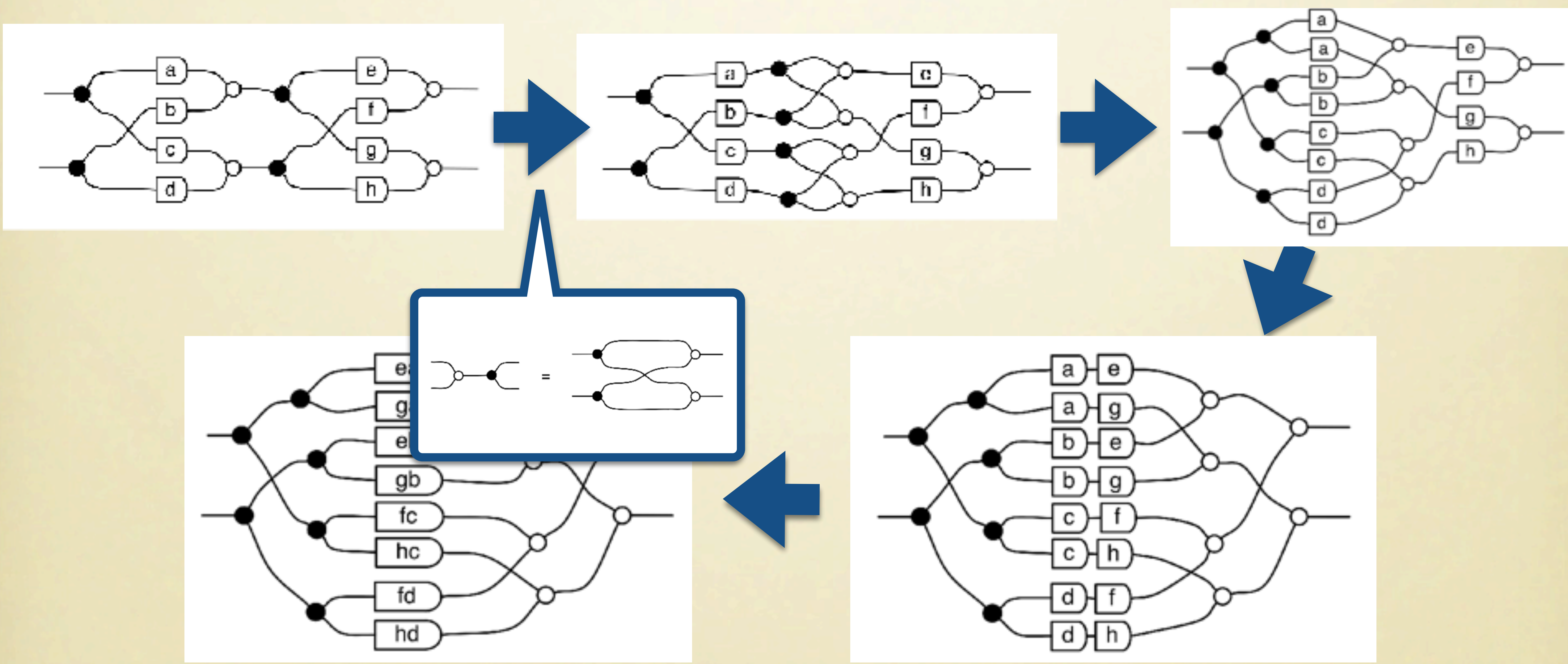

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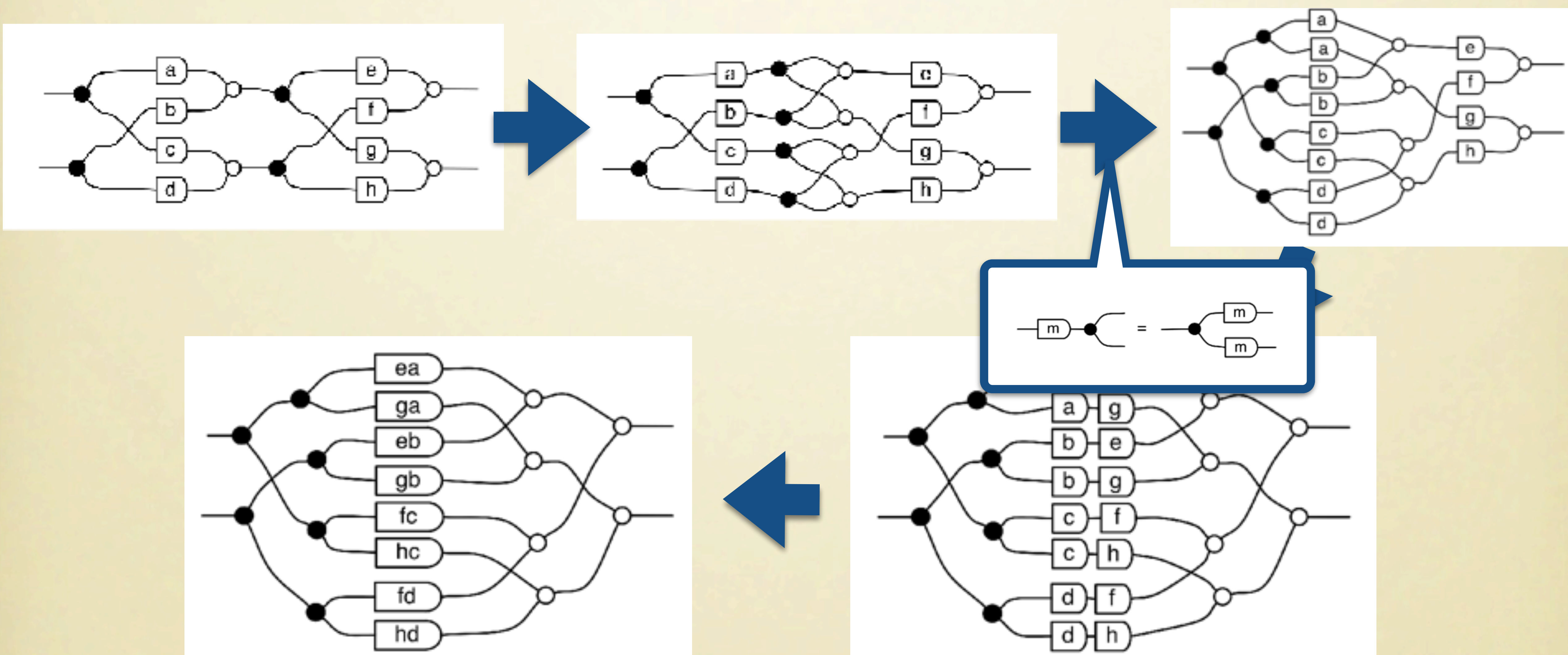

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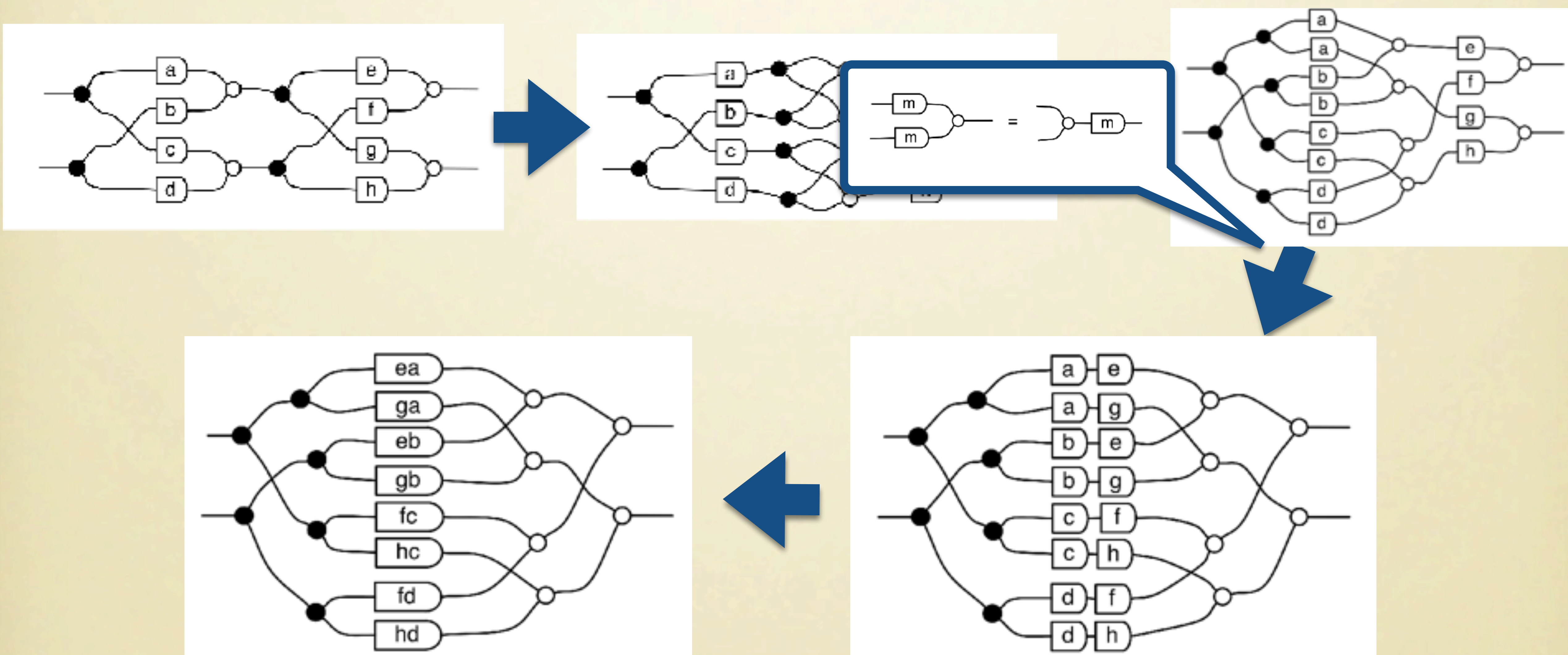

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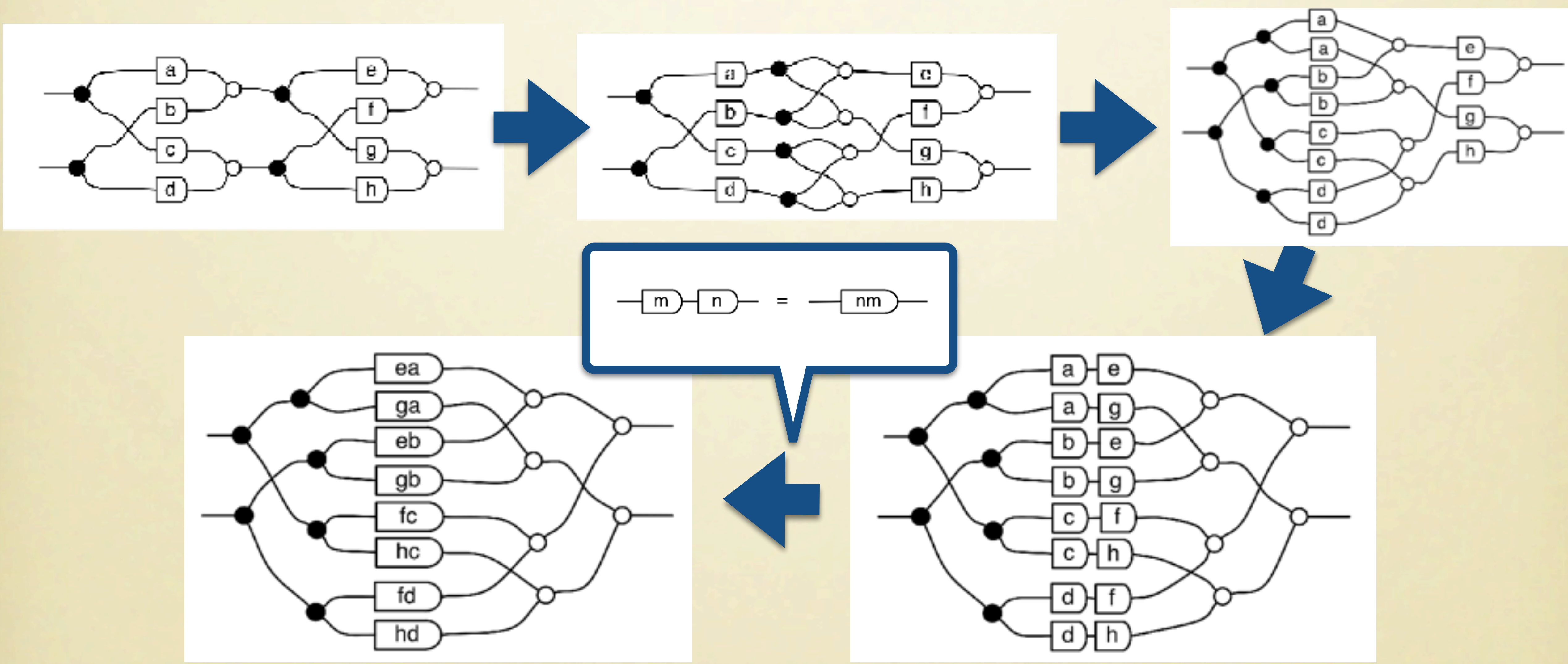

Graphical Matrix Algebra

Theorem

There is a 1-1 correspondence between:

- String Diagrams with n left wires and m right wires *modulo the equations of bialgebras*.
- $\mathbb{N}$-matrices with n columns and m rows.

Beyond Natural Numbers

We can easily reason about matrices over any semiring **R**, not just $\mathbb{N}$.

The difference is that we cannot treat scalars as syntactic sugar, but need to be introduced explicitly in the string diagram syntax and equations.

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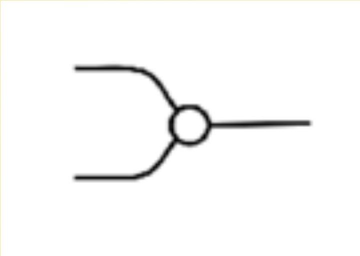

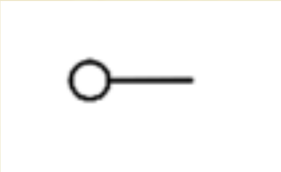

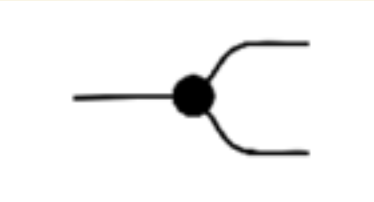

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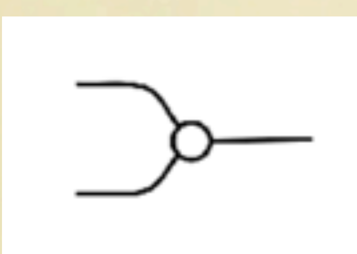
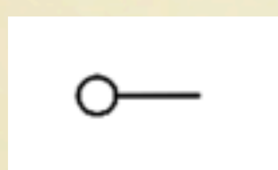
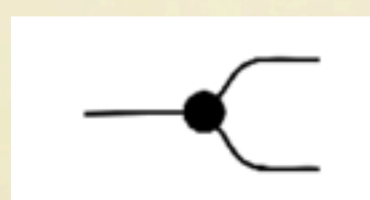
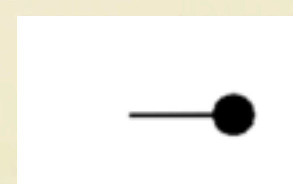
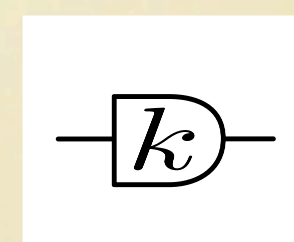
$k \in \mathsf{R}$











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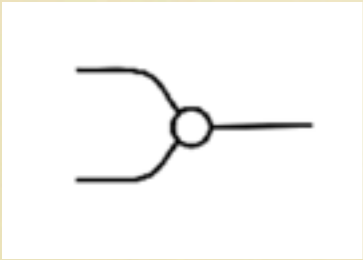
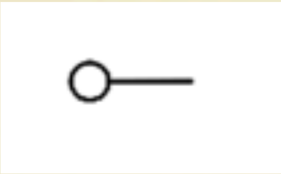
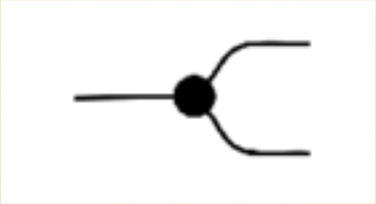
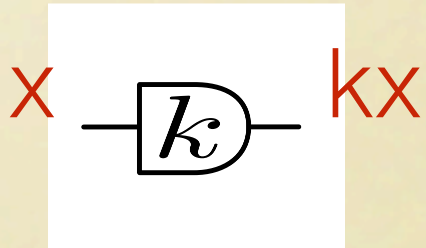




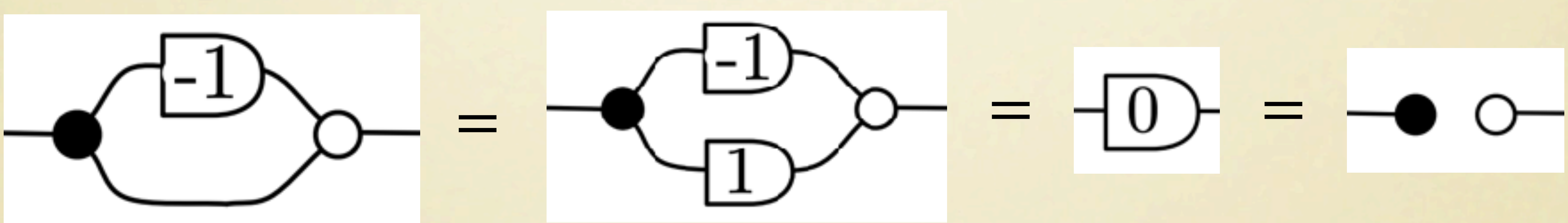




Beyond Natural Numbers

When we deal with an actual ring, the equational theory is not just bialgebras anymore, it's **Hopf algebras**.

The scalar -1 plays the role of the **antipode**. Indeed:



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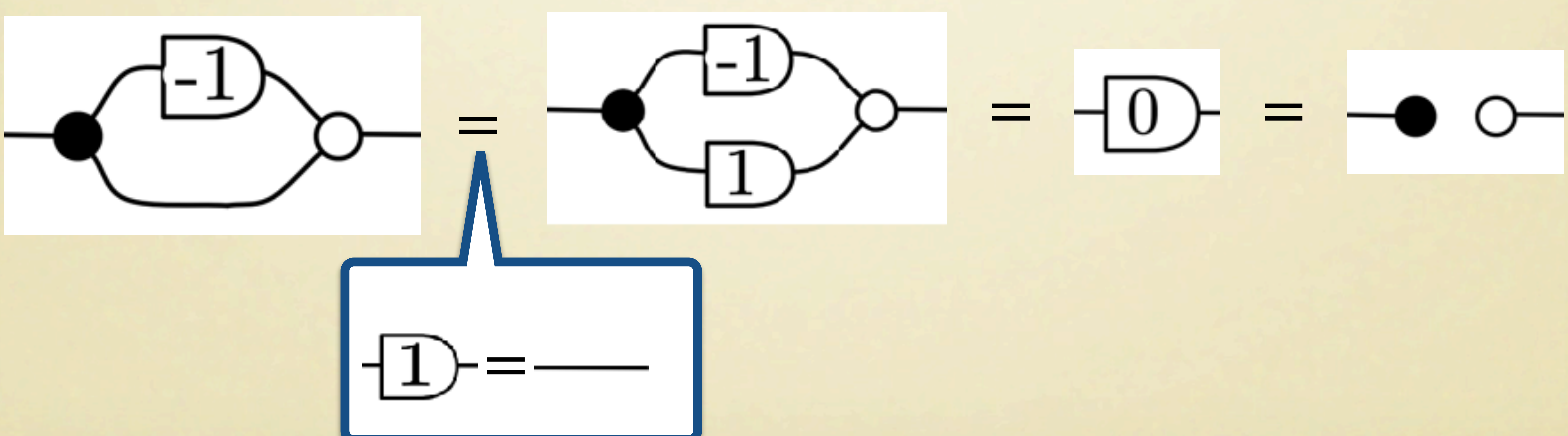
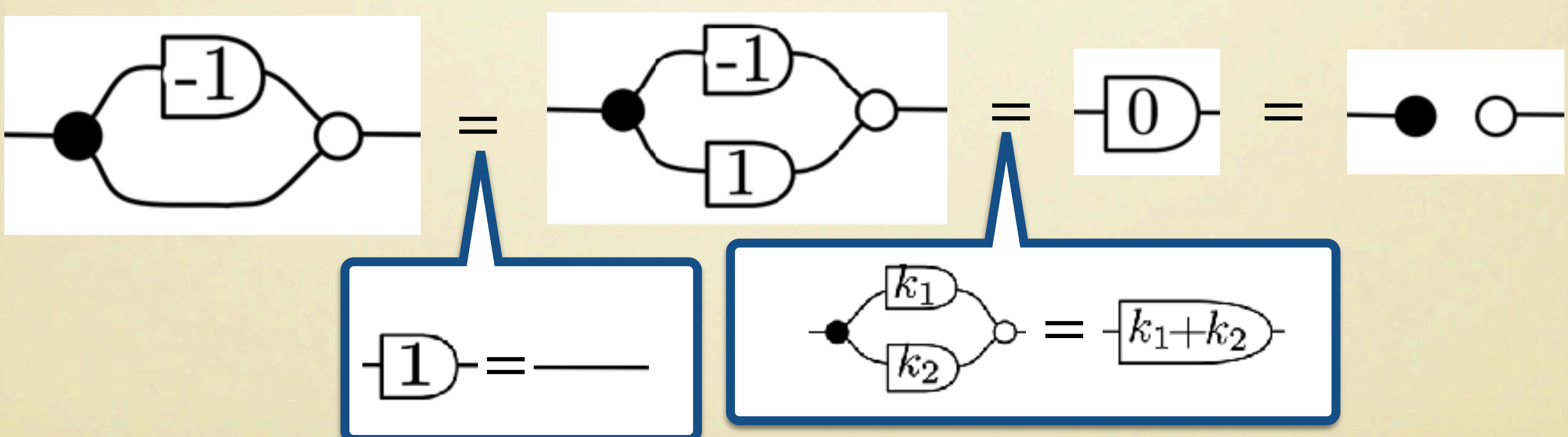


Beyond Natural Numbers

When we deal with an actual ring, the equational theory is not just bialgebras anymore, it's **Hopf algebras**.

The scalar -1 plays the role of the **antipode**. Indeed:



Graphical Matrix Algebra (Revisited)

Theorem

There is a 1-1 correspondence between:

- String Diagrams with n left wires and m right wires *modulo the equations of* R*-Hopf Algebras*.
- R-matrices with n columns and m rows.

Full Linear Algebra:
the calculus of relations

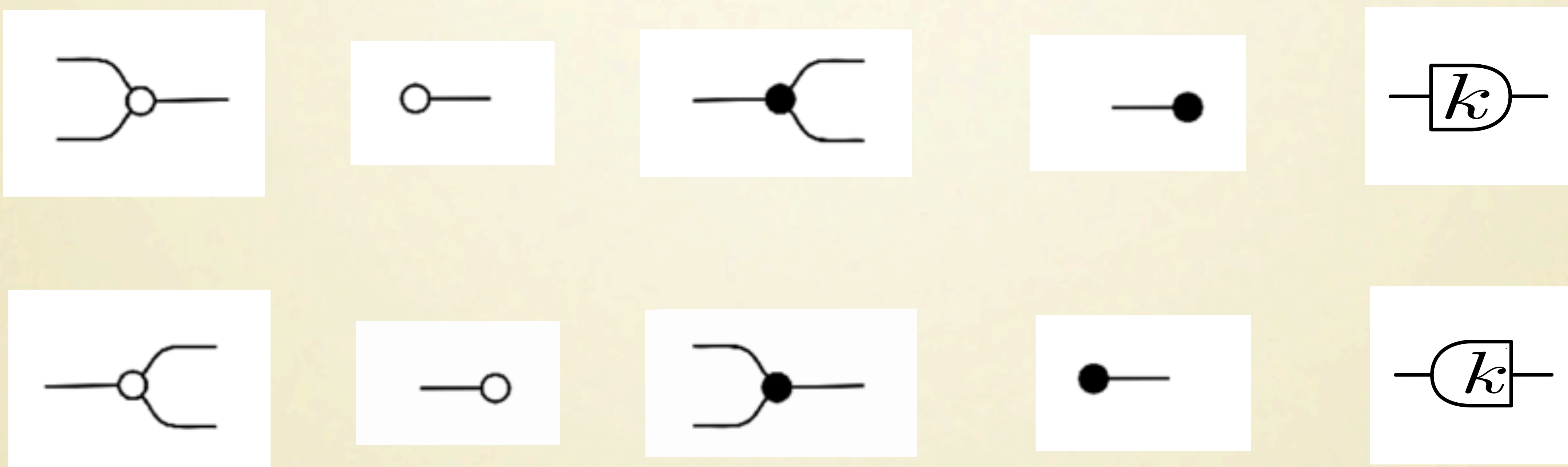
Subspaces

So far our string diagrams can only represent linear maps: this is a limited picture of what linear algebra is about.

How do we extend our calculus to be able to reason diagrammatically about **subspaces**?

Extending Our Calculus

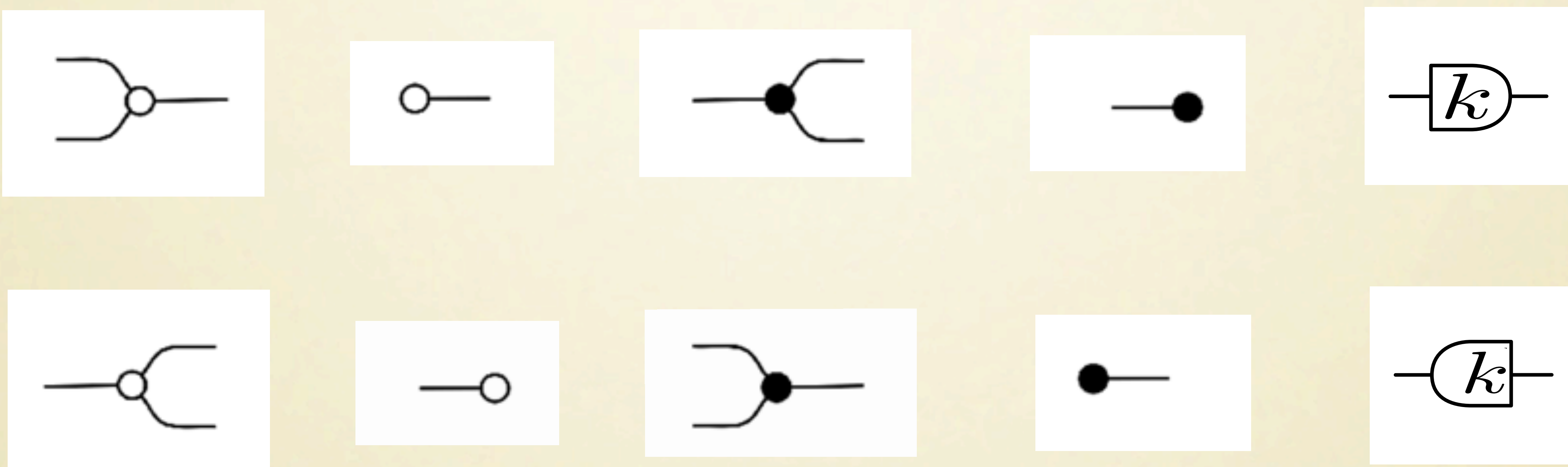
Operations



Semantics: linear ~~functions~~ relations (= subspaces)

Extending Our Calculus

Operations $k \in \mathbb{R}$

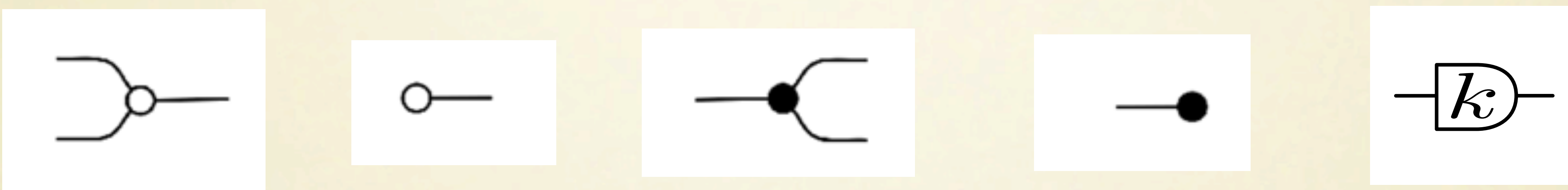


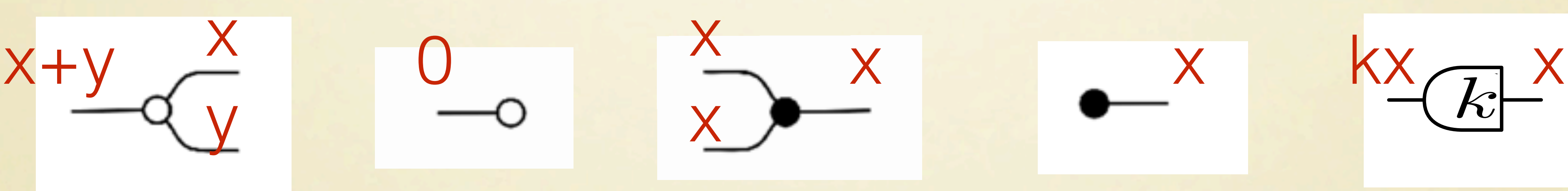
Semantics: linear ~~functions~~ relations (= subspaces)

Extending Our Calculus

Operations

$k \in \mathrm{R}$

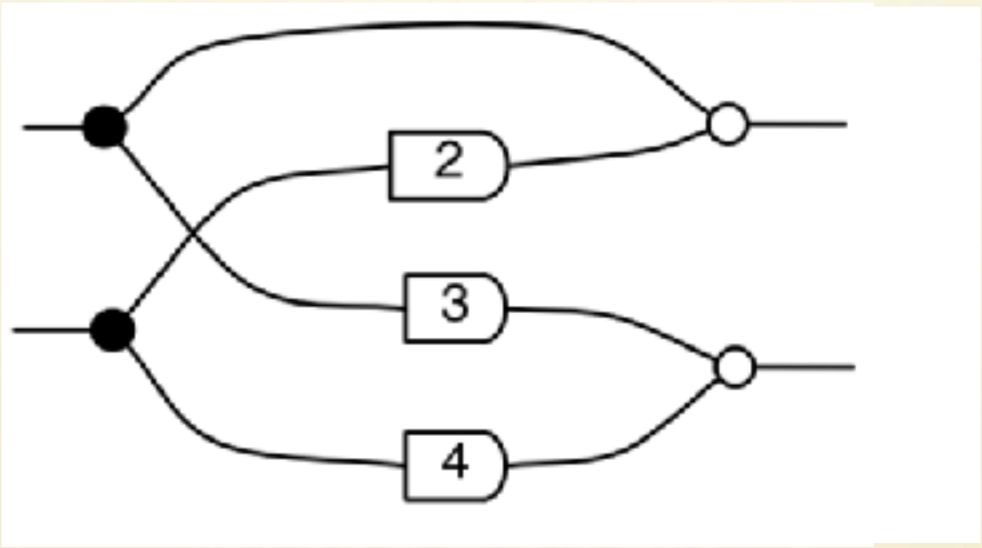




Semantics: linear ~~functions~~ relations (= subspaces)

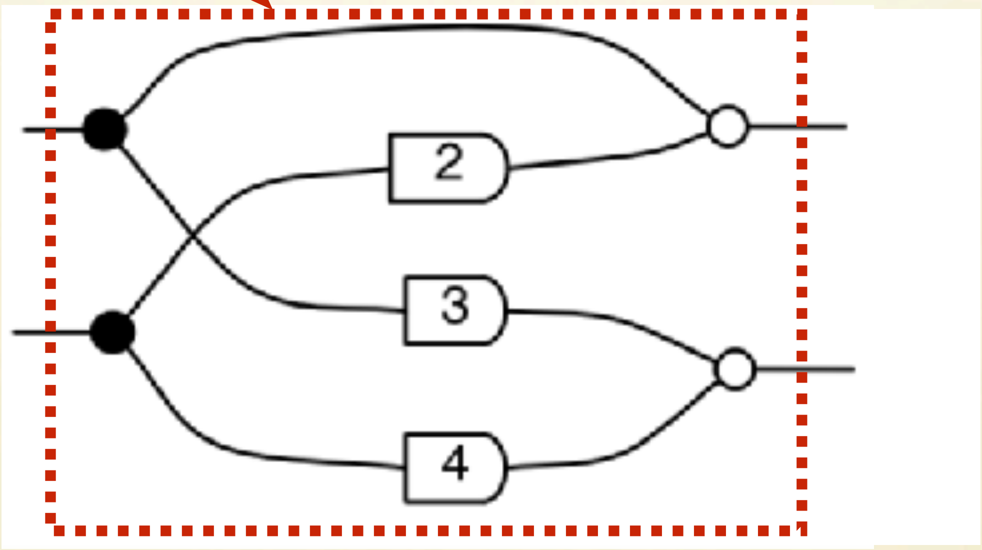
Kernel of a matrix

Kernel of a matrix



Kernel of a matrix

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$



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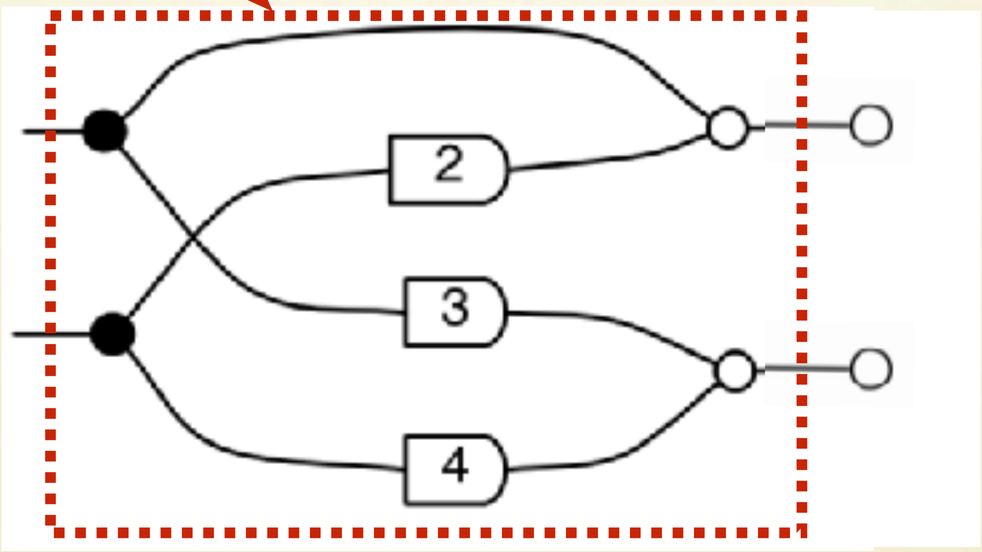

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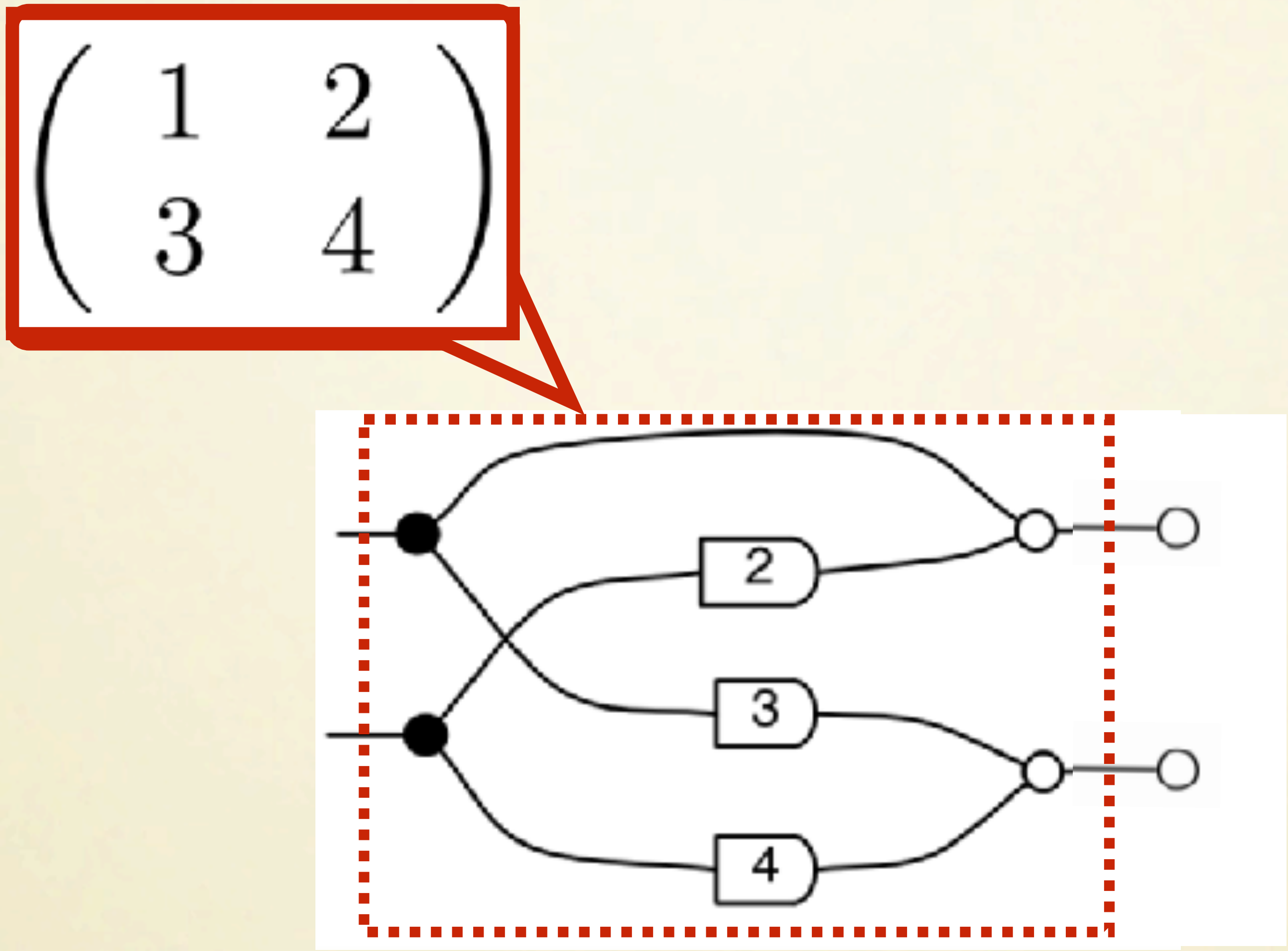
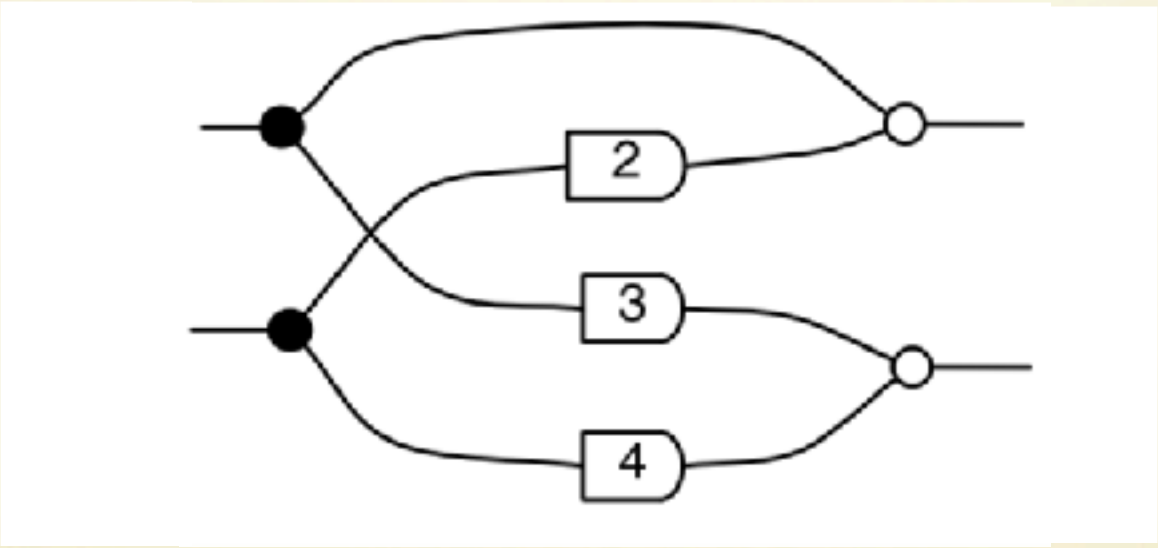


$$\{v \mid Mv = 0\}$$

Image of a matrix



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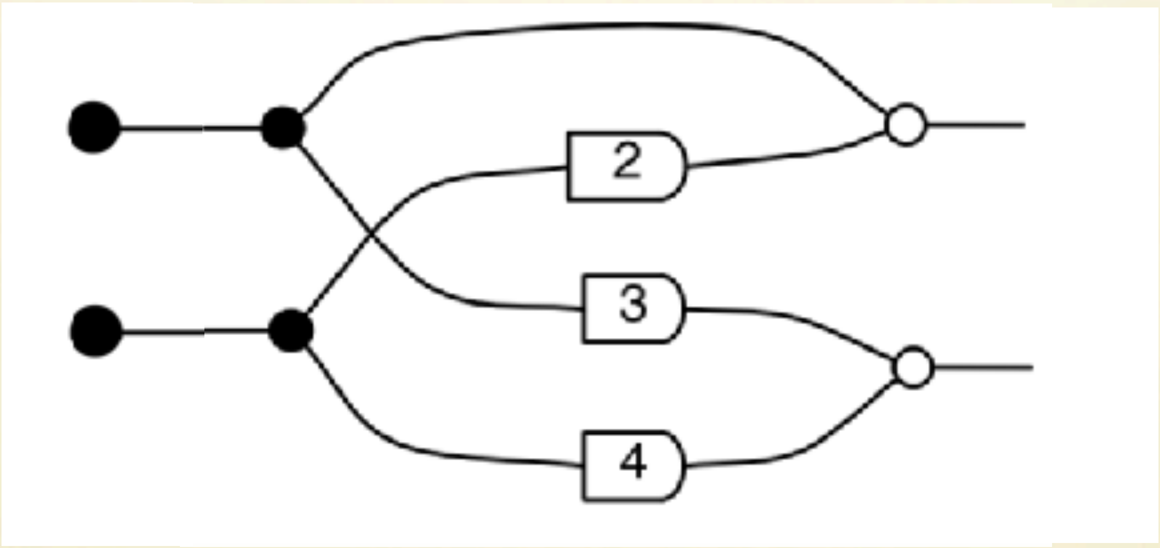

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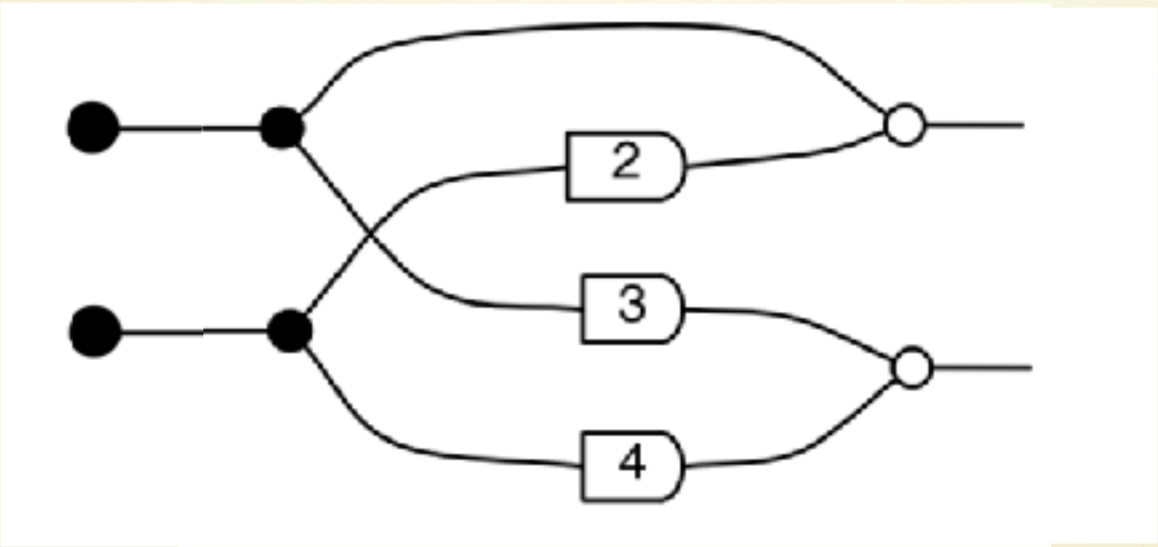
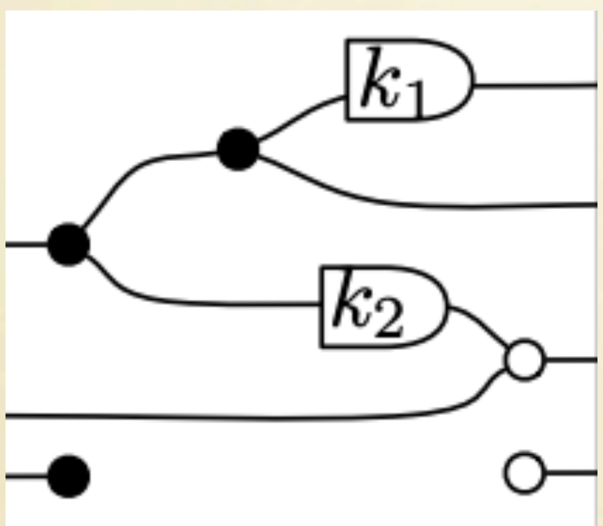
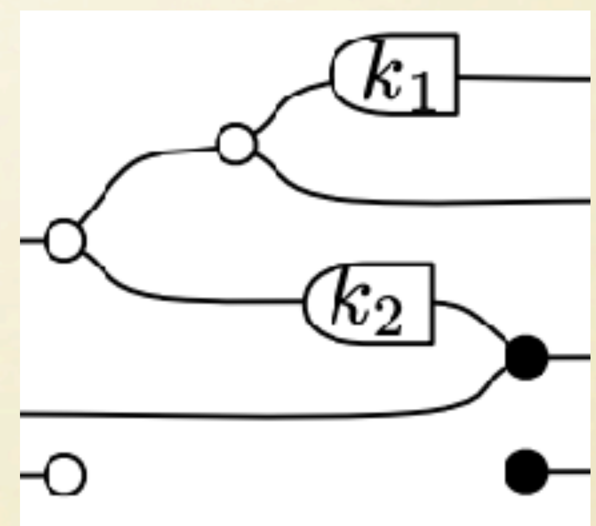


$$\{Mv \mid v \in \mathrm{R}^n\}$$

Transpose

Transpose corresponds to
flip scalars and colour-swap.

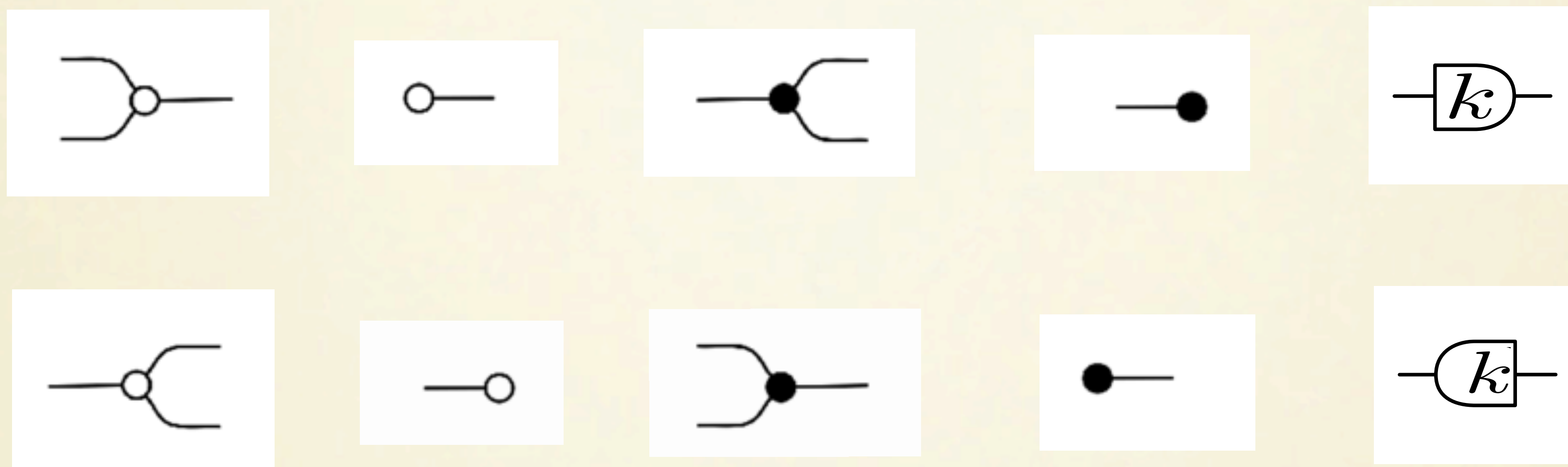




$$M = \begin{pmatrix} k_1 & 0 & 0 \\ 1 & 0 & 0 \\ k_2 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$M^T = \begin{pmatrix} k_1 & 1 & k_2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

What about equations?



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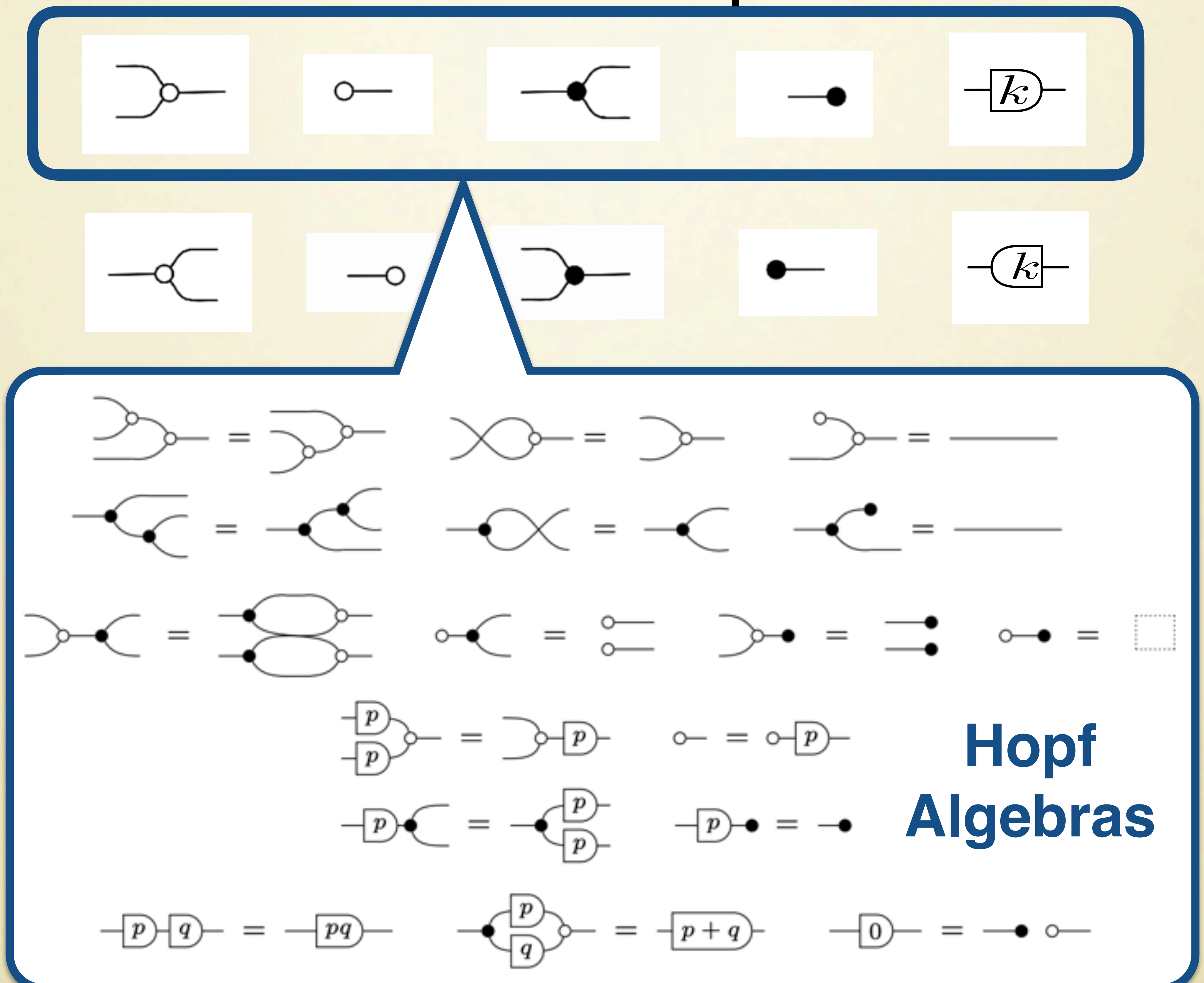

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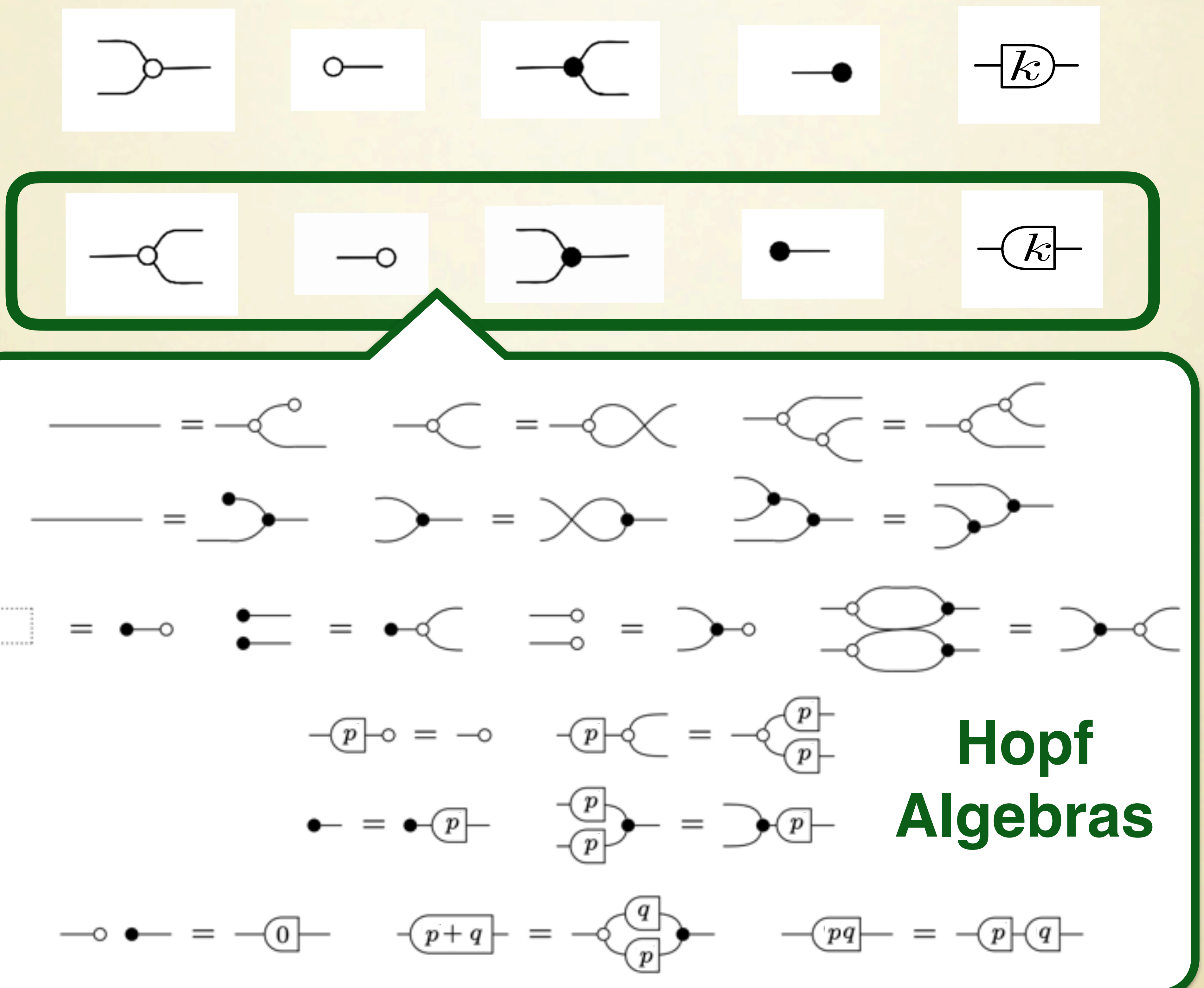

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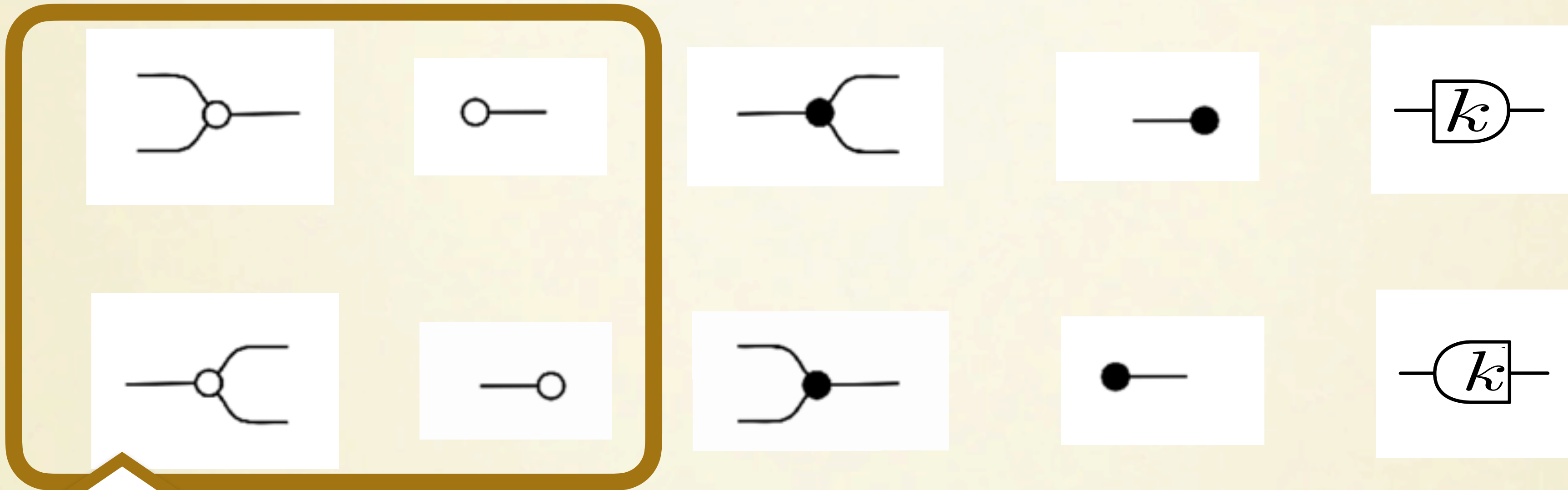
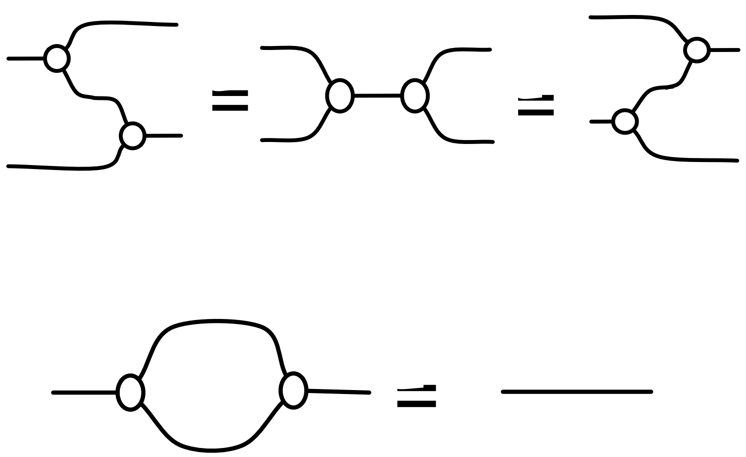
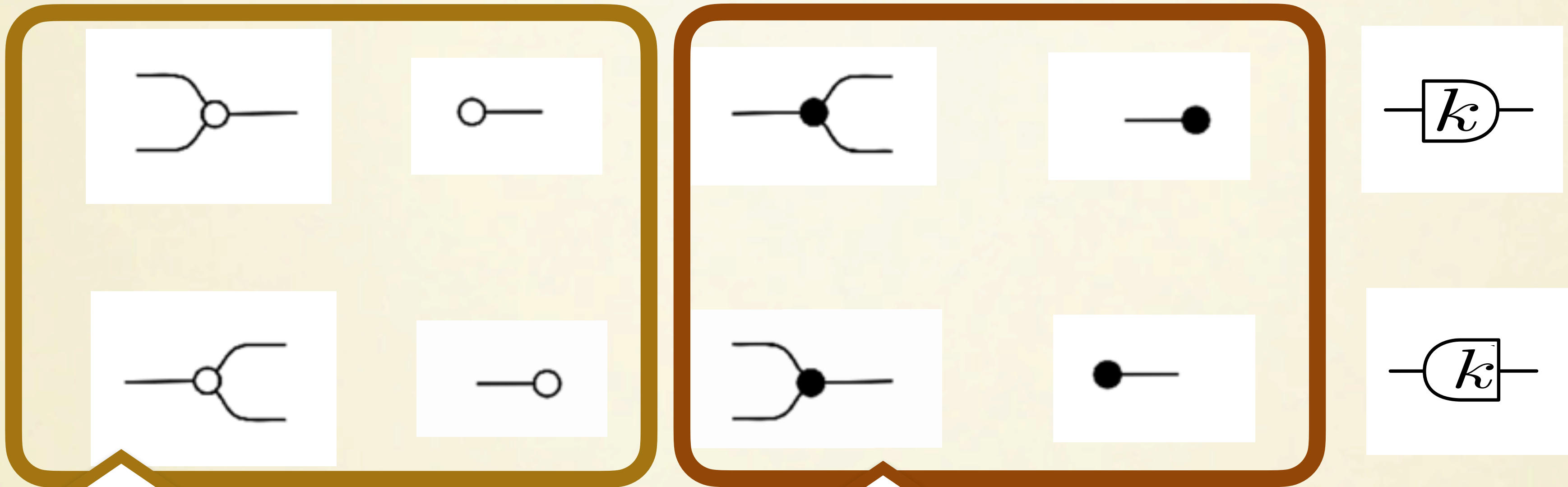
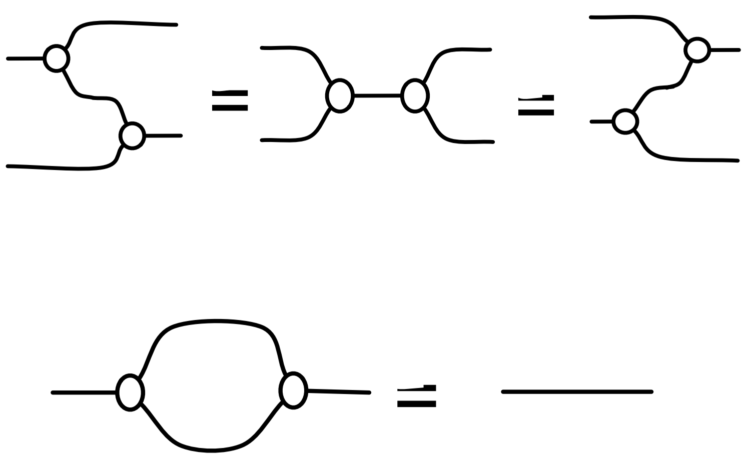




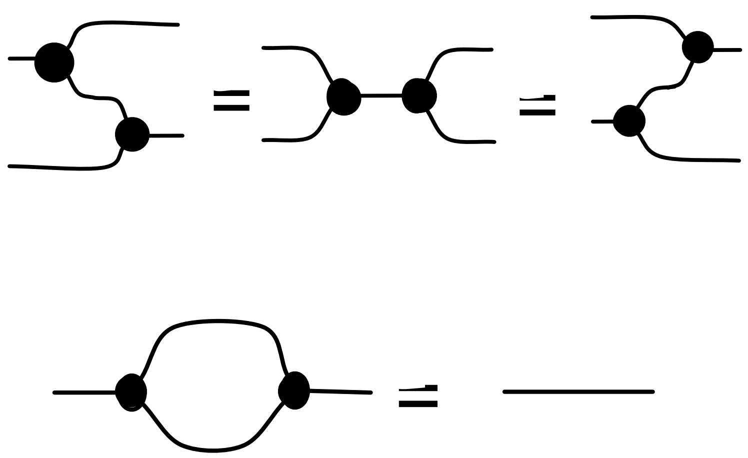
Frobenius Algebras

What about equations?





Frobenius Algebras



Frobenius Algebras

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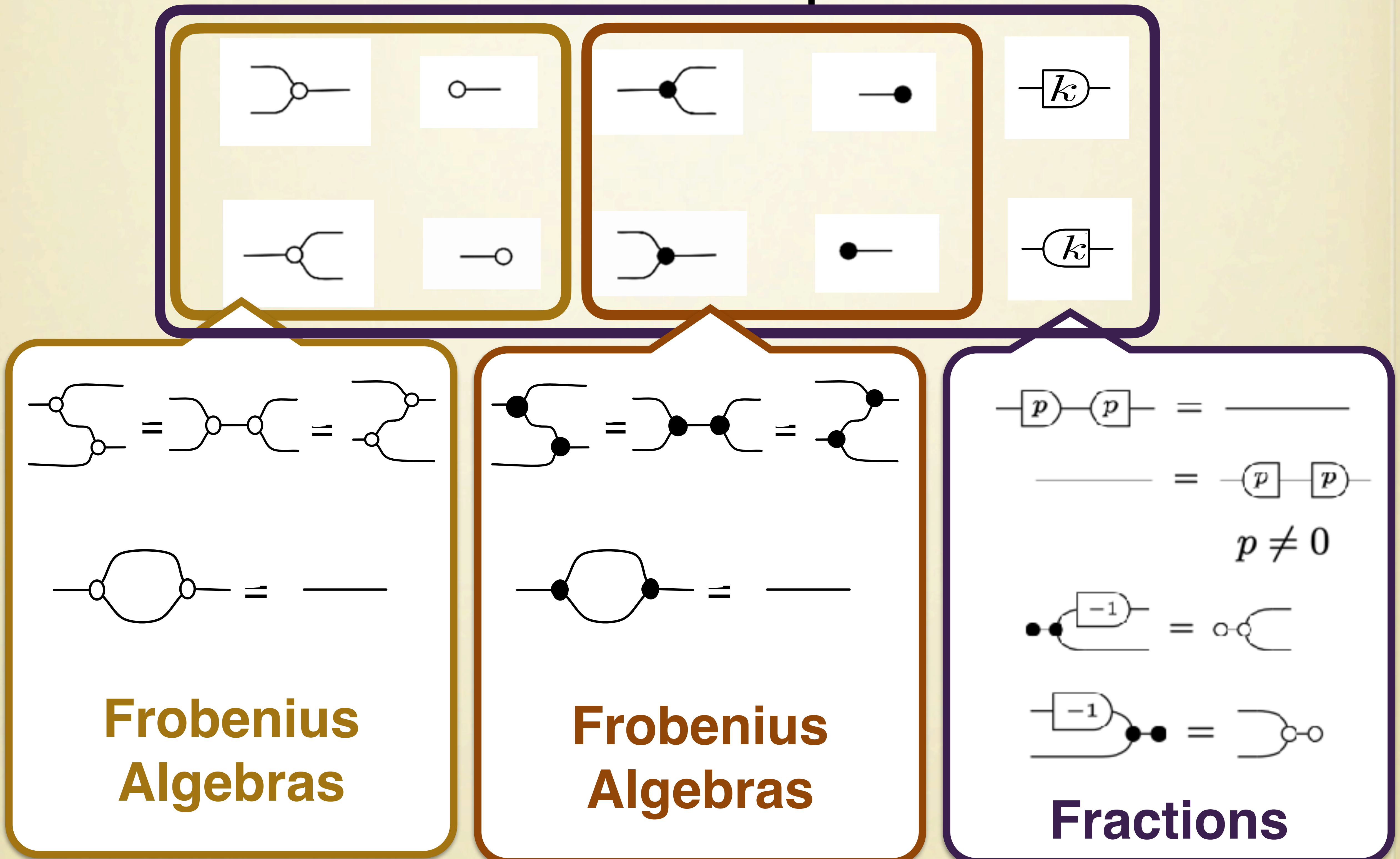
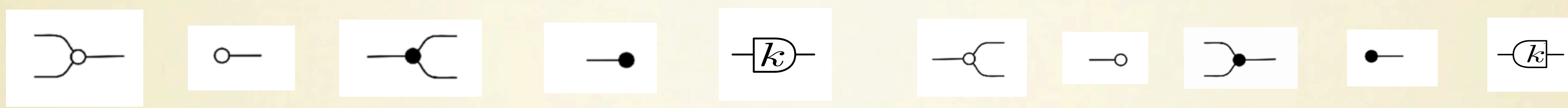
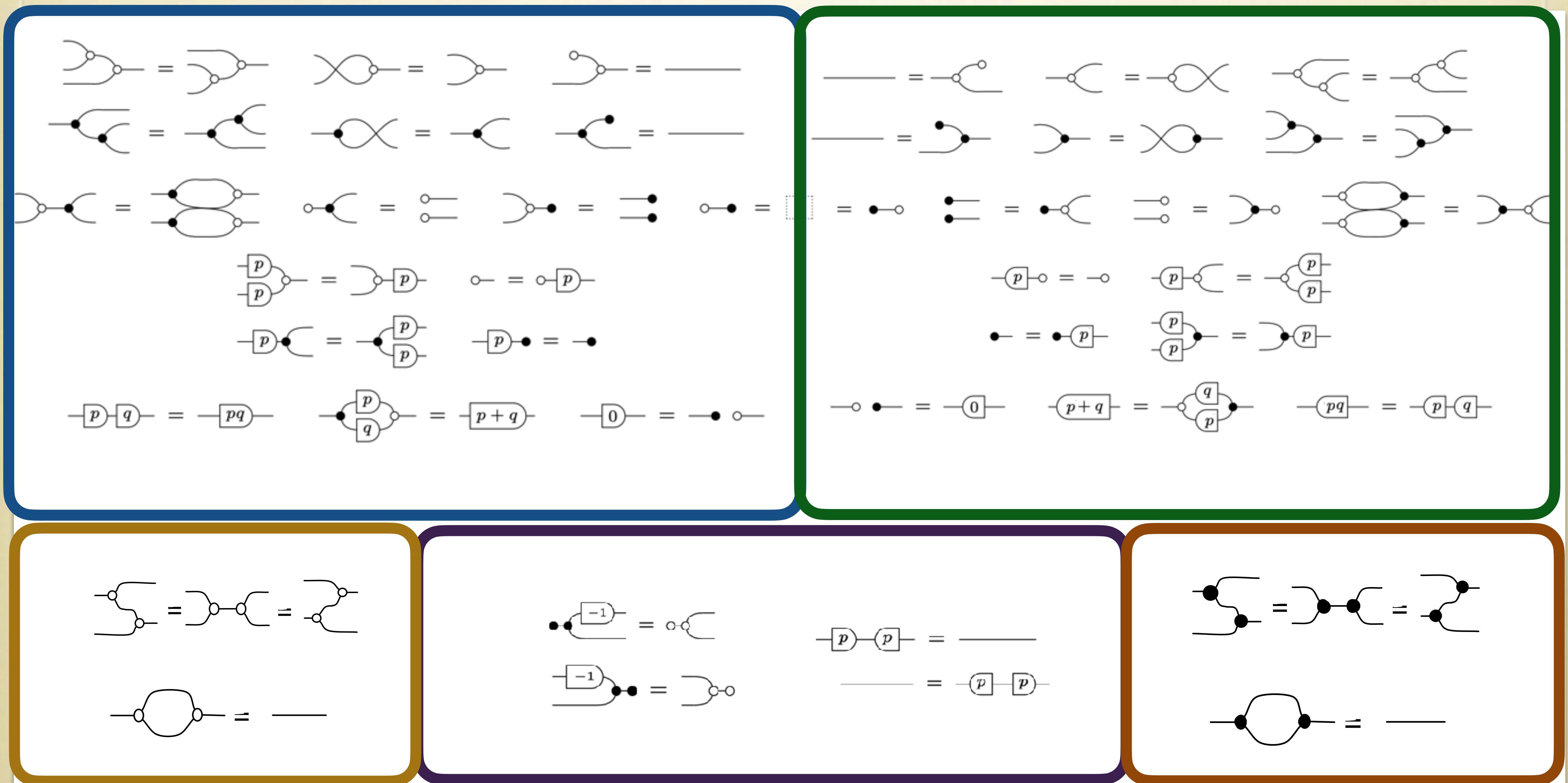


The Full Picture: Interacting Hopf Algebras





Graphical Linear Algebra

Theorem

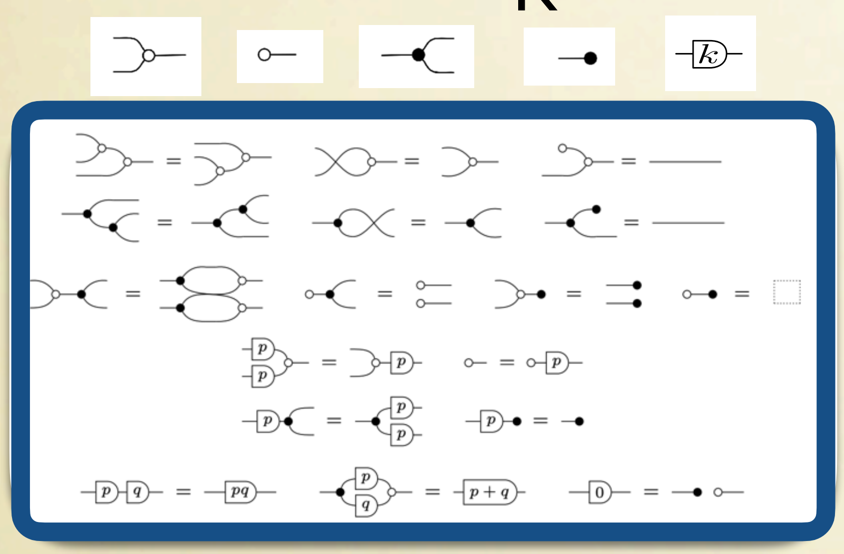
There is a 1-1 correspondence between:

- String Diagrams with n left wires and m right wires *modulo the equations of interacting* R*-Hopf algebras*
- Subspaces of $\mathsf{k}^n \times \mathsf{k}^m$ *(where* k *is the field of fractions of* R*)*

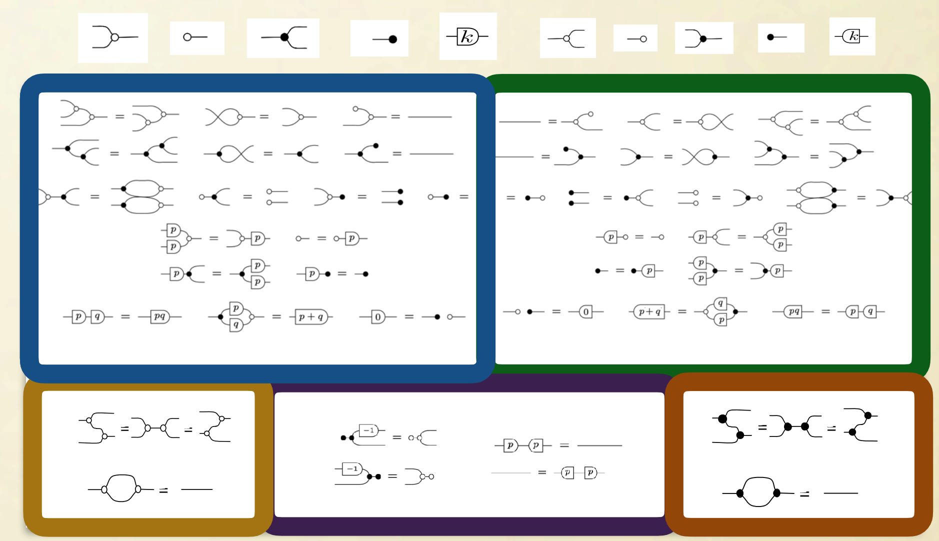
The categorical substrate

String diagrams are the morphisms of
symmetric monoidal categories.

Hopf_R



$\mathsf{IntHopf}_R$



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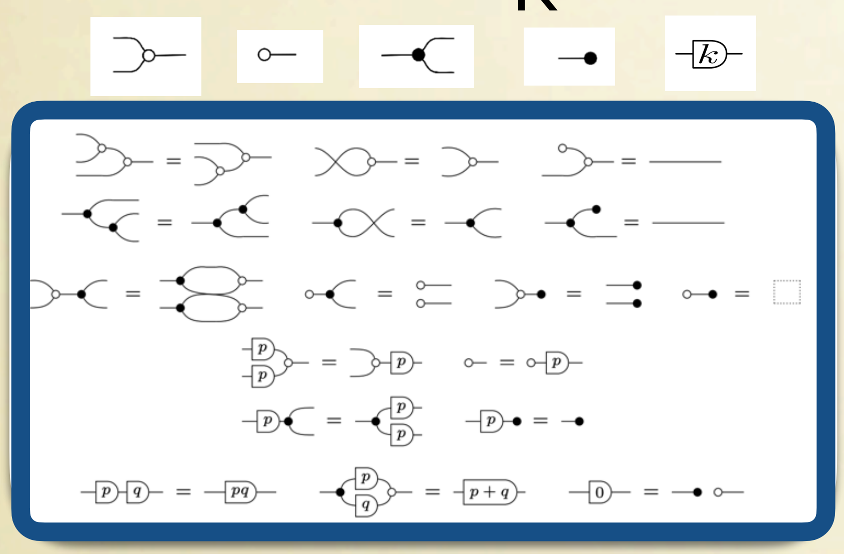

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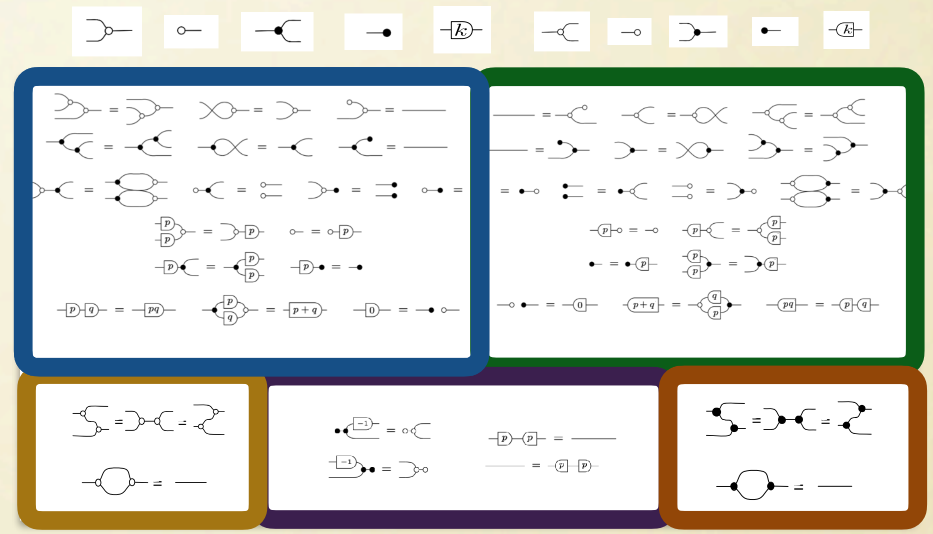

The semantic interpretations are symmetric monoidal **functors**

$\mathrm{Hopf}_R \to \mathrm{Mat}_R$

$\mathrm{IntHopf}_R \to \mathrm{SubVect}_k$

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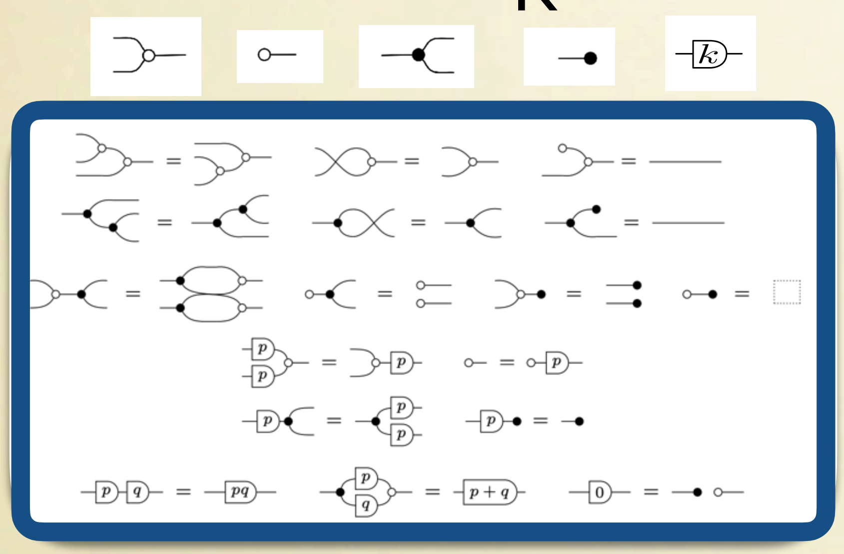
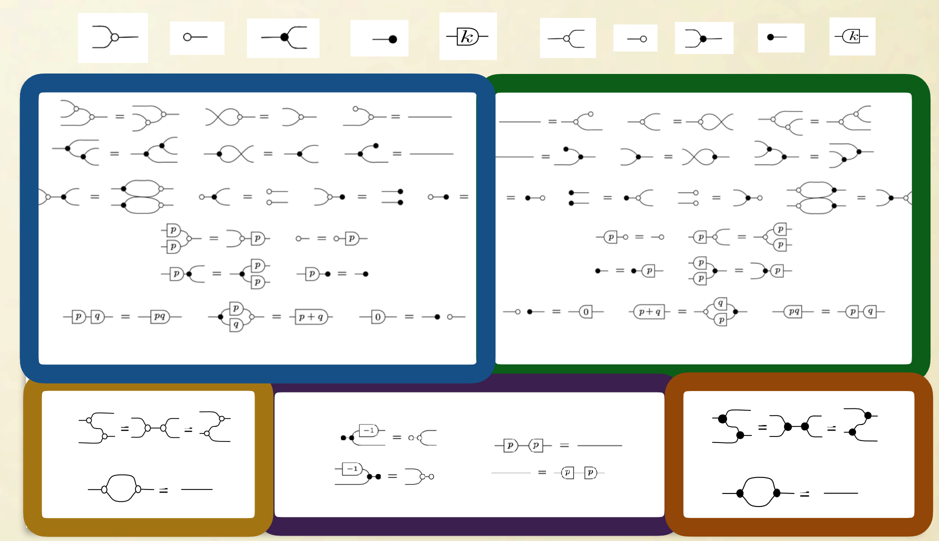


$\mathrm{IntHopf}_R$



The semantic interpretations are symmetric monoidal **functors**

$\mathrm{Hopf}_R \to \mathrm{Mat}_R$ $\mathrm{IntHopf}_R \to \mathrm{SubVect}_k$

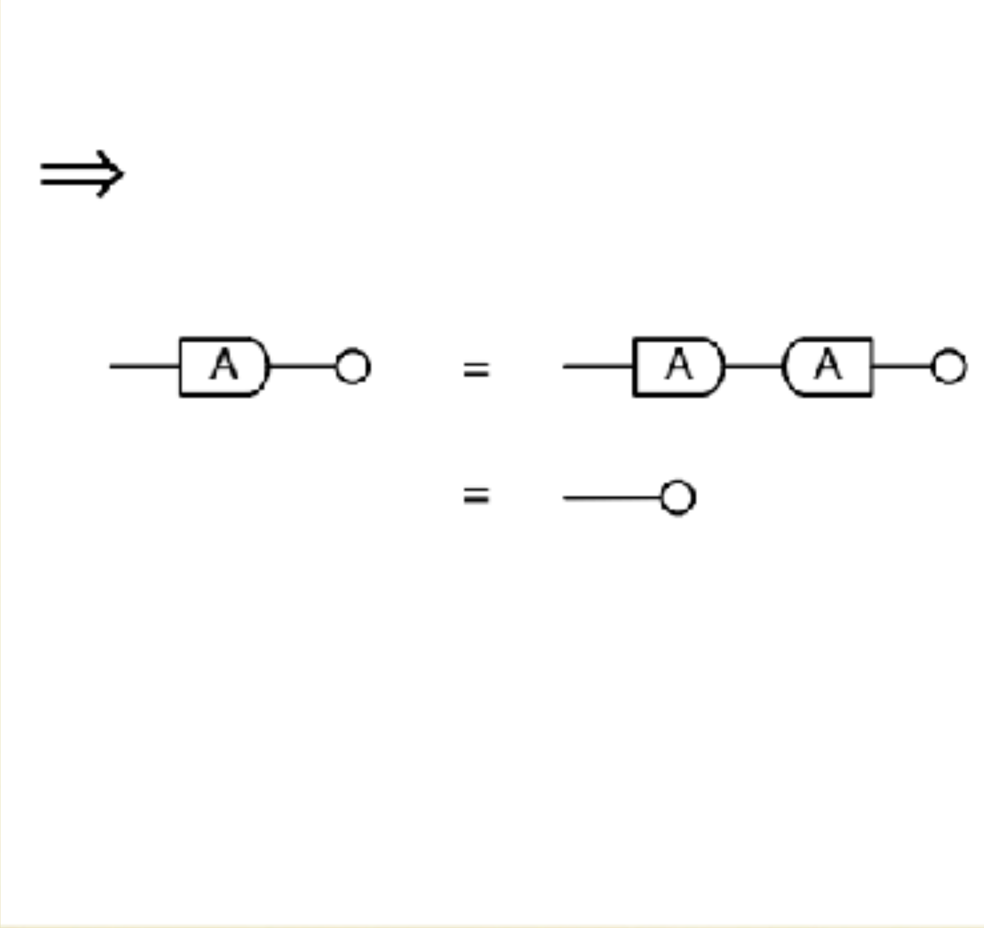
The 1-1 correspondences mean these functors are **isomorphisms**

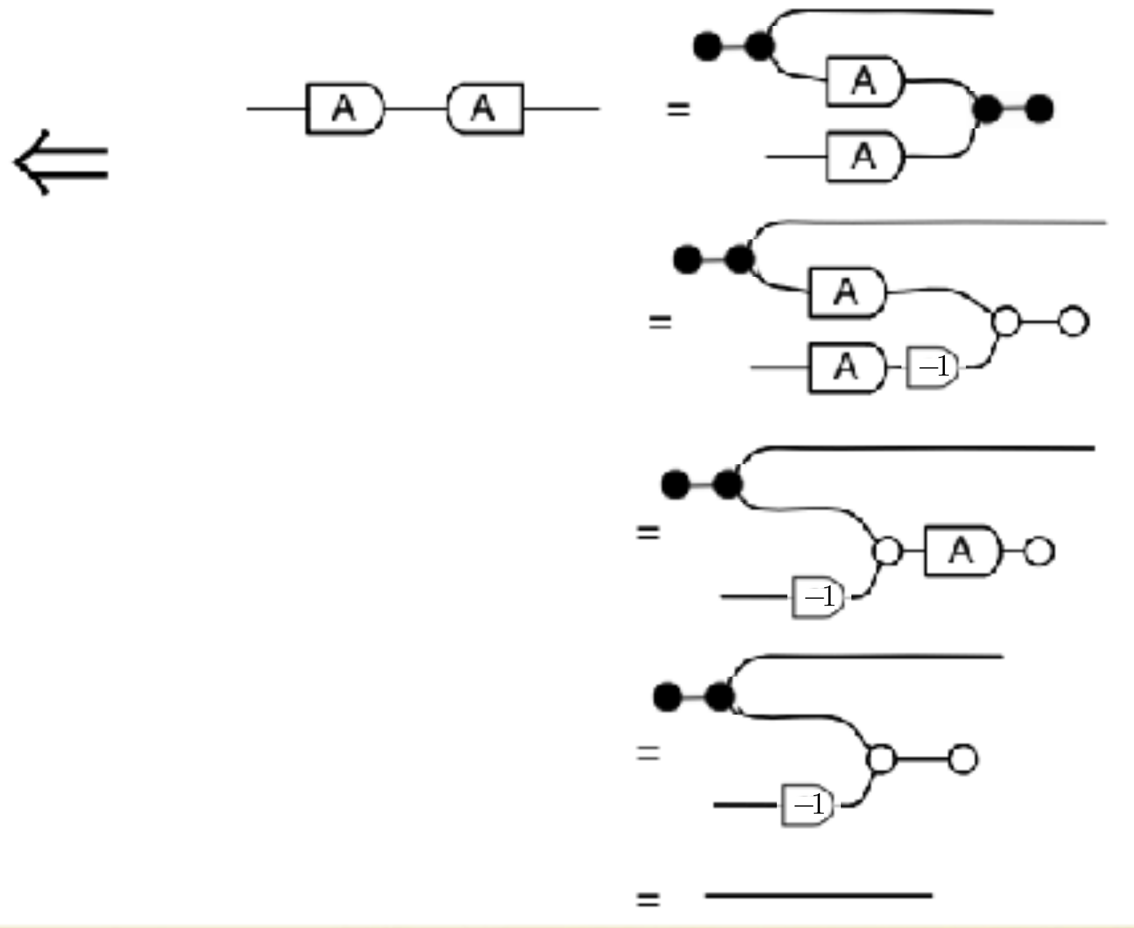
$\mathrm{Hopf}_R \cong \mathrm{Mat}_R$ $\mathrm{IntHopf}_R \cong \mathrm{SubVect}_k$

A proof in GLA

Proposition: a matrix A is injective iff its kernel is 0.

Proof





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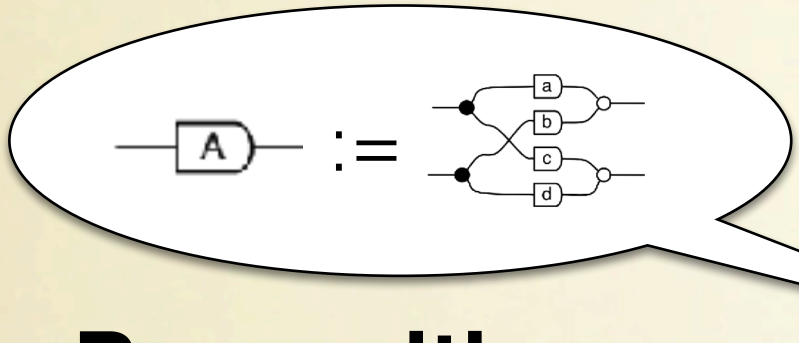

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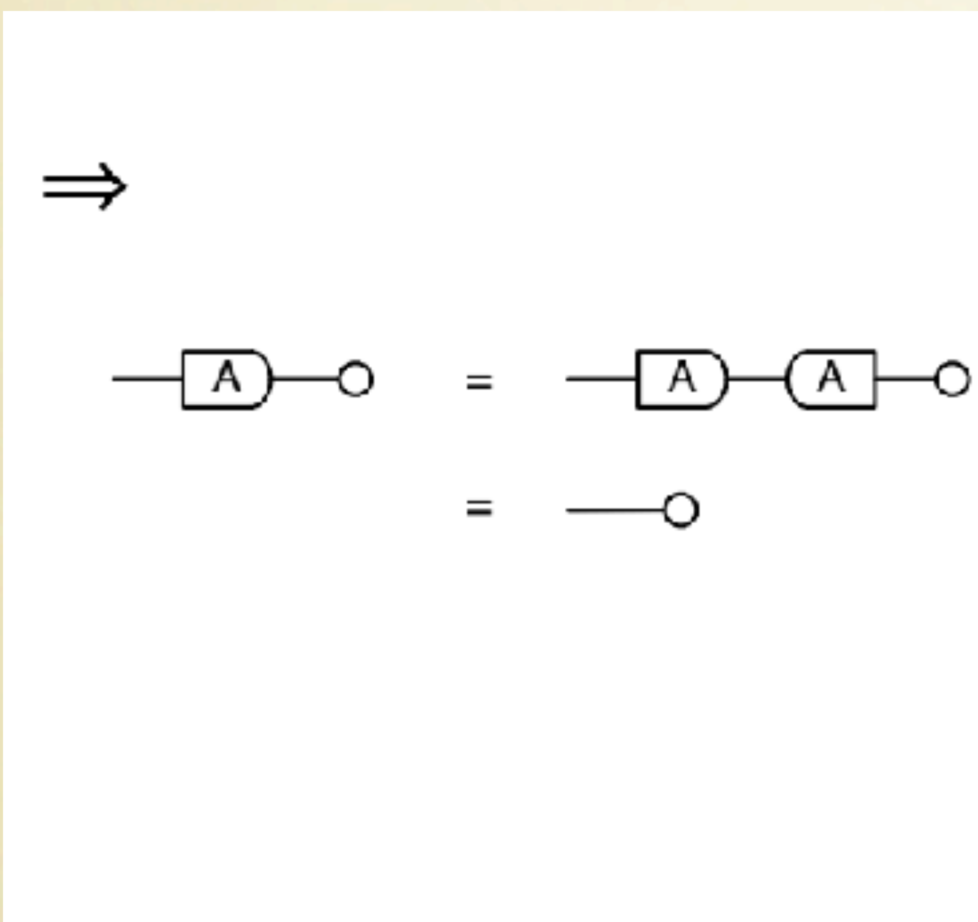

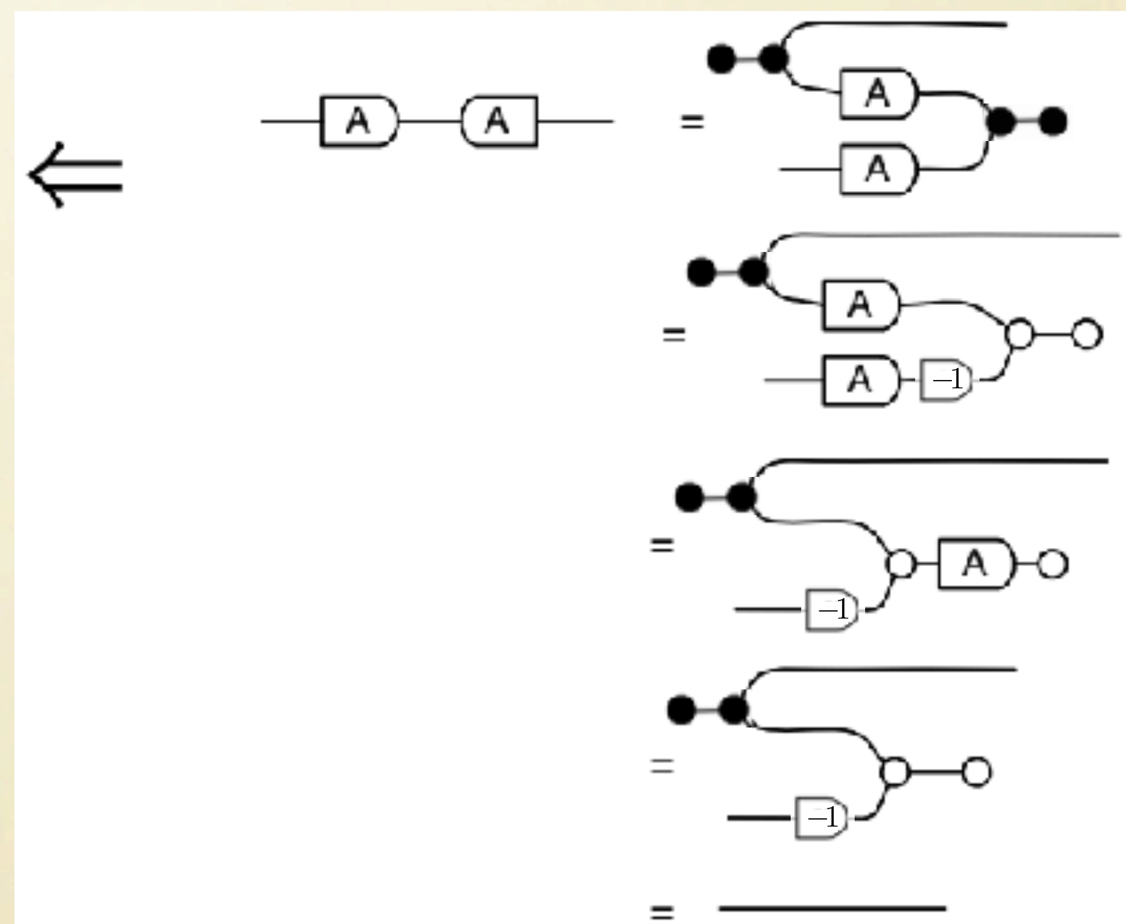

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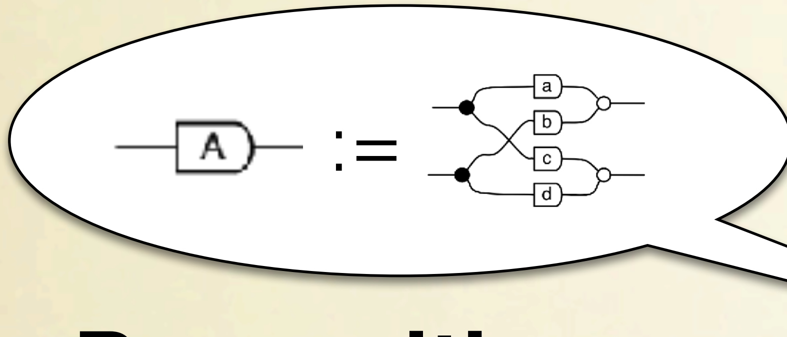

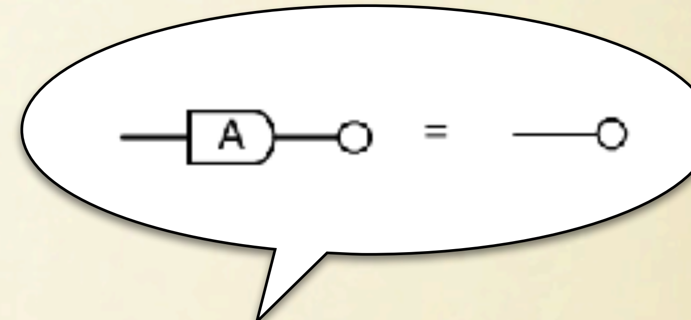

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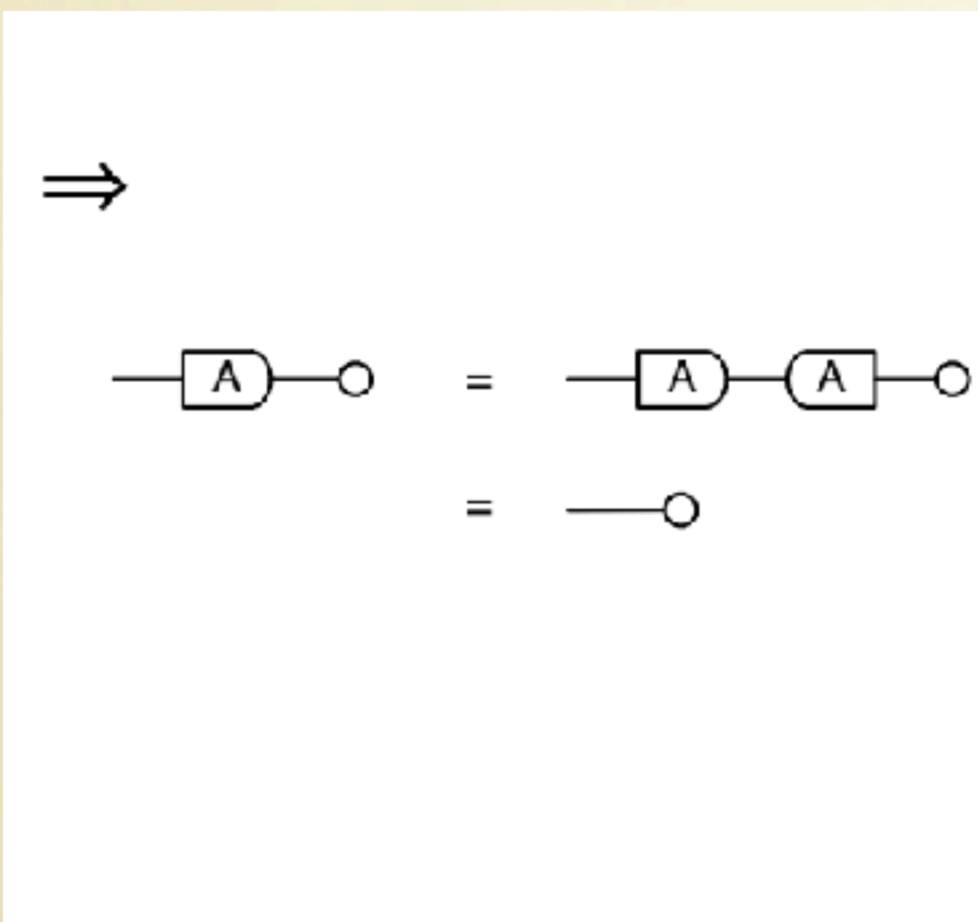

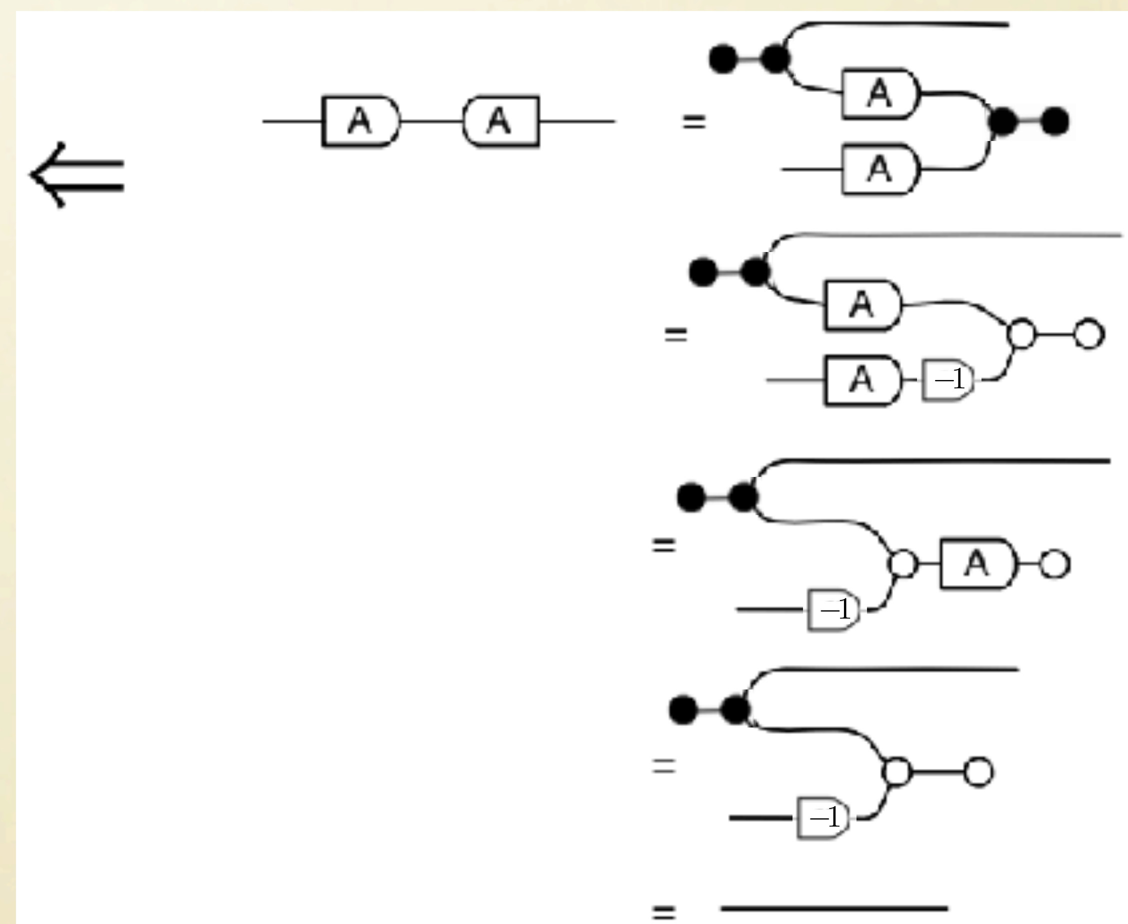

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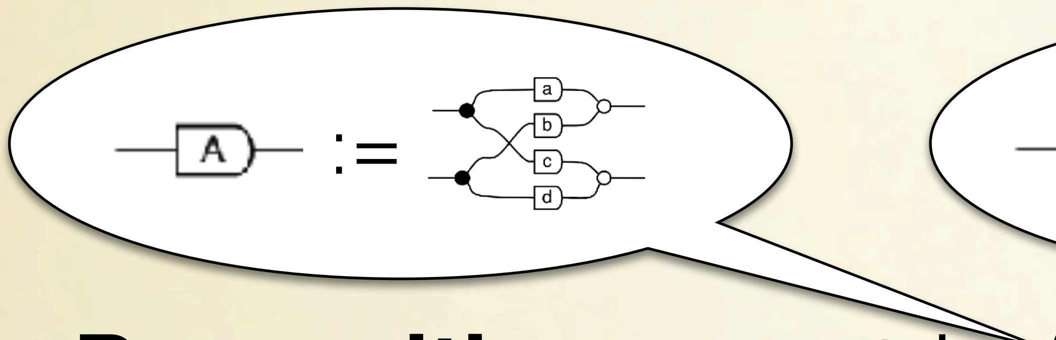

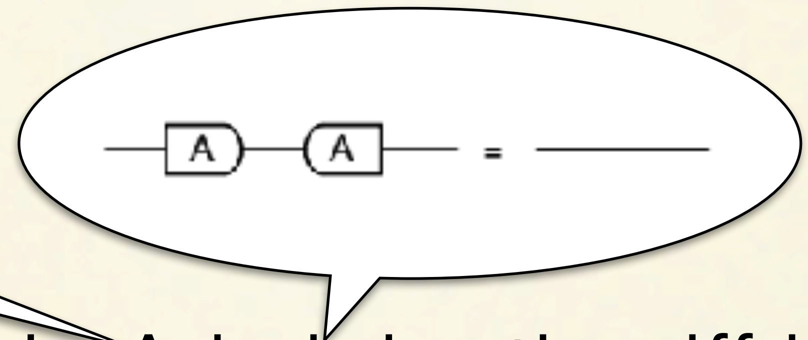

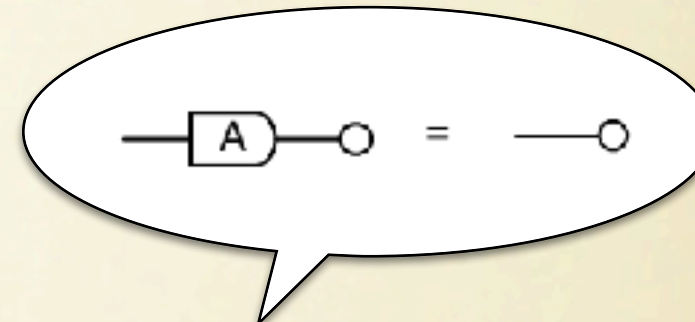

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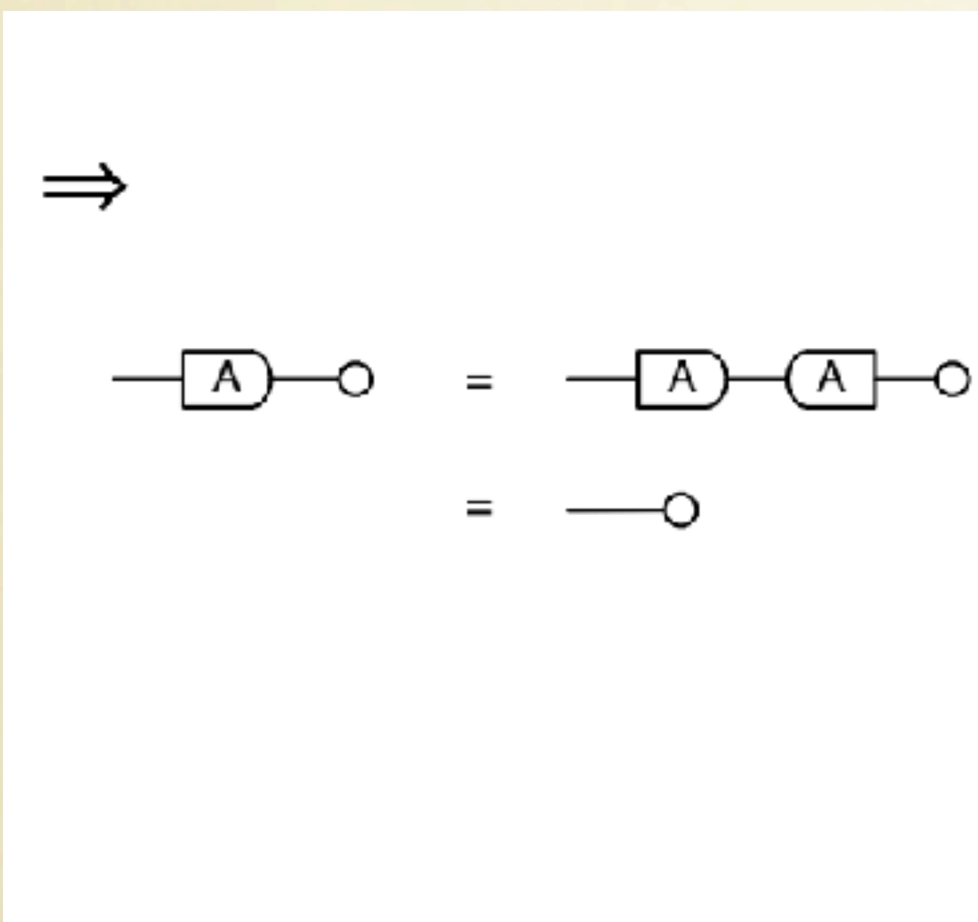

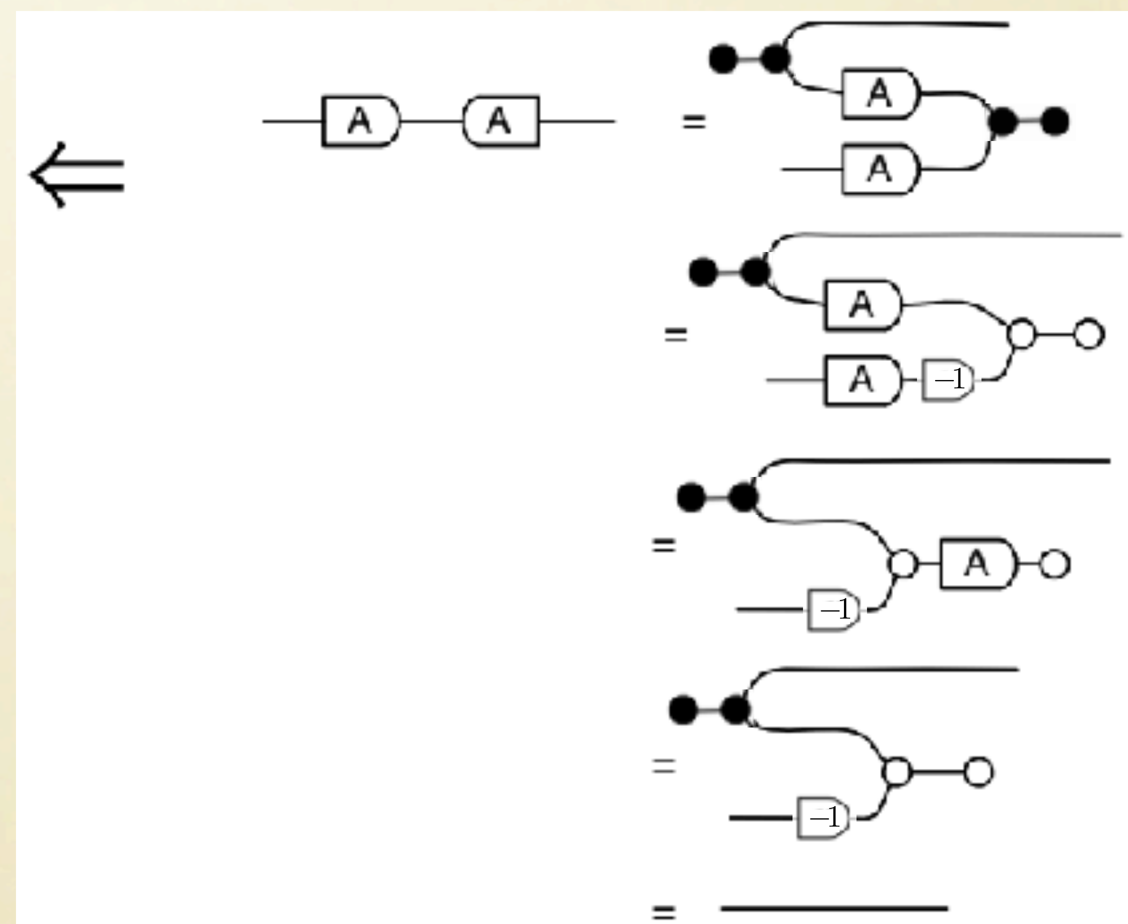

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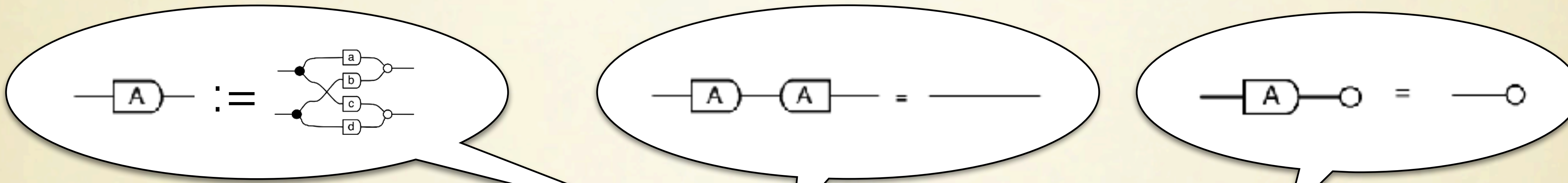

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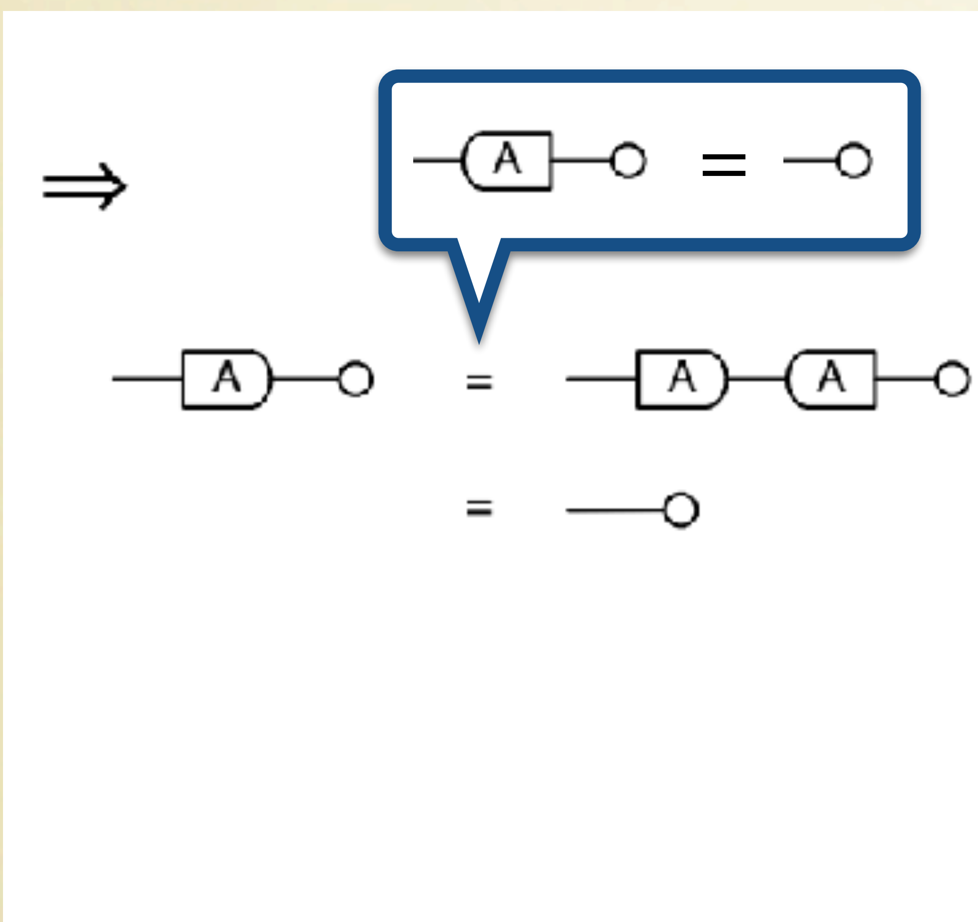

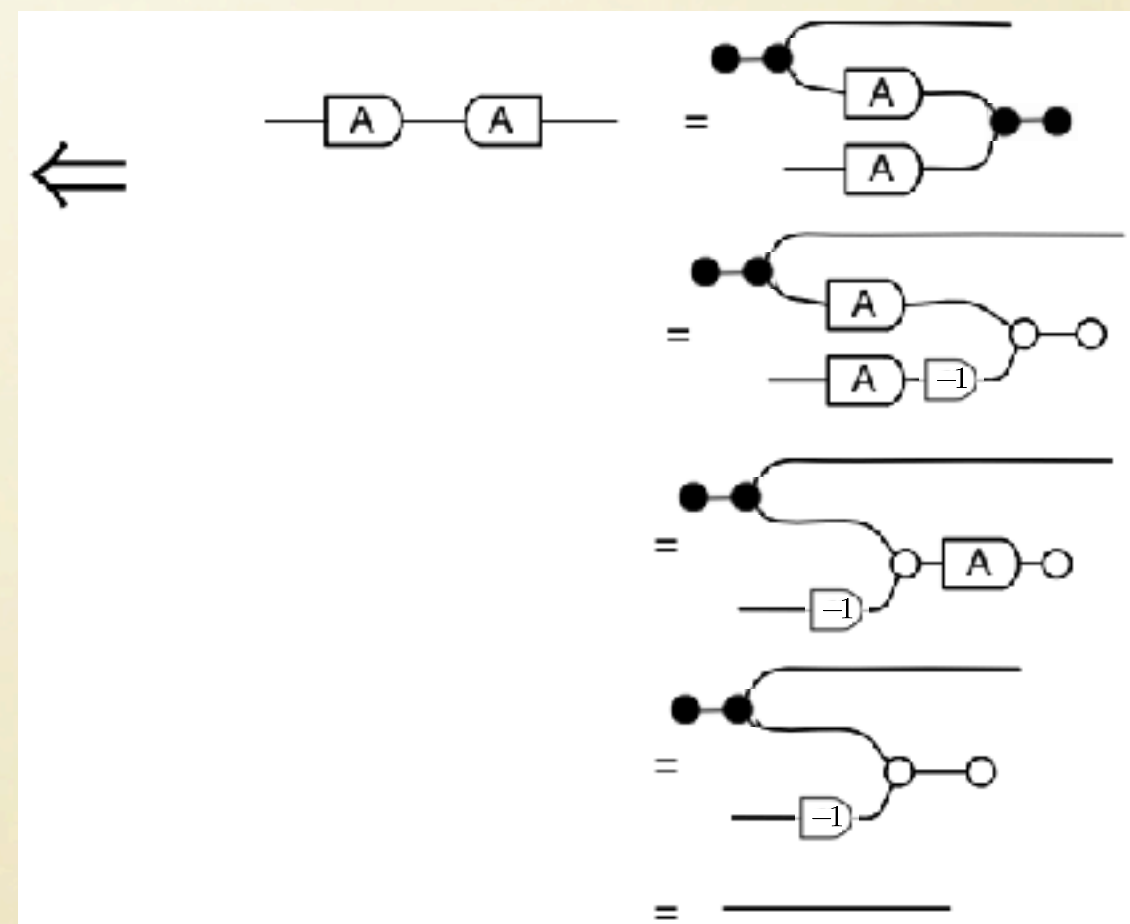

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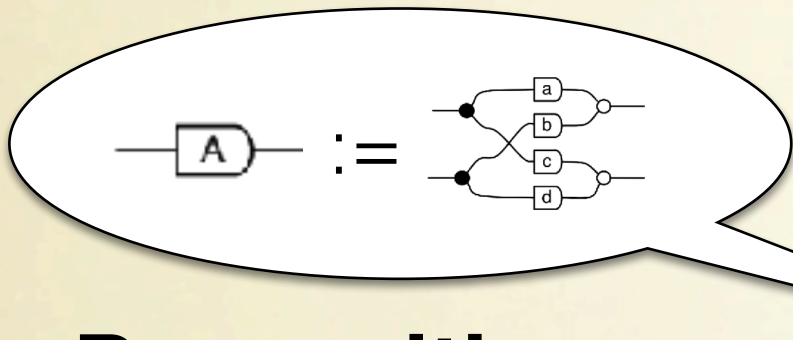

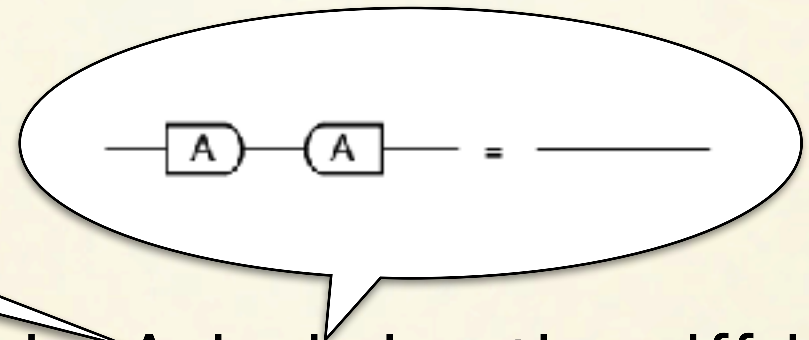

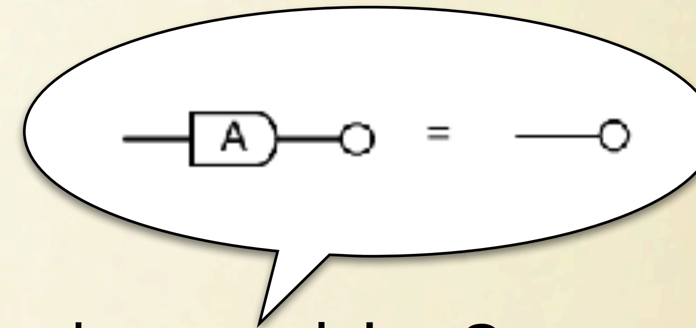

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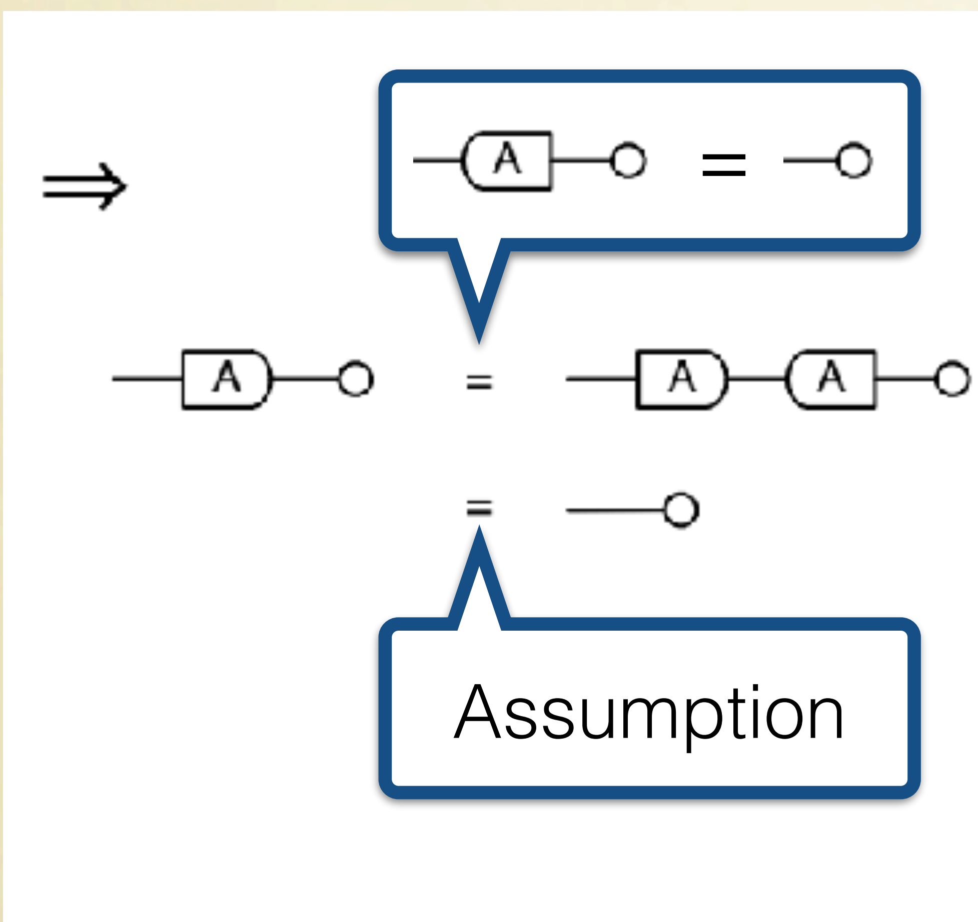

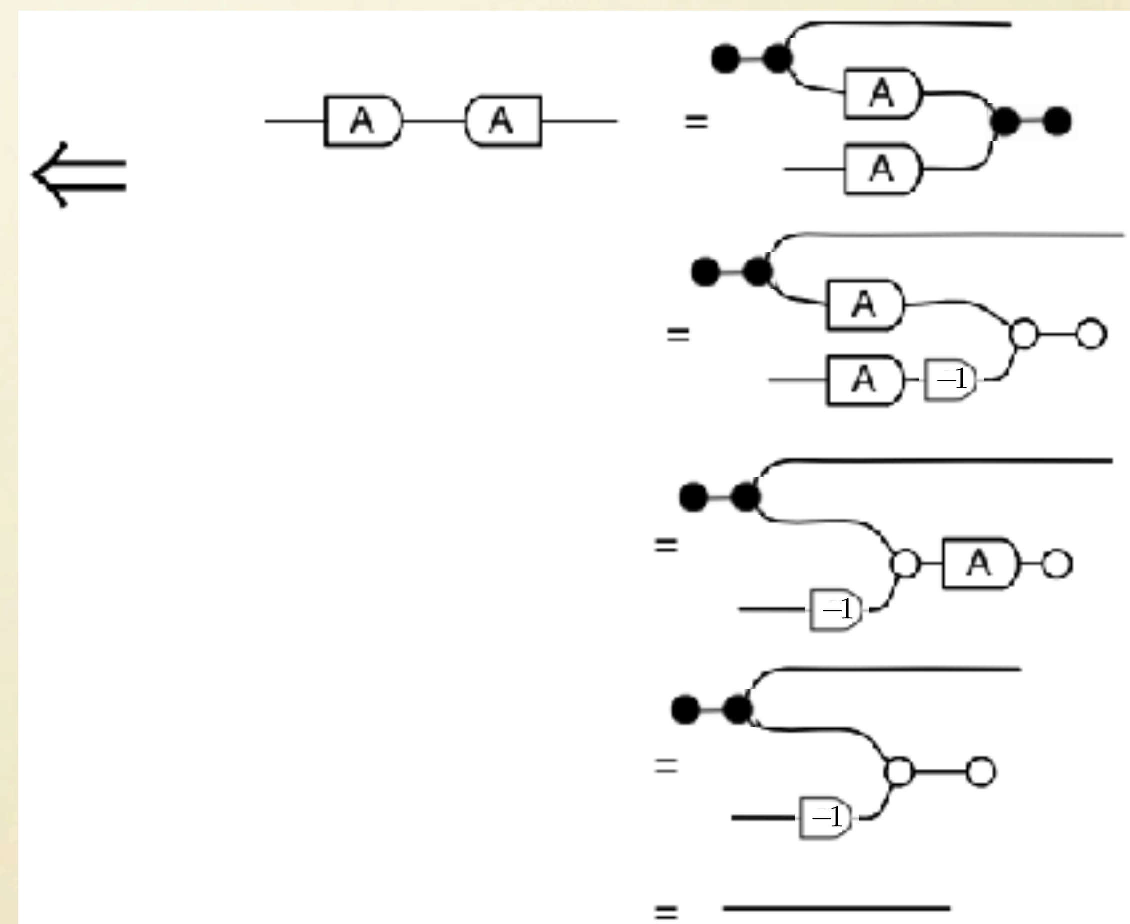

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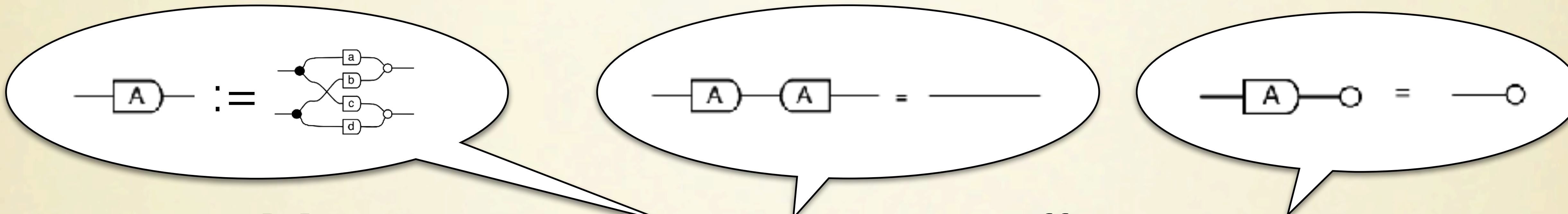

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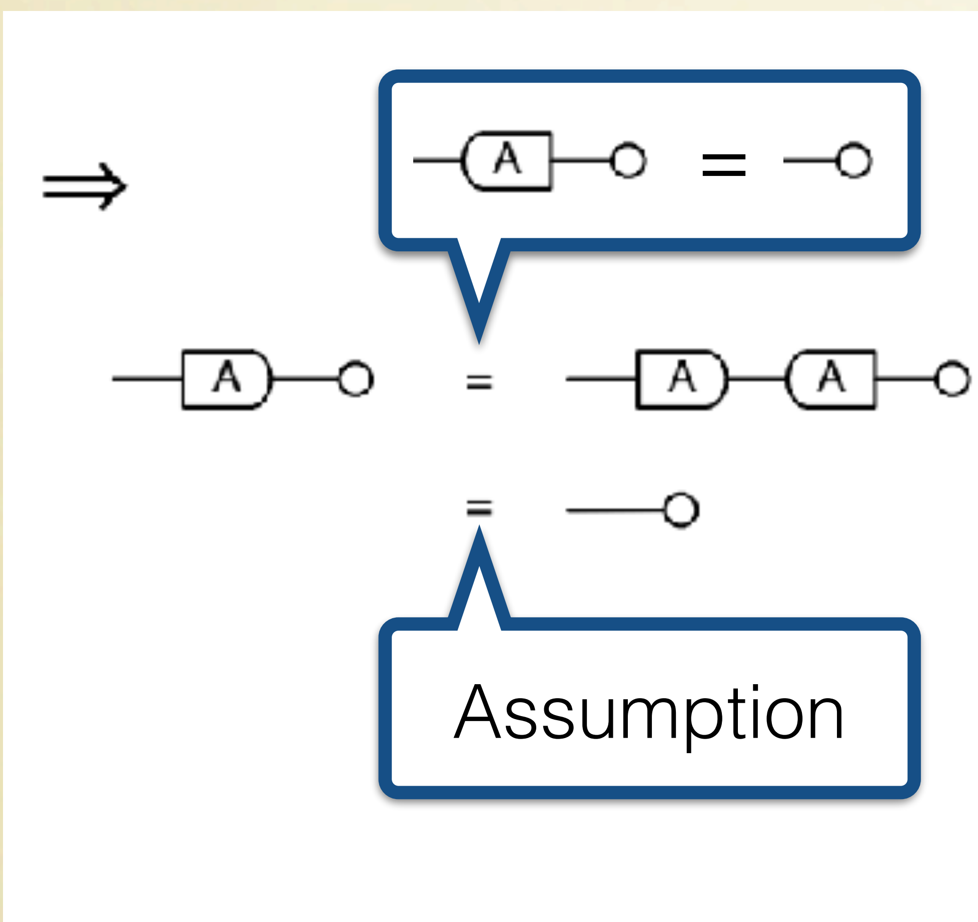

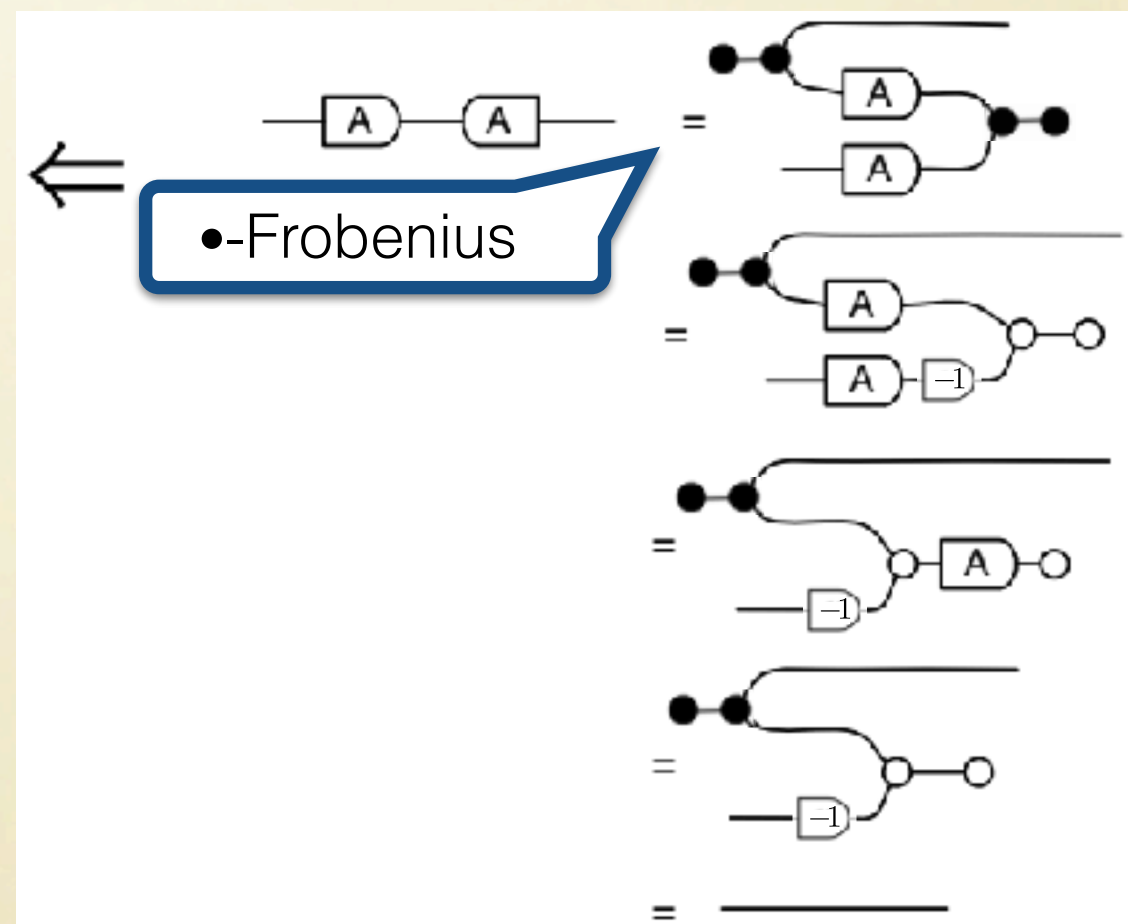

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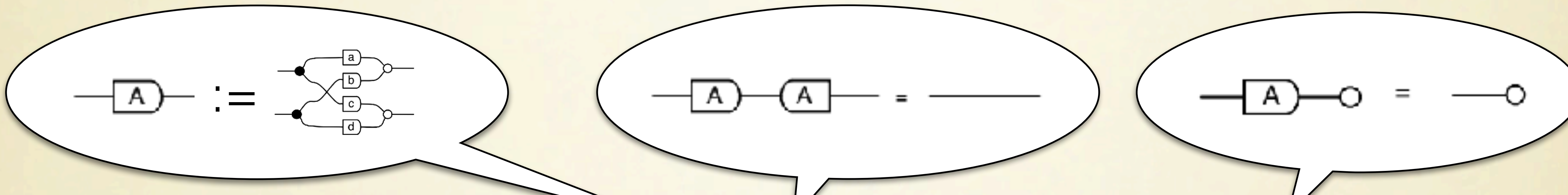

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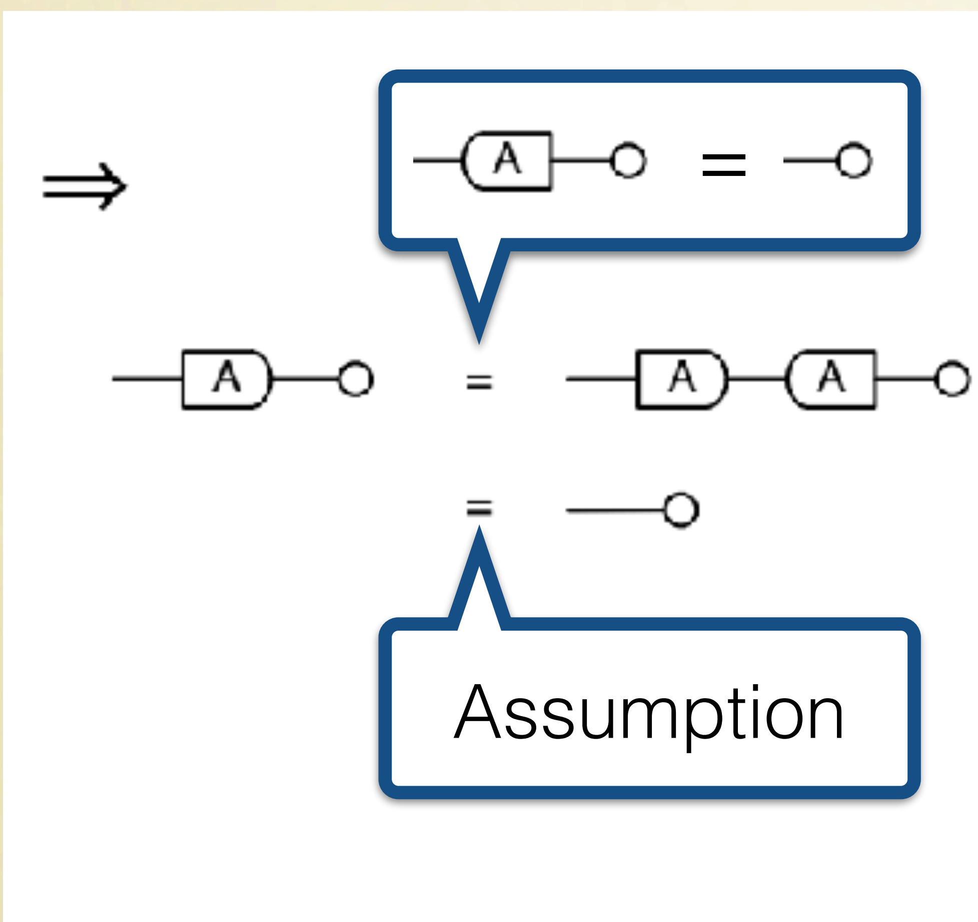

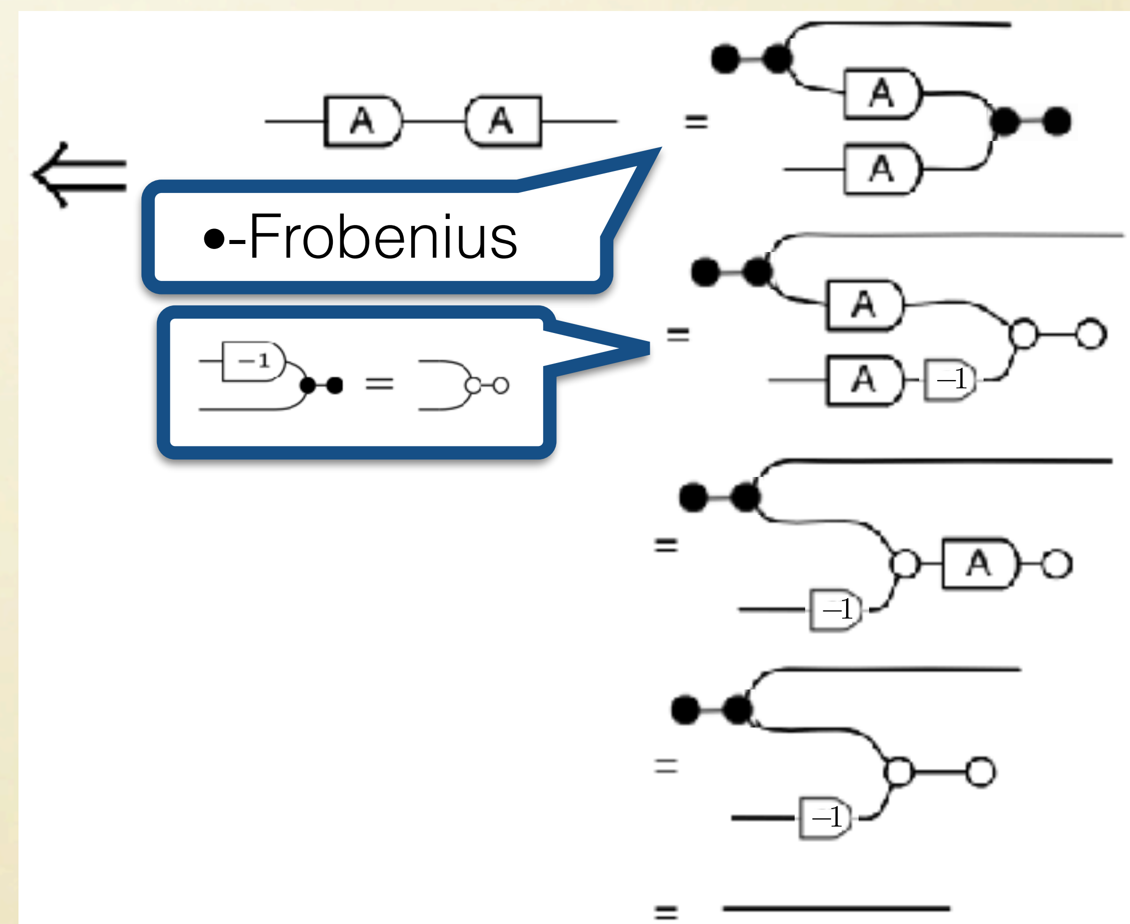

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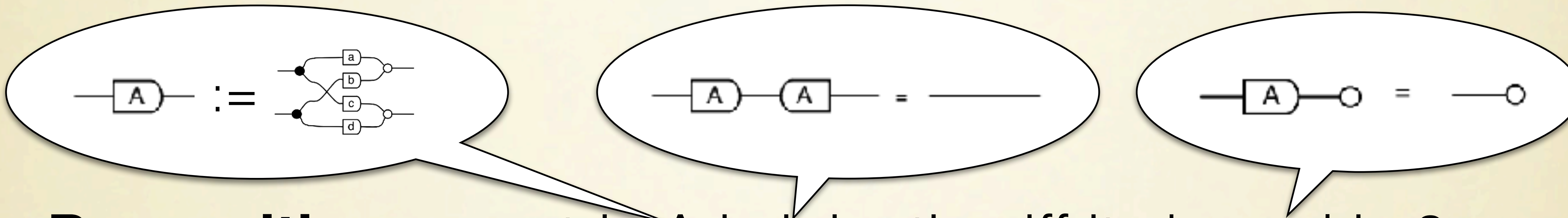

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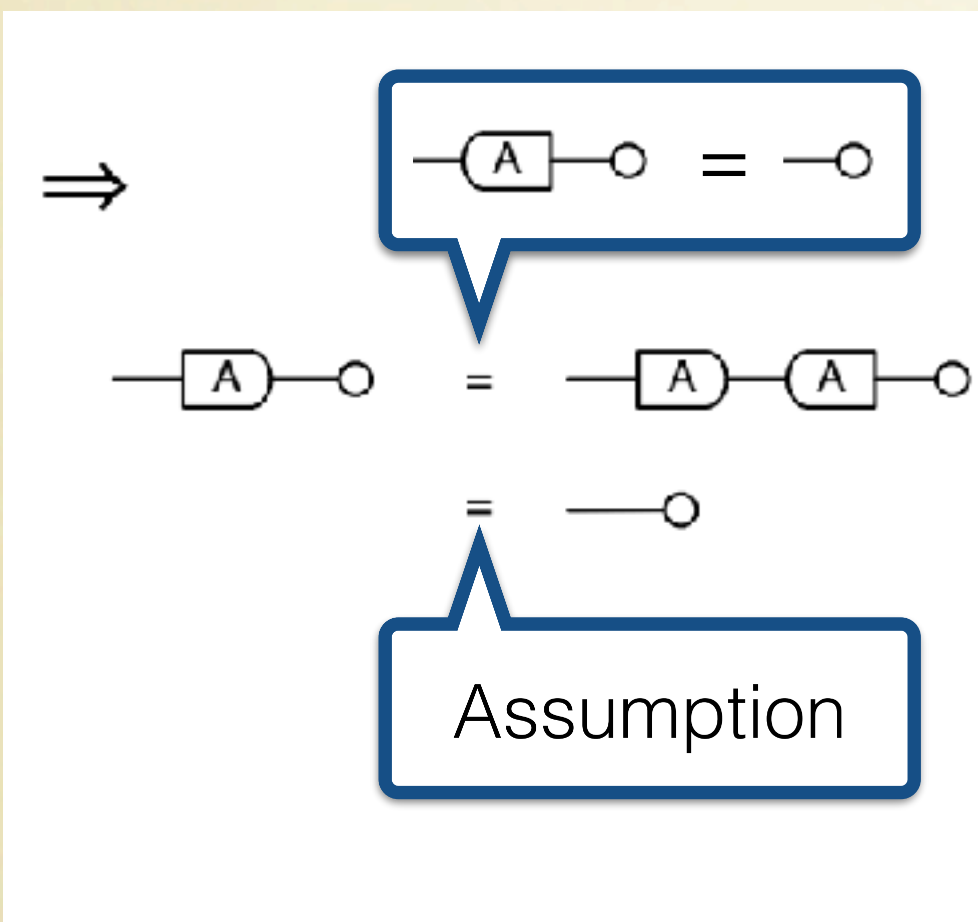
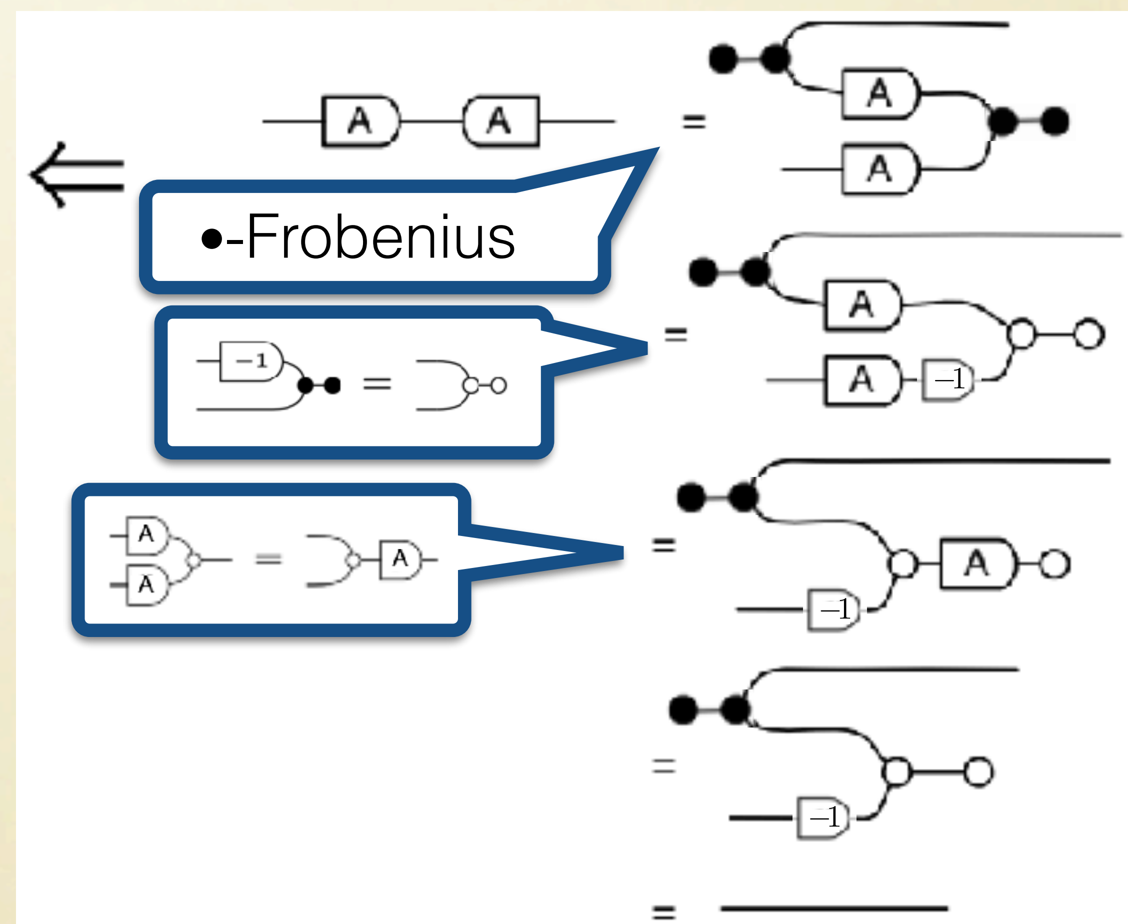
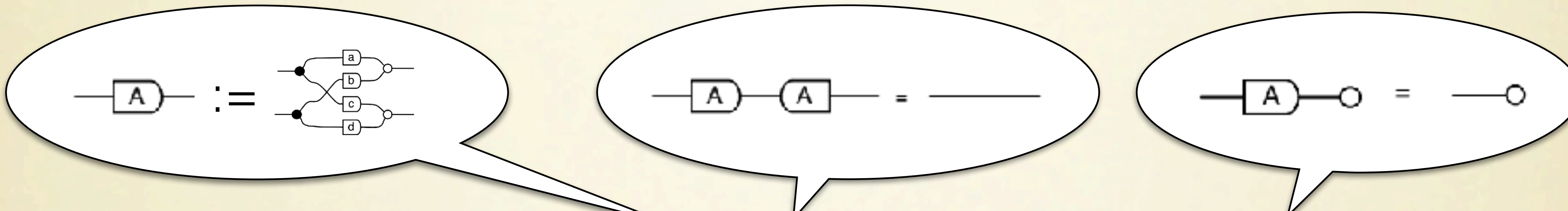




A proof in GLA



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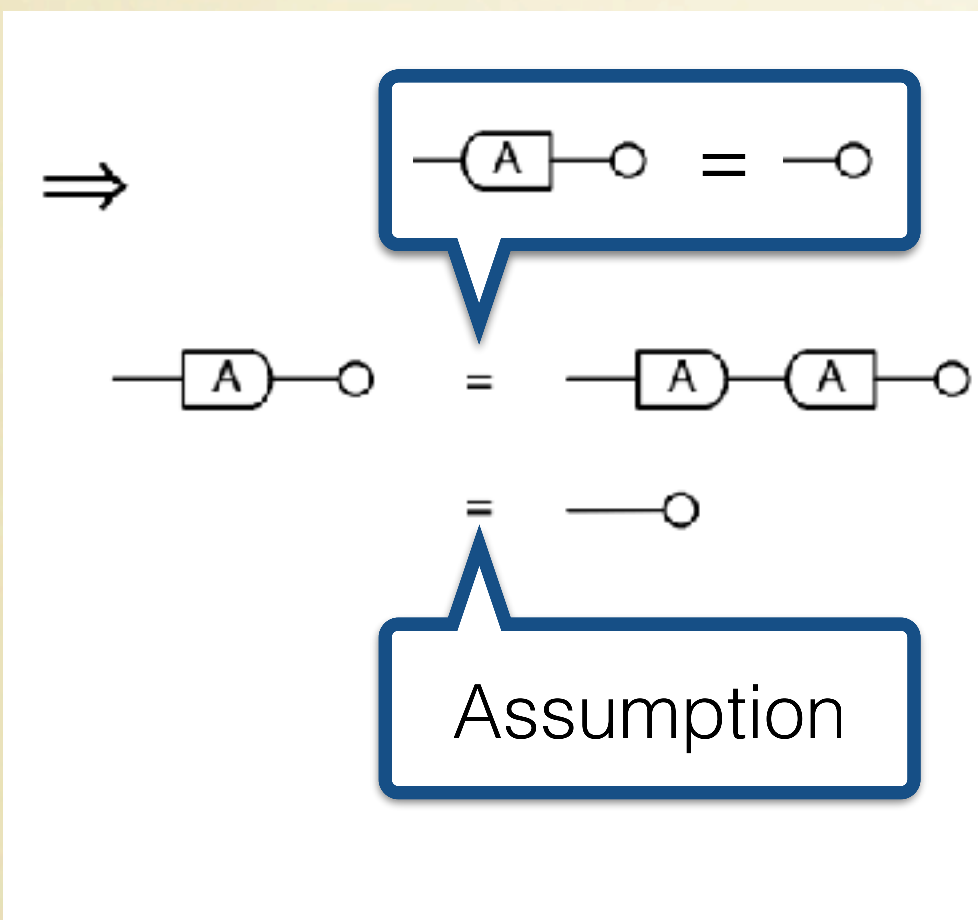
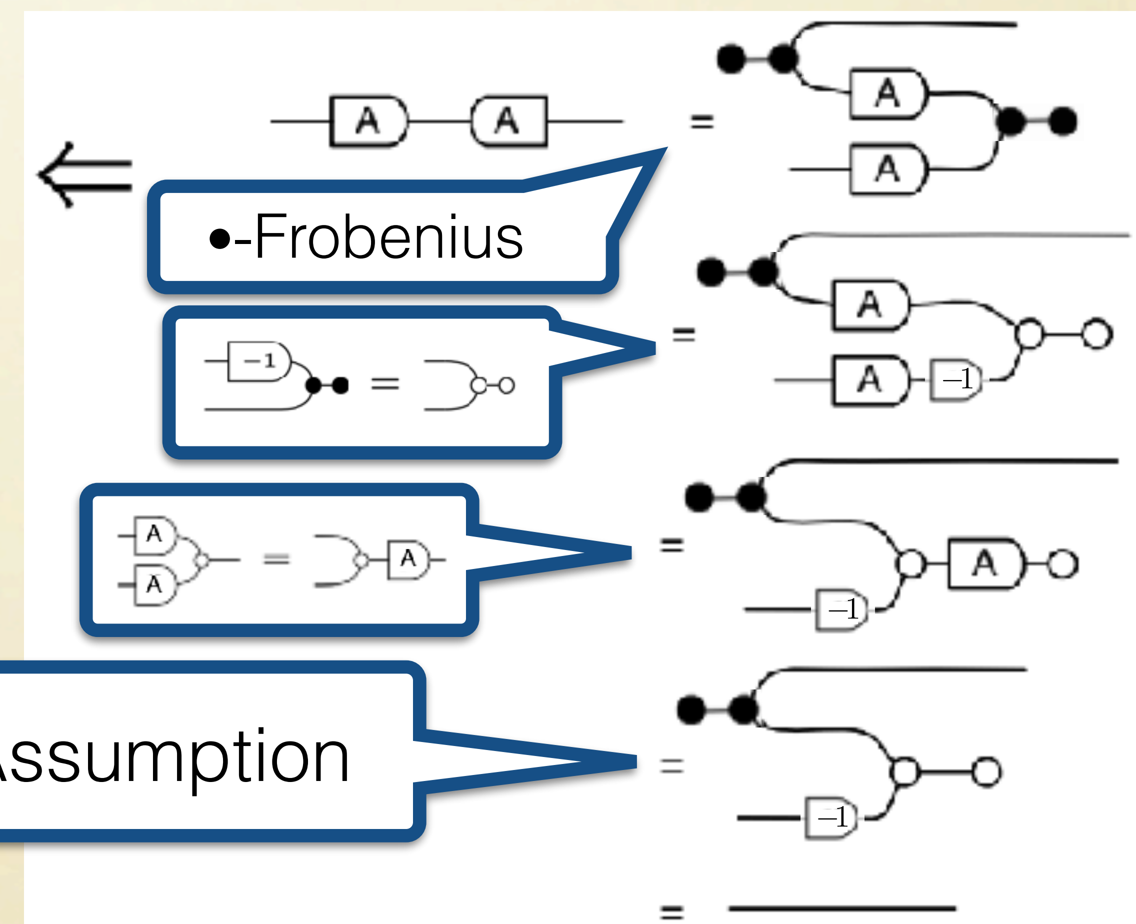
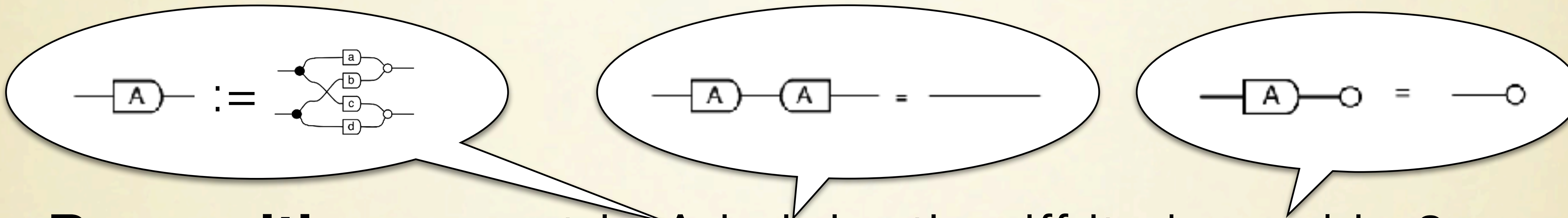


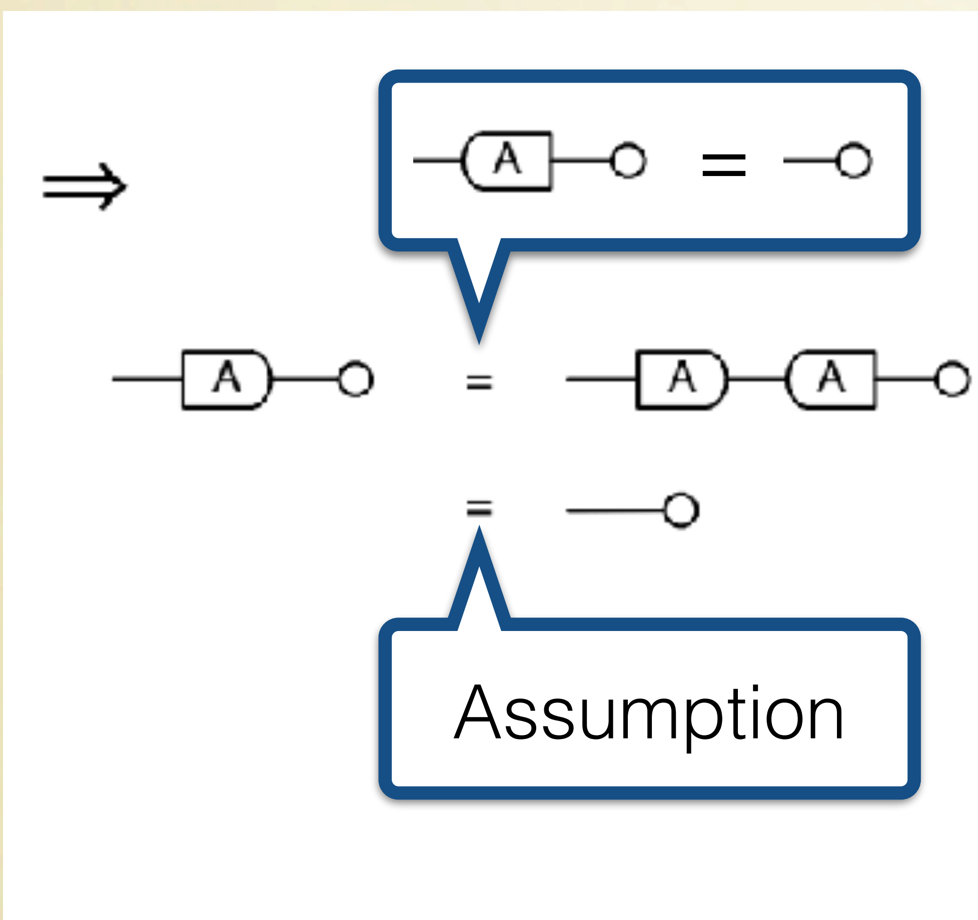


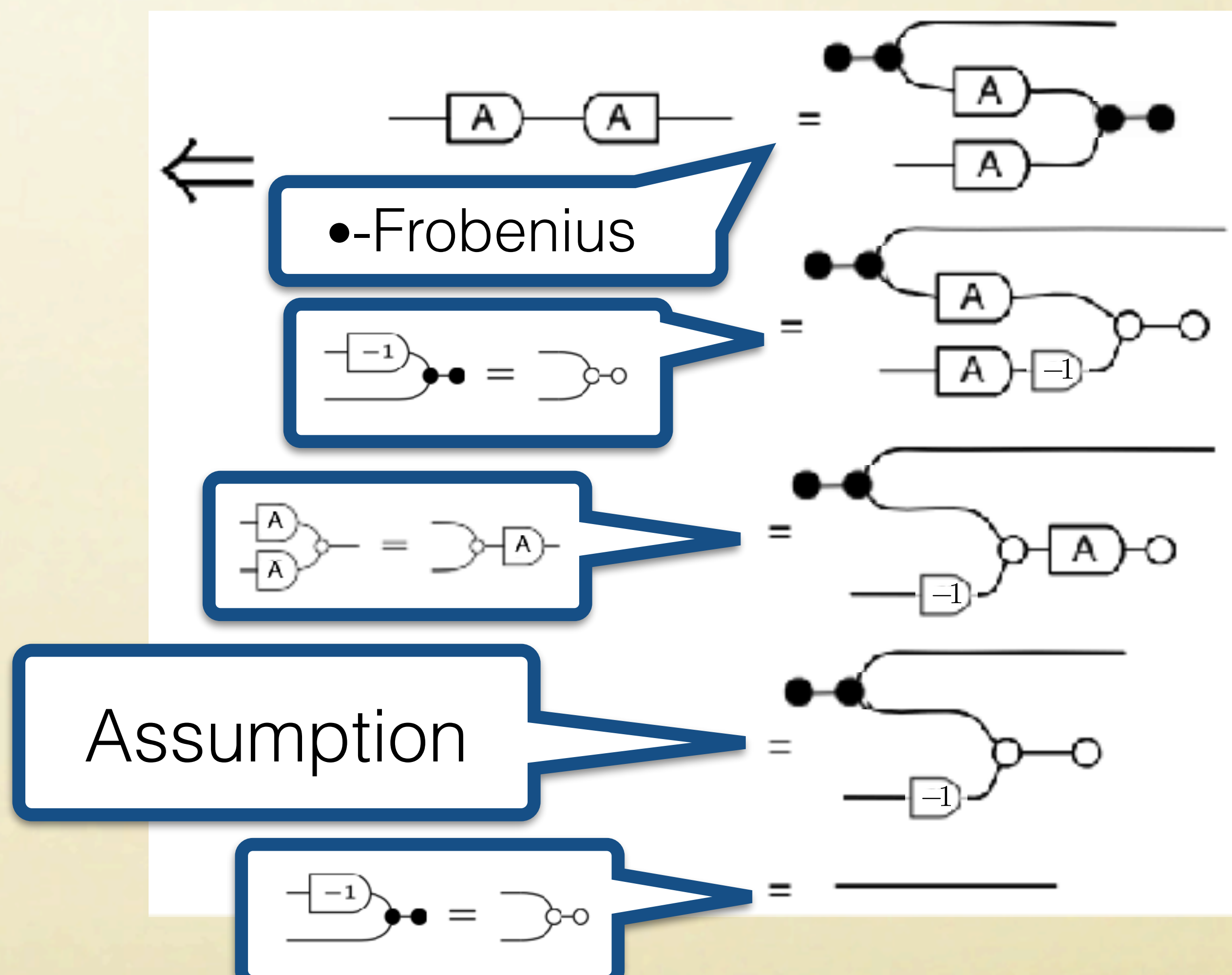
A proof in GLA



Proposition: a matrix A is injective iff its kernel is 0.

Proof

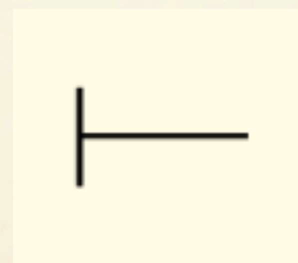




Extensions

Graphical Affine Algebra

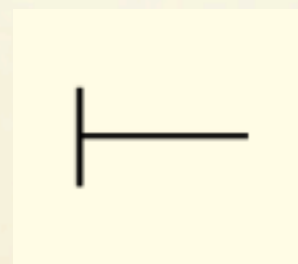
Extra operation



Extensions

Graphical Affine Algebra

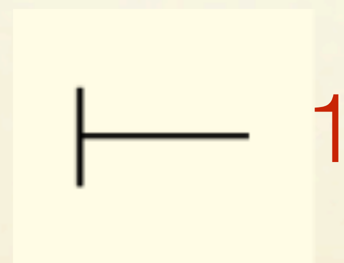
Extra operation

 1

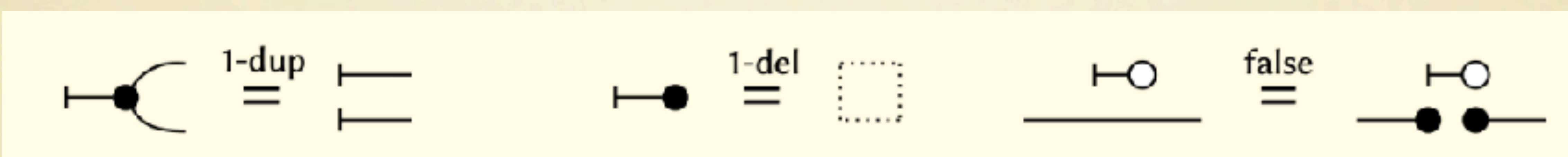
Extensions

Graphical Affine Algebra

Extra operation



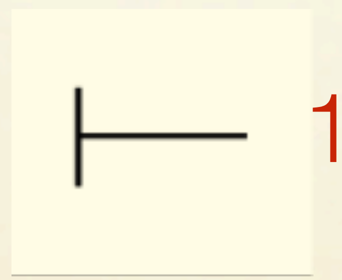
Extra equations



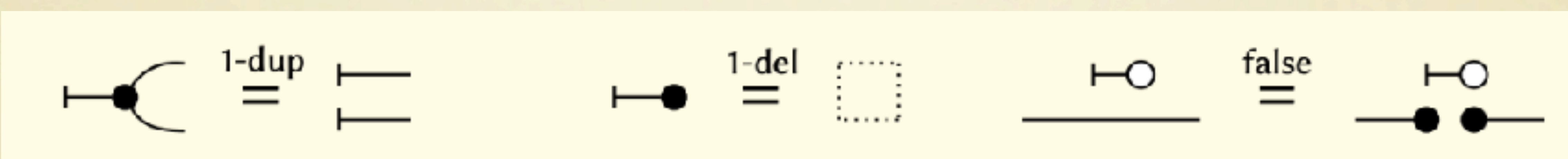
Extensions

Graphical Affine Algebra

Extra operation



Extra equations



In 1-1 correspondence with **affine** subspaces

Extensions

Graphical Quadratic Algebra

Extra operation

Extensions

Graphical Quadratic Algebra

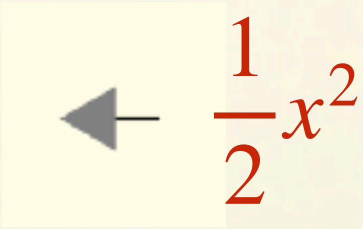
Extra operation

$$\leftarrow \frac{1}{2}x^2$$

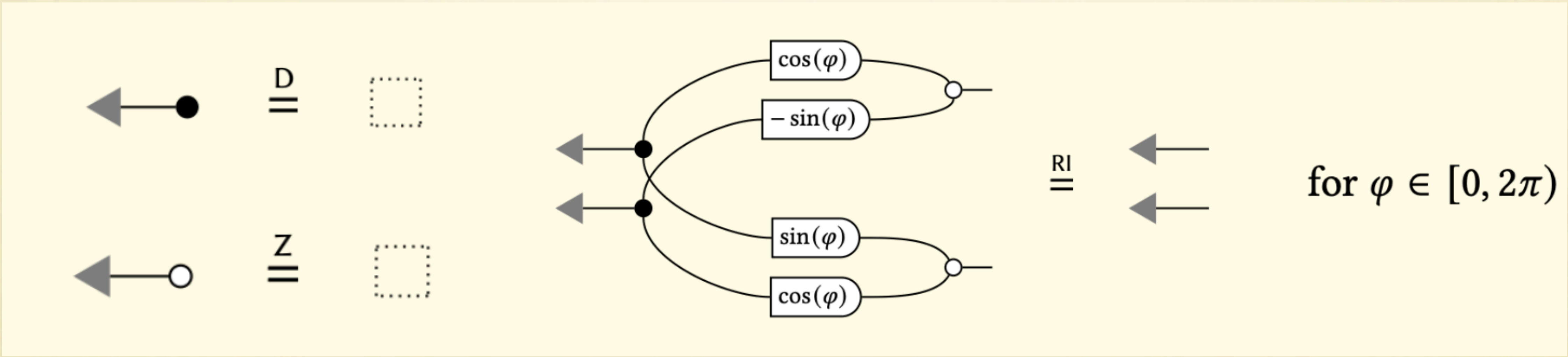
Extensions

Graphical Quadratic Algebra

Extra operation



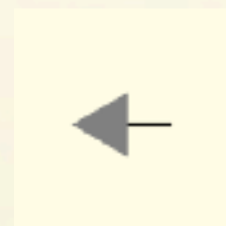
Extra equations



Extensions

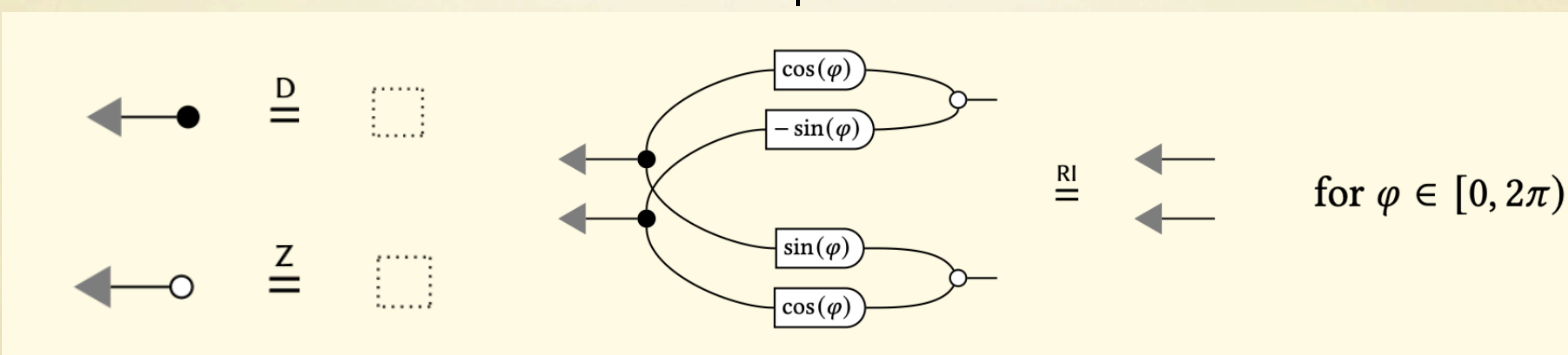
Graphical Quadratic Algebra

Extra operation



$\frac{1}{2}x^2$

Extra equations

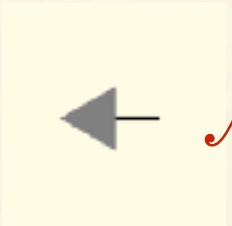


In 1-1 corresponence with **quadratic** subspaces (solutions of quadratric optimisation problems).

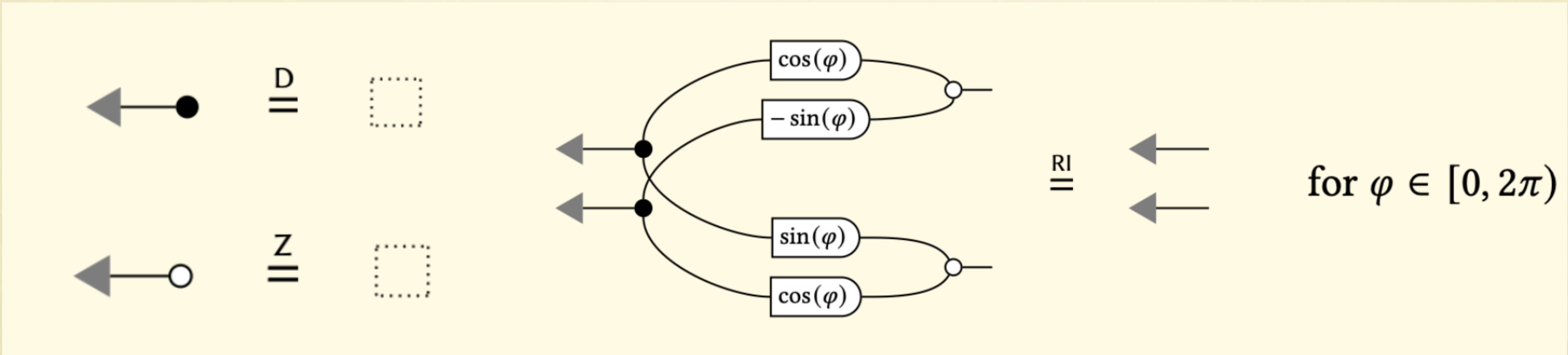
Extensions

Graphical Quadratic Algebra

Extra operation

 $\mathcal{N}(0,1)$

Extra equations



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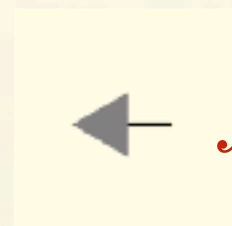
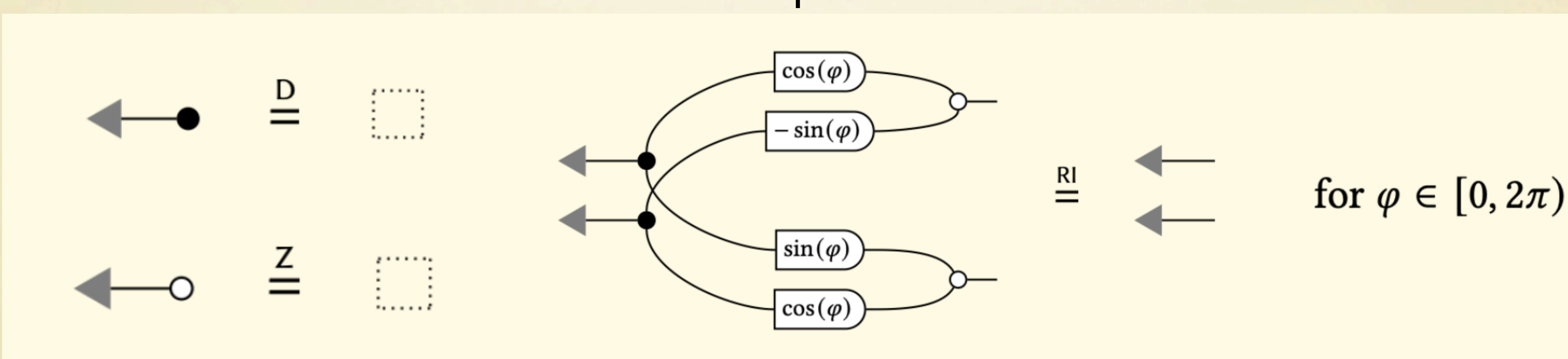
Graphical Quadratic Algebra

Extra operation



$\mathcal{N}(0,1)$

Extra equations

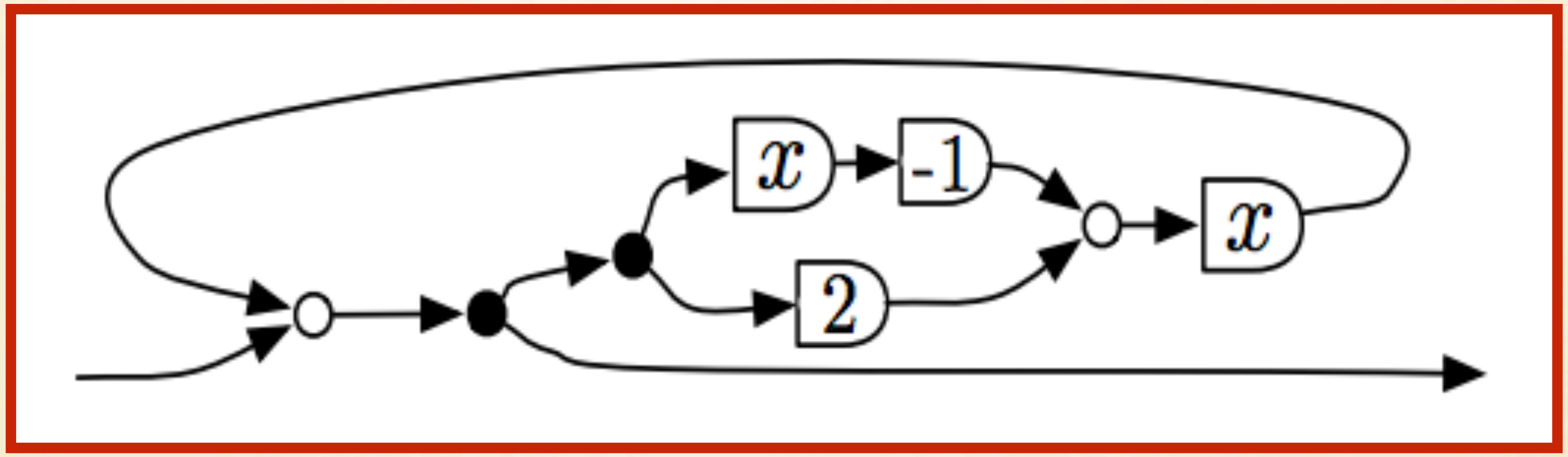


In 1-1 corresponence with **quadratic** subspaces (solutions of quadratric optimisation problems).

Matrix fragment: 1-1 with **Gaussian probability**

Graphical Linear Algebra in Action

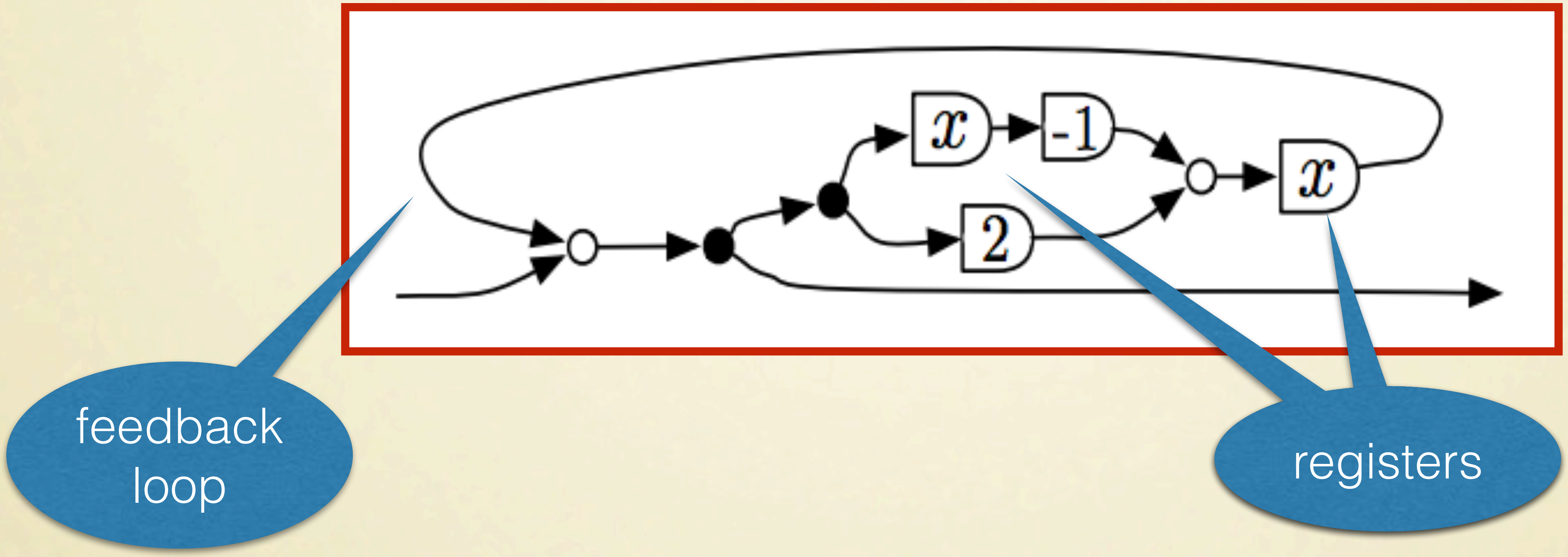
Control Theory in GLA



Signal Flow Graphs

(Shannon 1942)

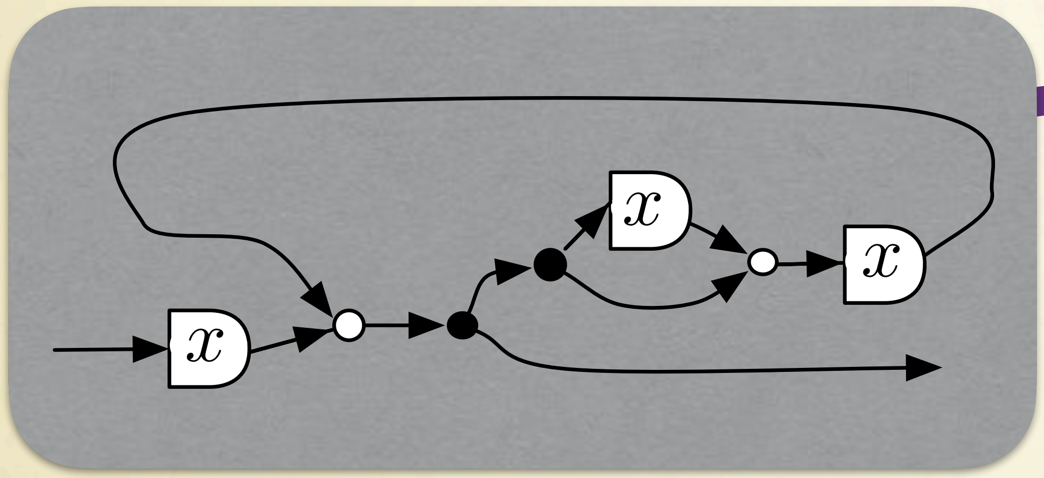
Control Theory in GLA

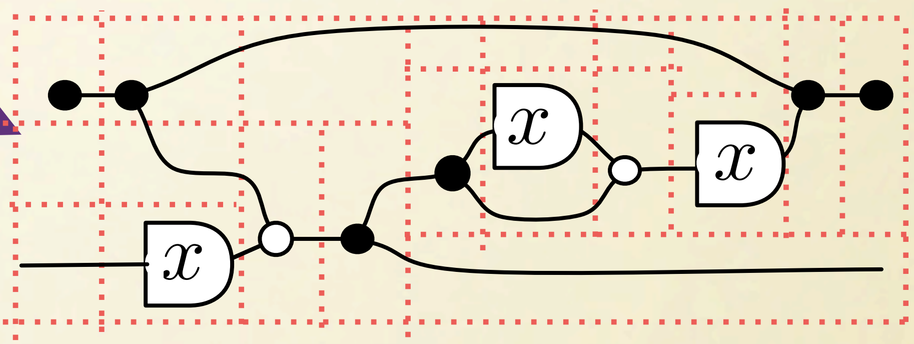


Signal Flow Graphs

(Shannon 1942)

Control Theory in GLA



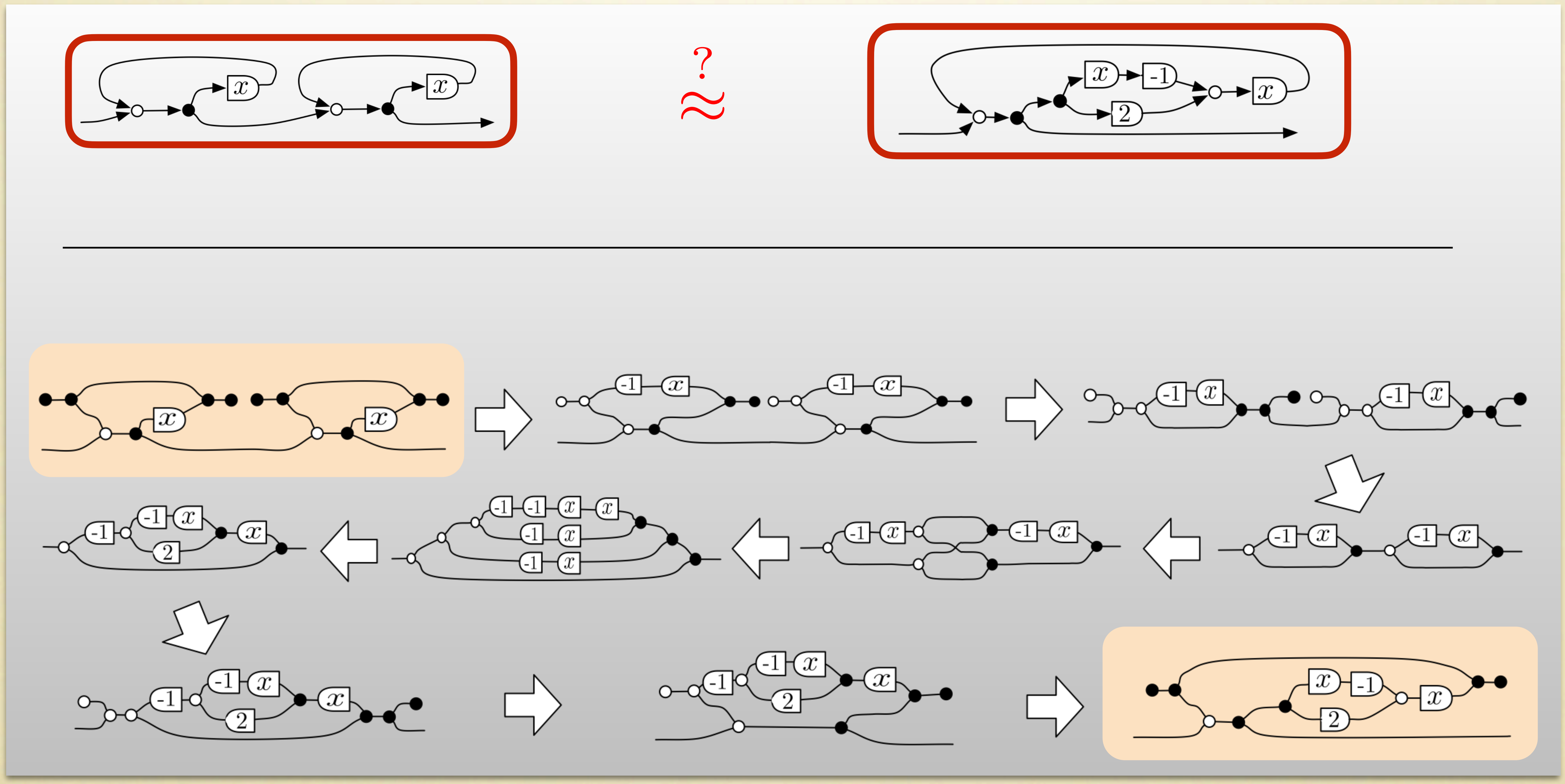


Every signal flow graph can be translated into a string diagram of GLA (on polynomial ring $\mathbb{R}[x]$).

The semantics of the string diagram reflects the behaviour of the signal flow graph.

Control Theory in GLA

Proof that two circuit diagrams represent the same dynamical system:



Control Theory in GLA

From a specification (a rational fraction) to its implementation as a linear dynamical system.

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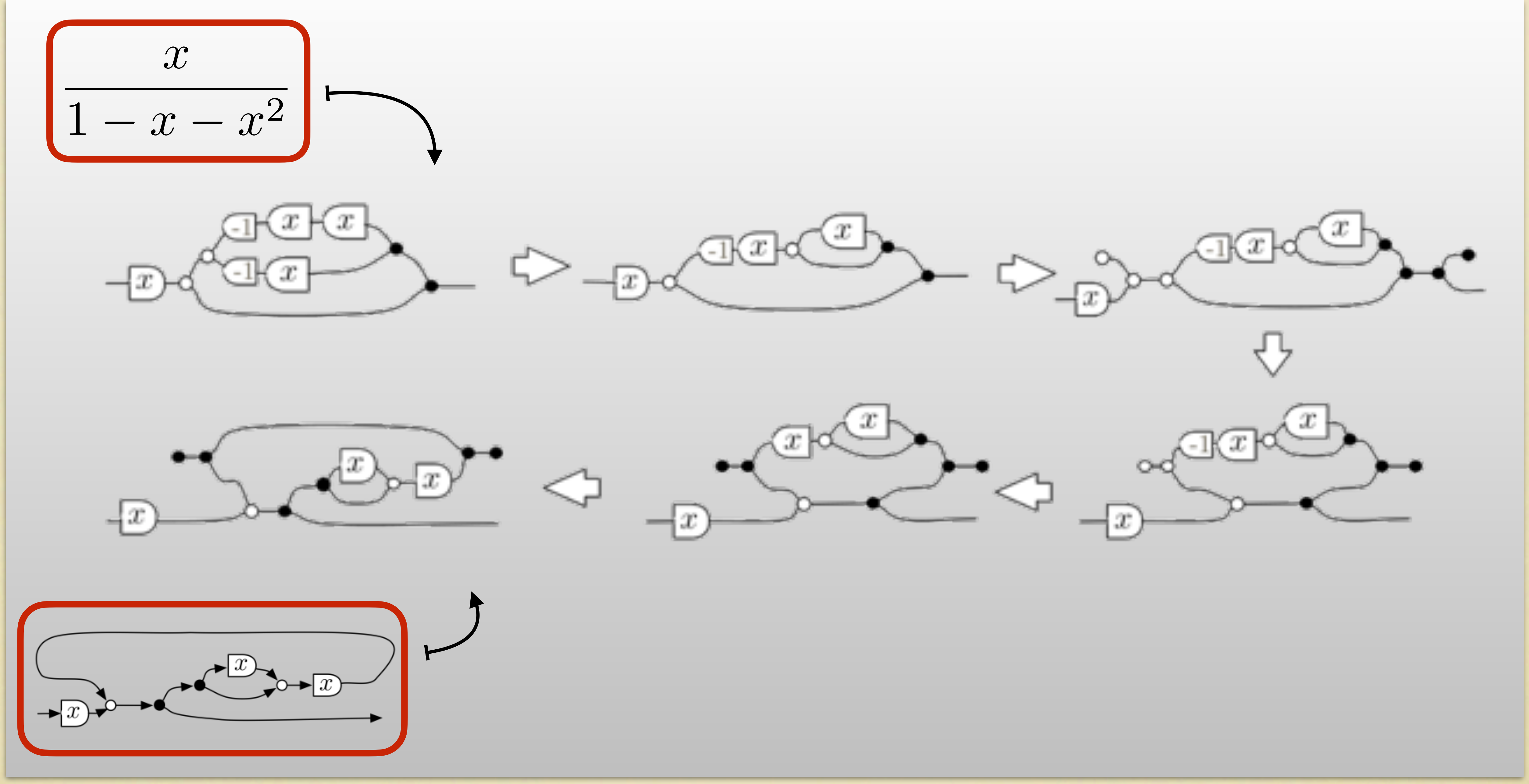

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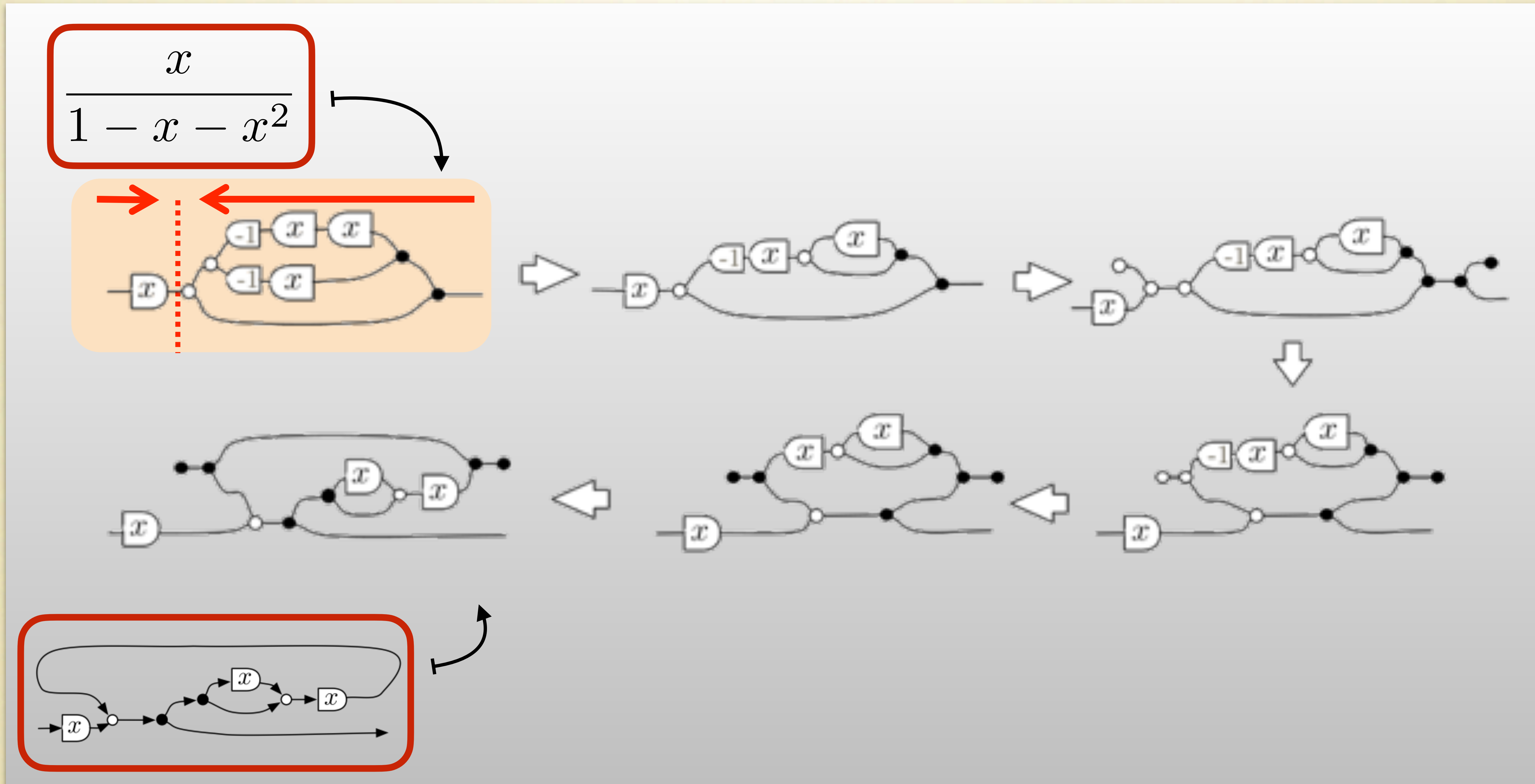

Control Theory in GLA

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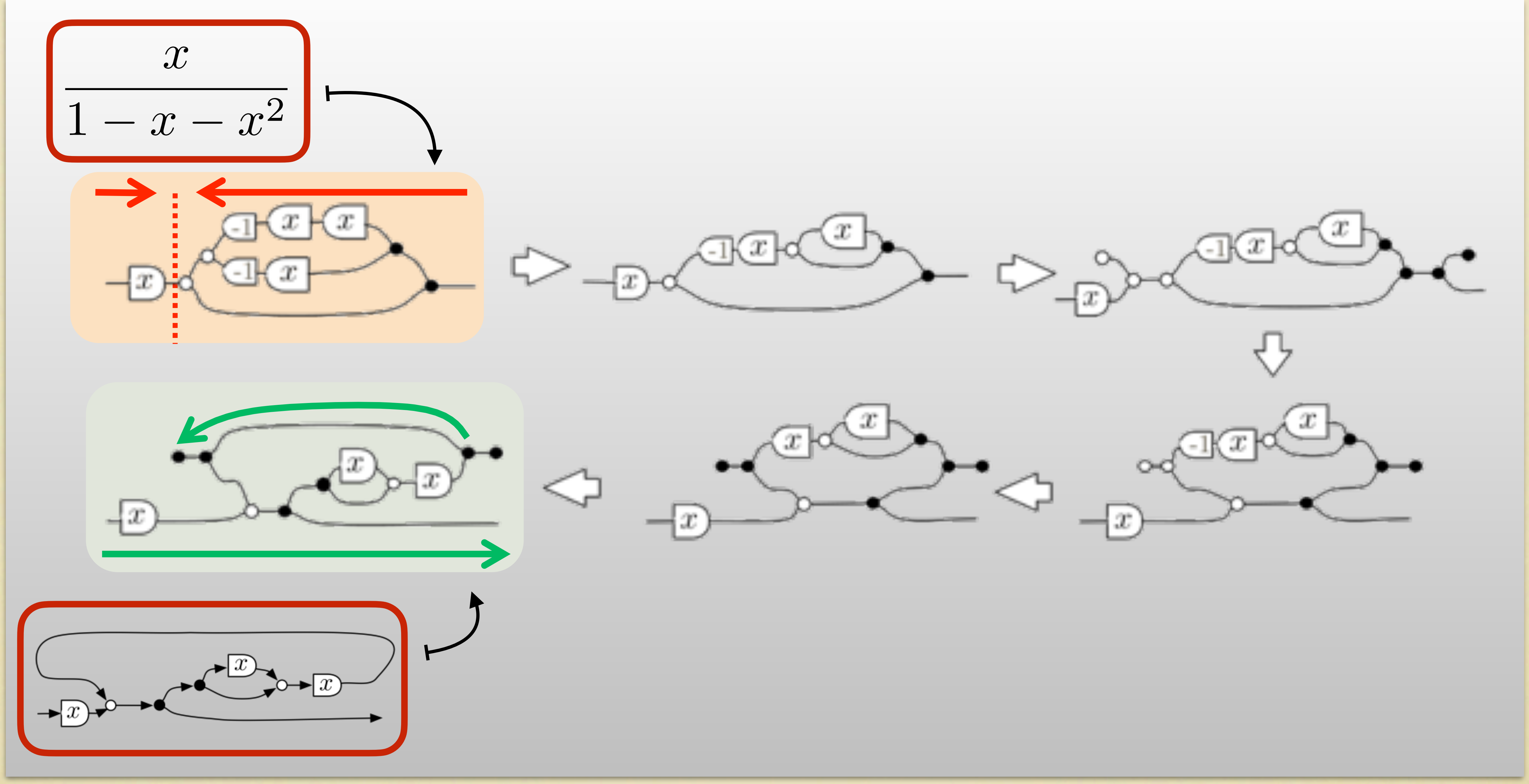
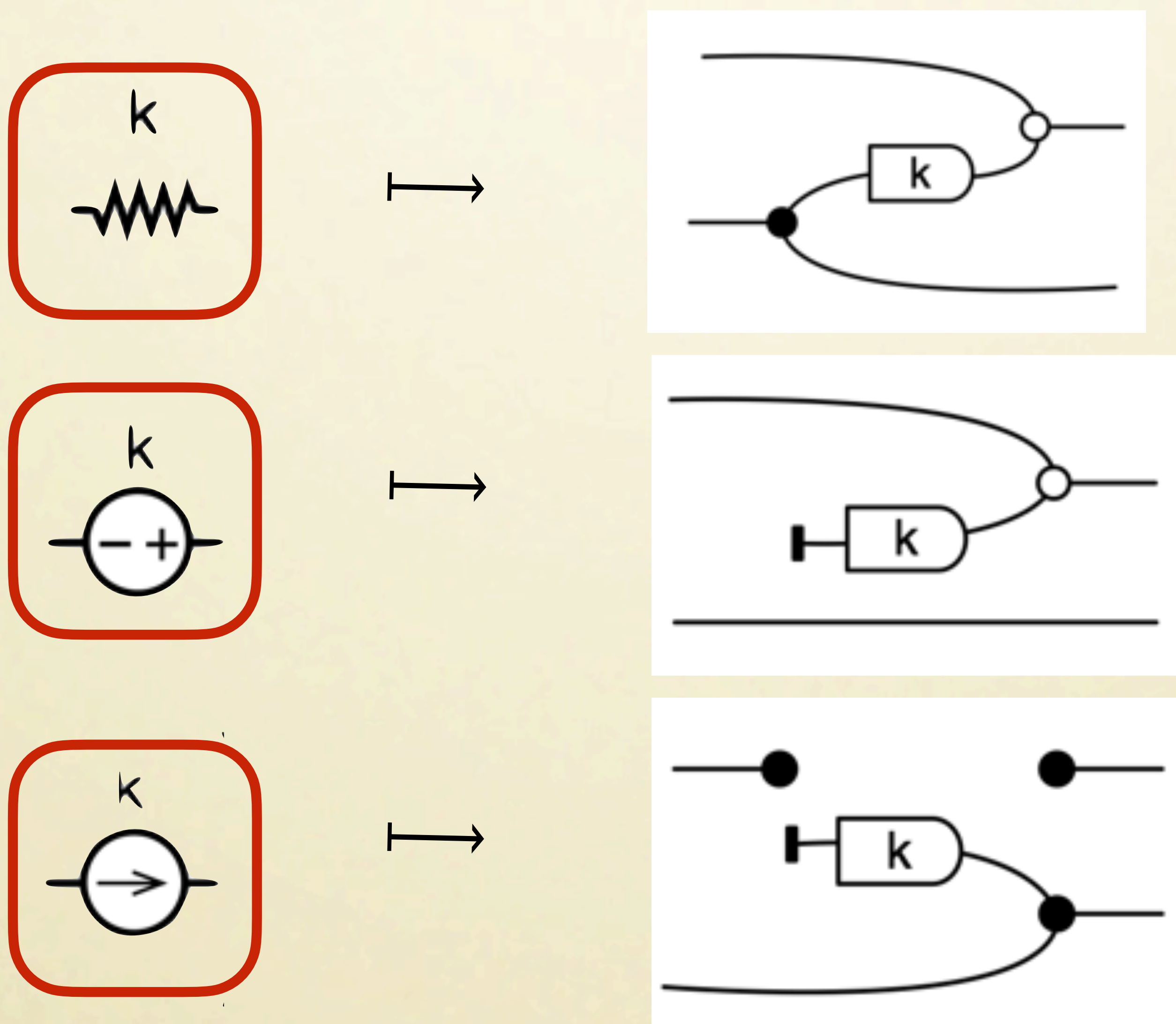


Electrical Circuits in GAA

Encode electrical components as string diagrams of Graphical Affine Algebra



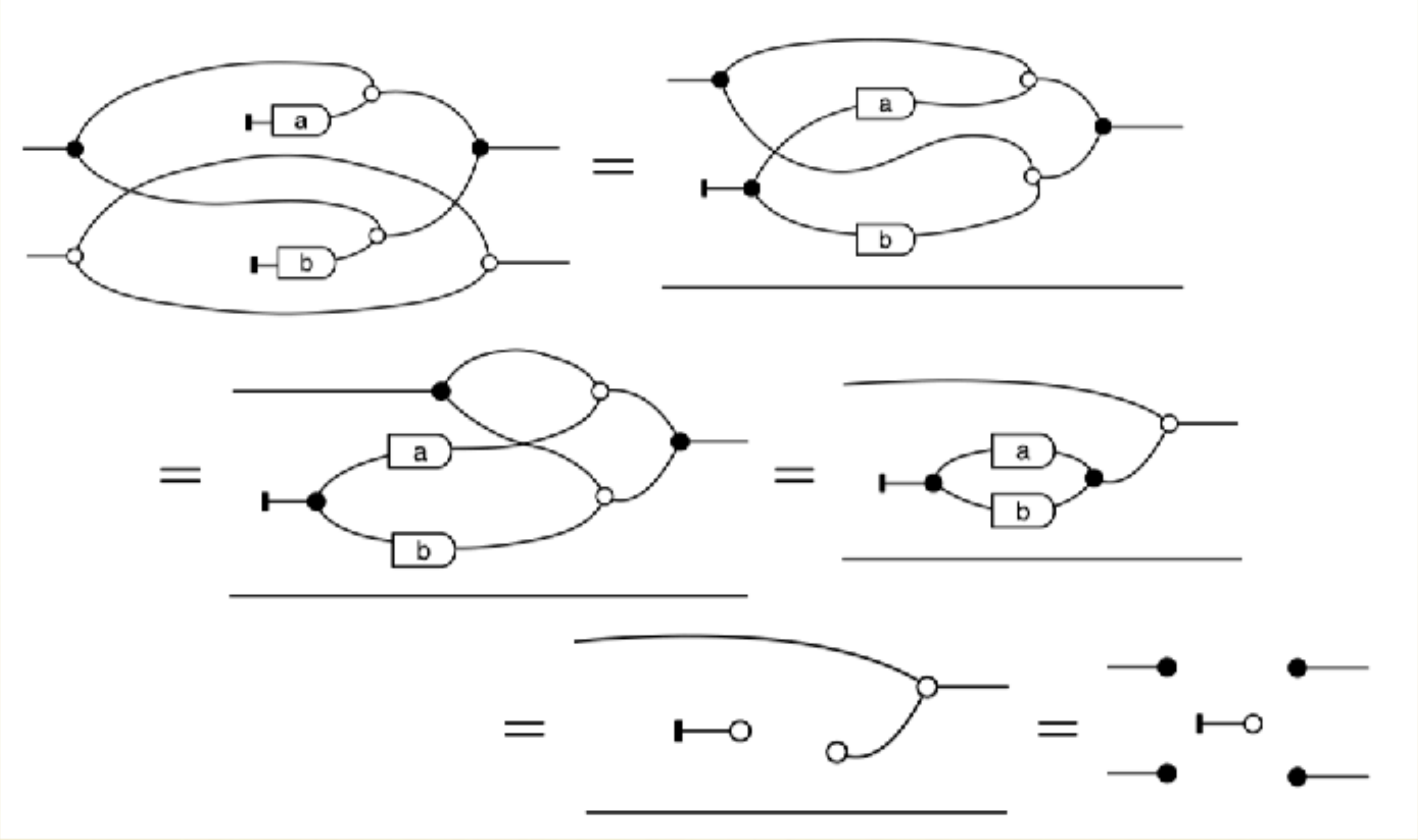
Resistor

Voltage

Current

Electrical Circuits in GAA

Proof that parallel voltage sources of different voltages are disallowed.



Electrical Circuits in GAA

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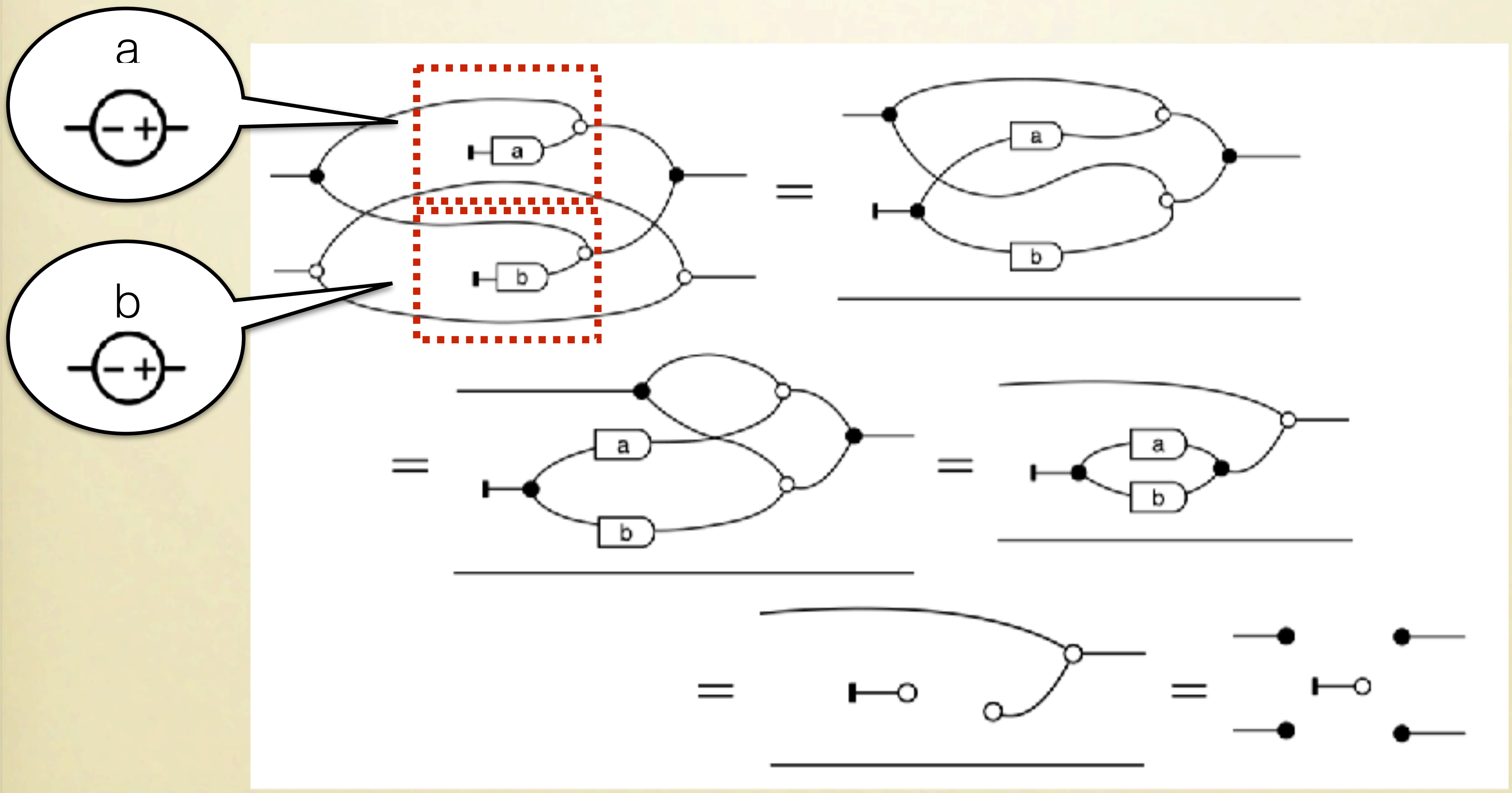

Electrical Circuits in GAA

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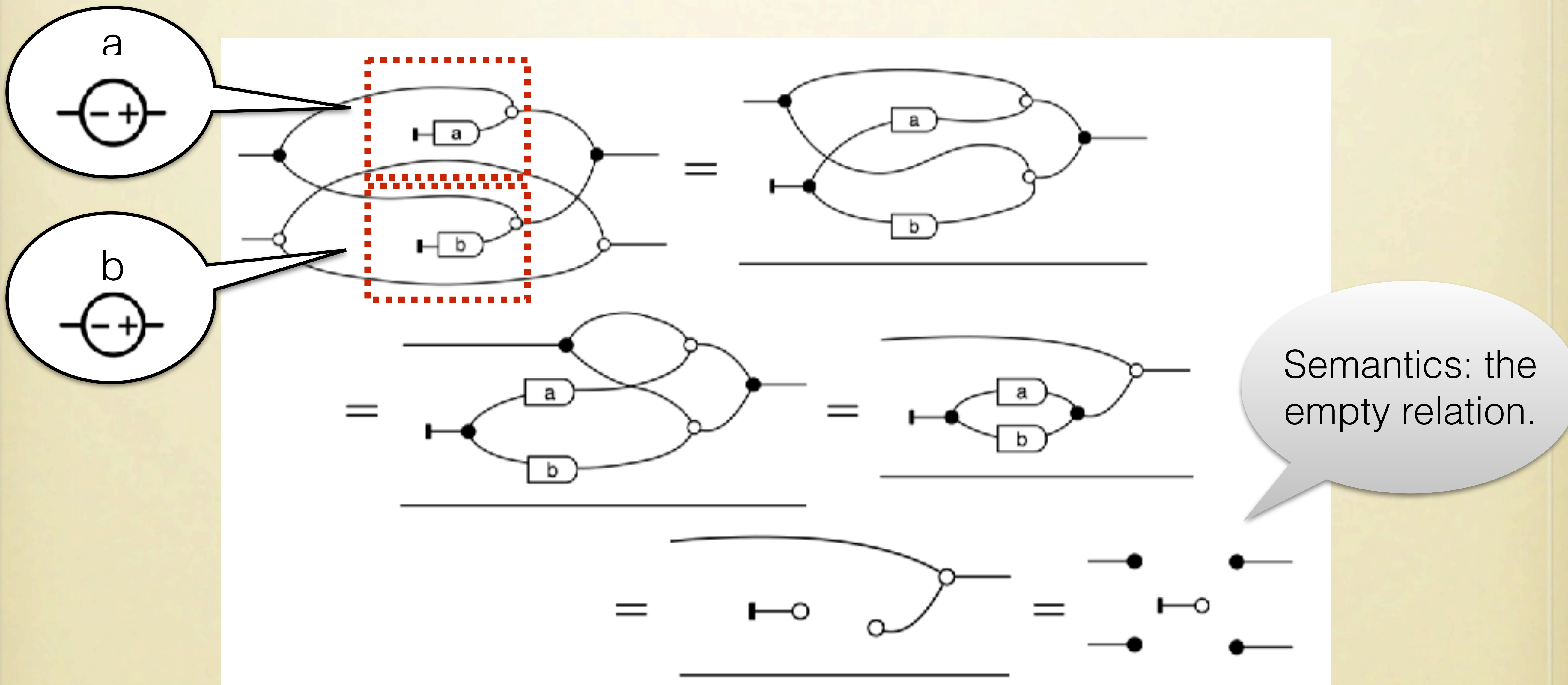
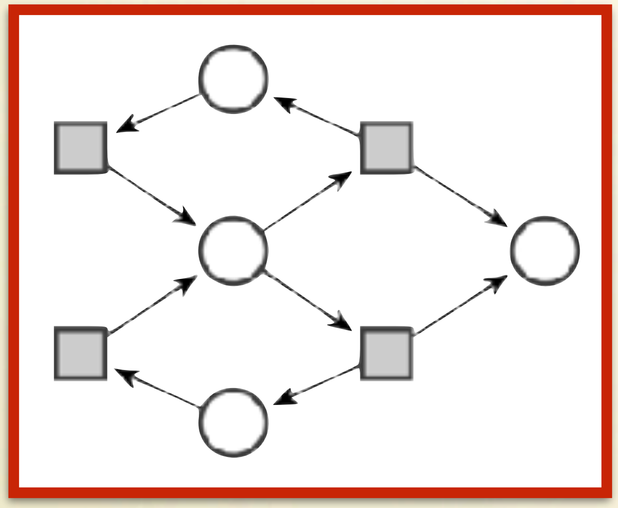
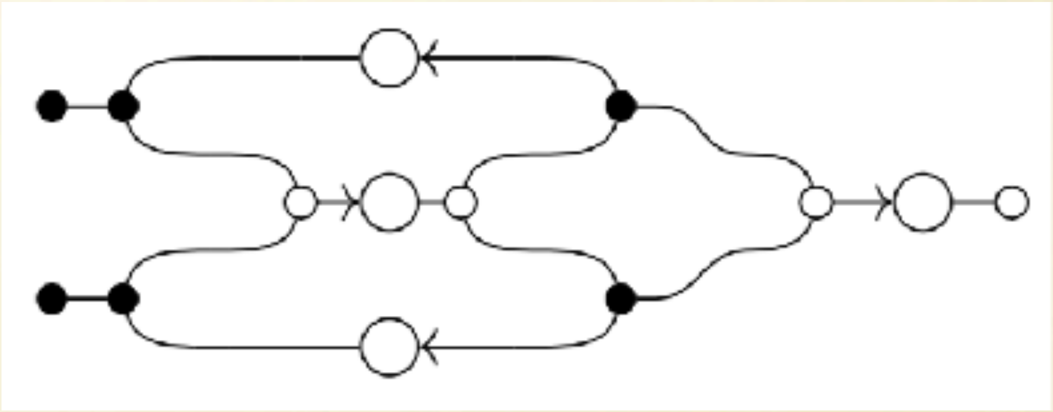


Petri Nets in GAA



$\longmapsto$



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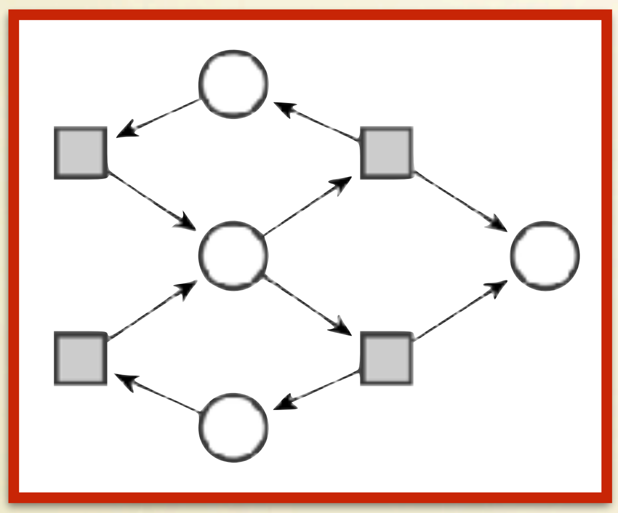
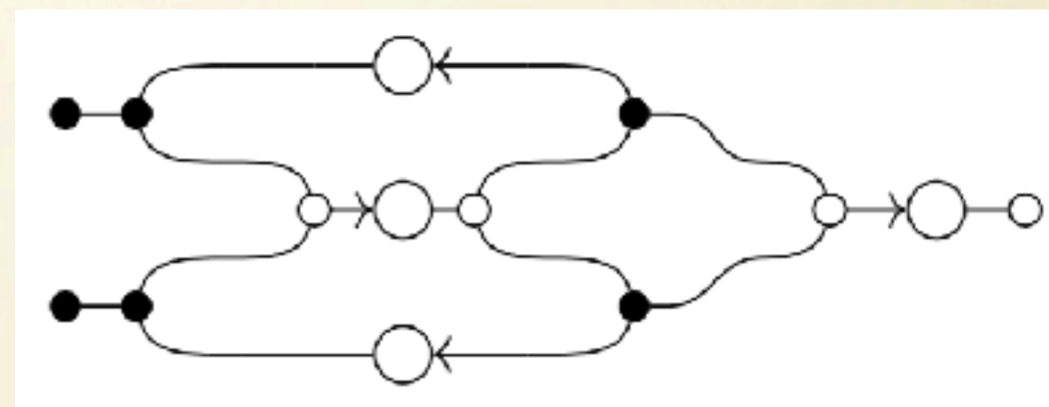
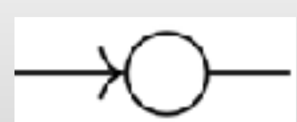


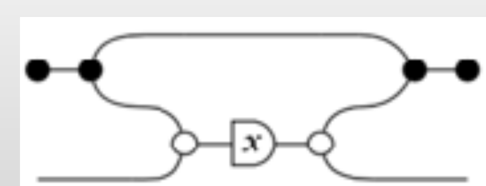
$\longmapsto$



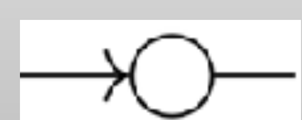
Firing semantics



$::=$



Banking semantics



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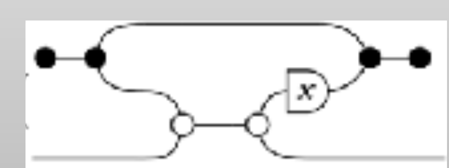
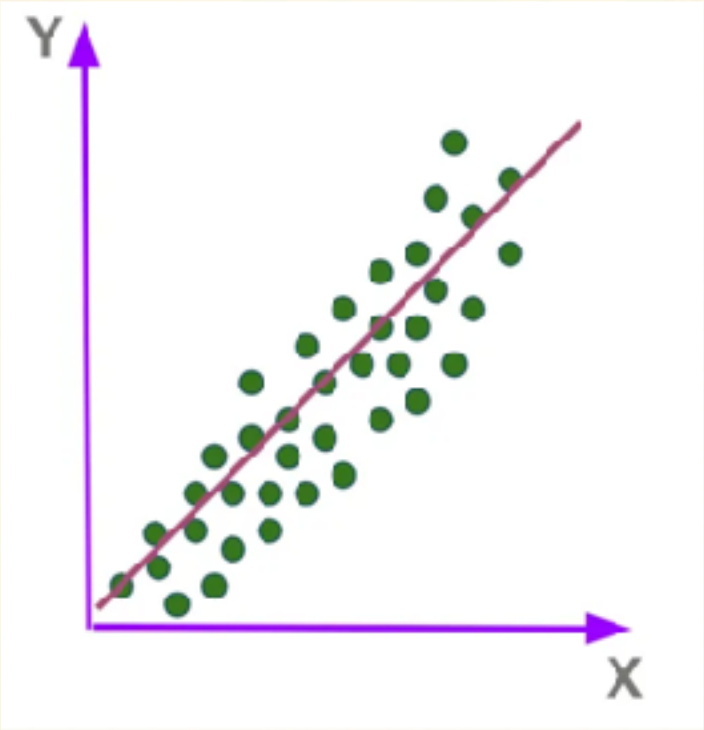


Linear Regression in GQA

Observations
(in green)

Dependent variable y



Independent variable x
(regressor)

Goal of linear regression:
to find a linear model that best fits a set of observations

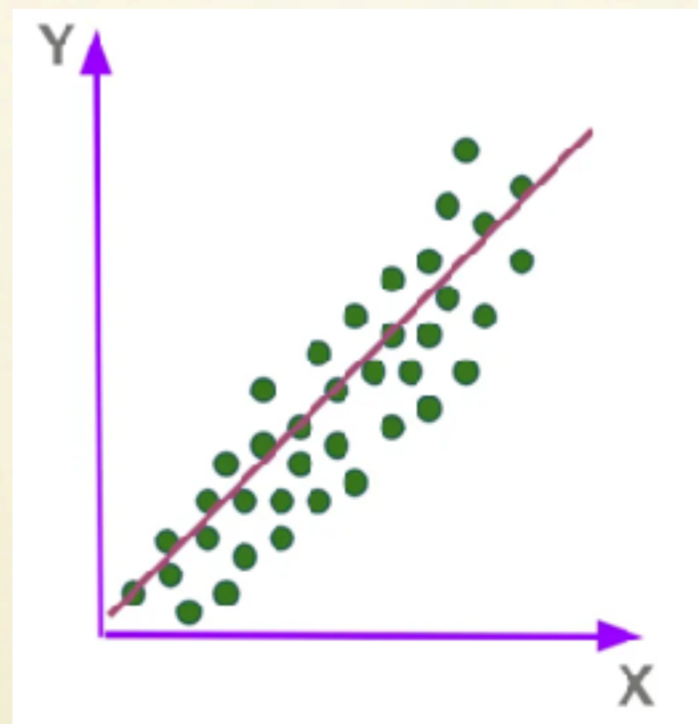
Linear Regression in GQA

Ordinary least-squares method

Compute x in $Ax = b$ minimising the errors $\|Ax - b\|^2$. Solution is A^+b.

Observations
(in green)

Dependent variable y



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Linear Regression in GQA

Ordinary least-squares method

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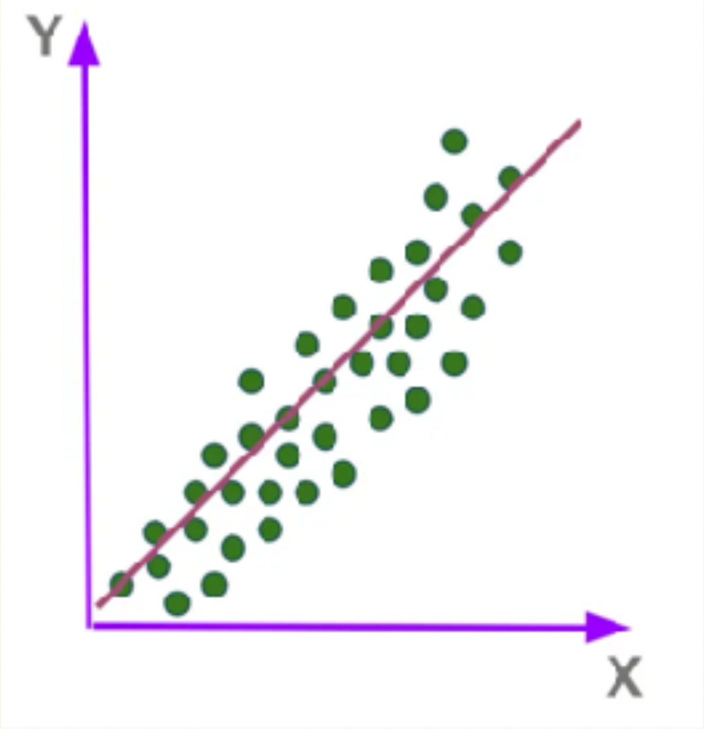
Observations (in green)

Dependent variable y



Independent variable x (regressor)

Stochastic method

Compute random variable x given $Ax + \epsilon = b$, where ϵ is the distribution of errors.

Goal of linear regression:
to find a linear model that best fits a set of observations

Linear Regression in GQA

Ordinary least-squares method

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Observations (in green)

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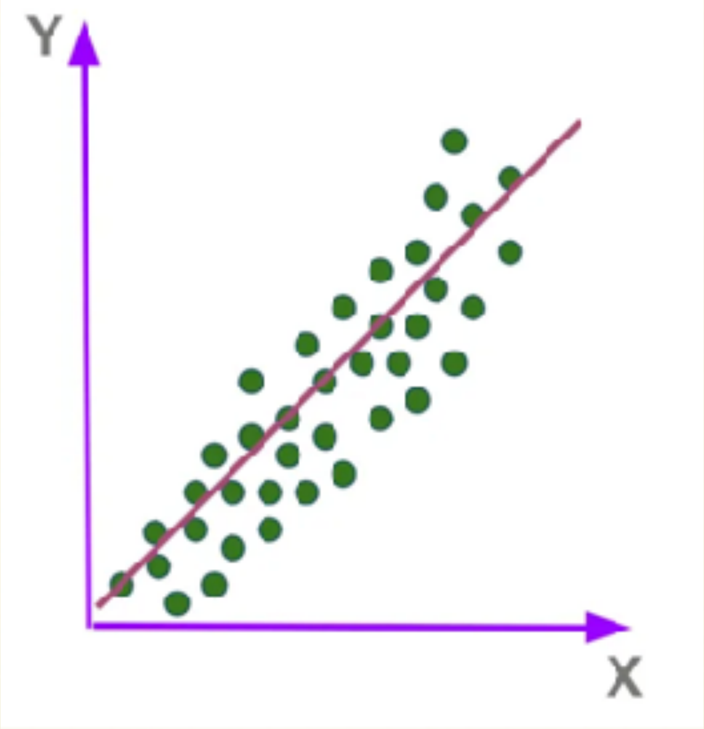


Independent variable x (regressor)

Stochastic method

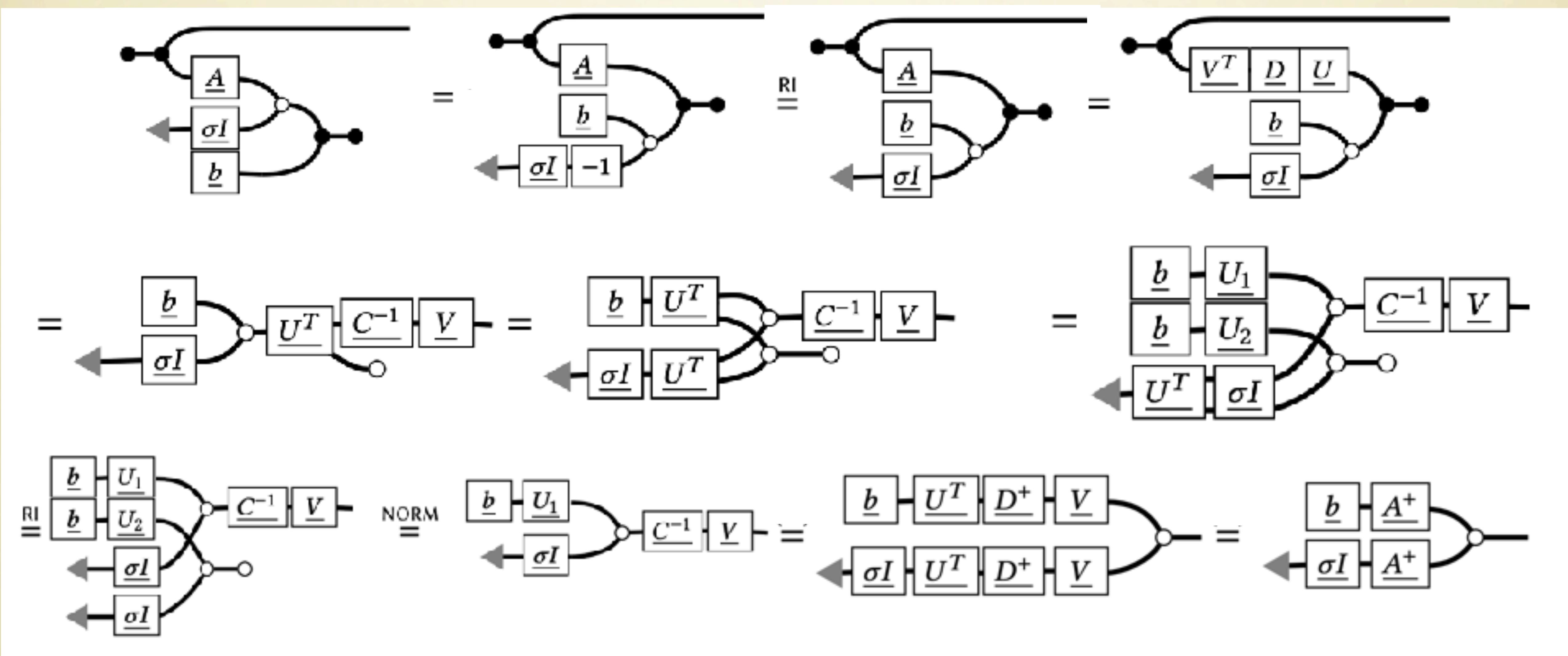
Compute random variable x given $Ax + \epsilon = b$, where ϵ is the distribution of errors.

If $\epsilon \sim \mathcal{N}(0\ ,\ \sigma^2 \cdot I)$, x has mean A^+b.

Goal of linear regression:
to find a linear model that best fits a set of observations

Linear Regression in GQA

Linear Regression in GQA

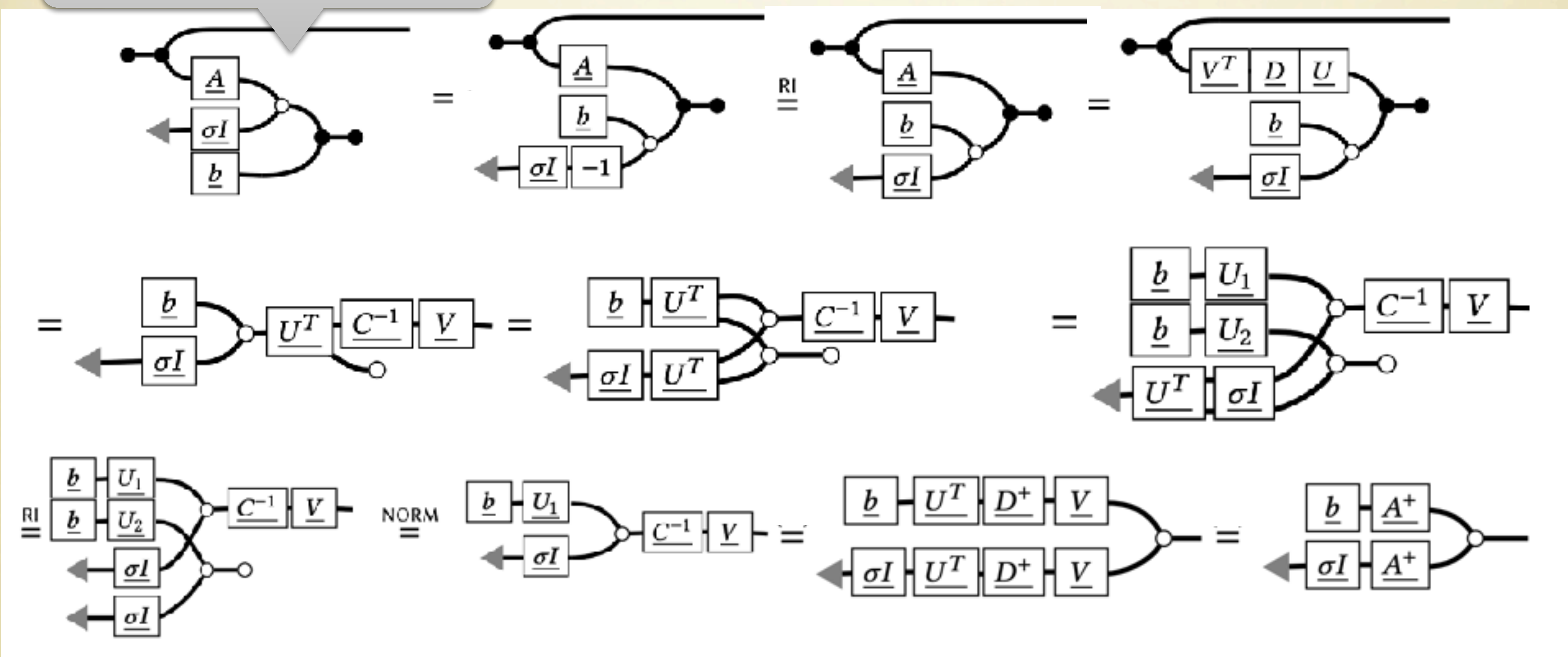


Linear Regression in GQA

What we observe:
random variable x

Our assumptions:

$$Ax + \epsilon = b$$
$$\epsilon \sim \mathcal{N}(0\ ,\ \sigma^2 \cdot I)$$



Linear Regression in GQA

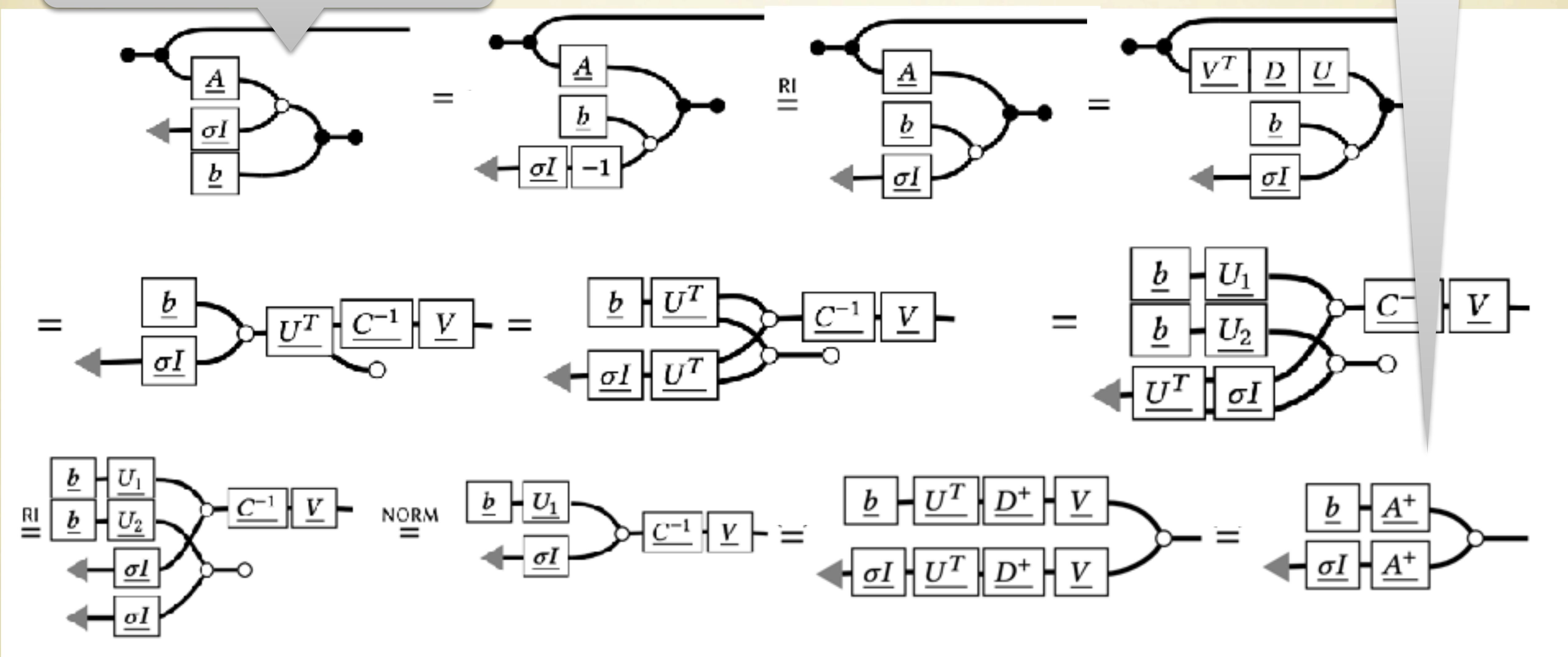
What we observe:
random variable x

Our assumptions:

$$Ax + \epsilon = b$$
$$\epsilon \sim \mathcal{N}(0\ ,\ \sigma^2 \cdot I)$$

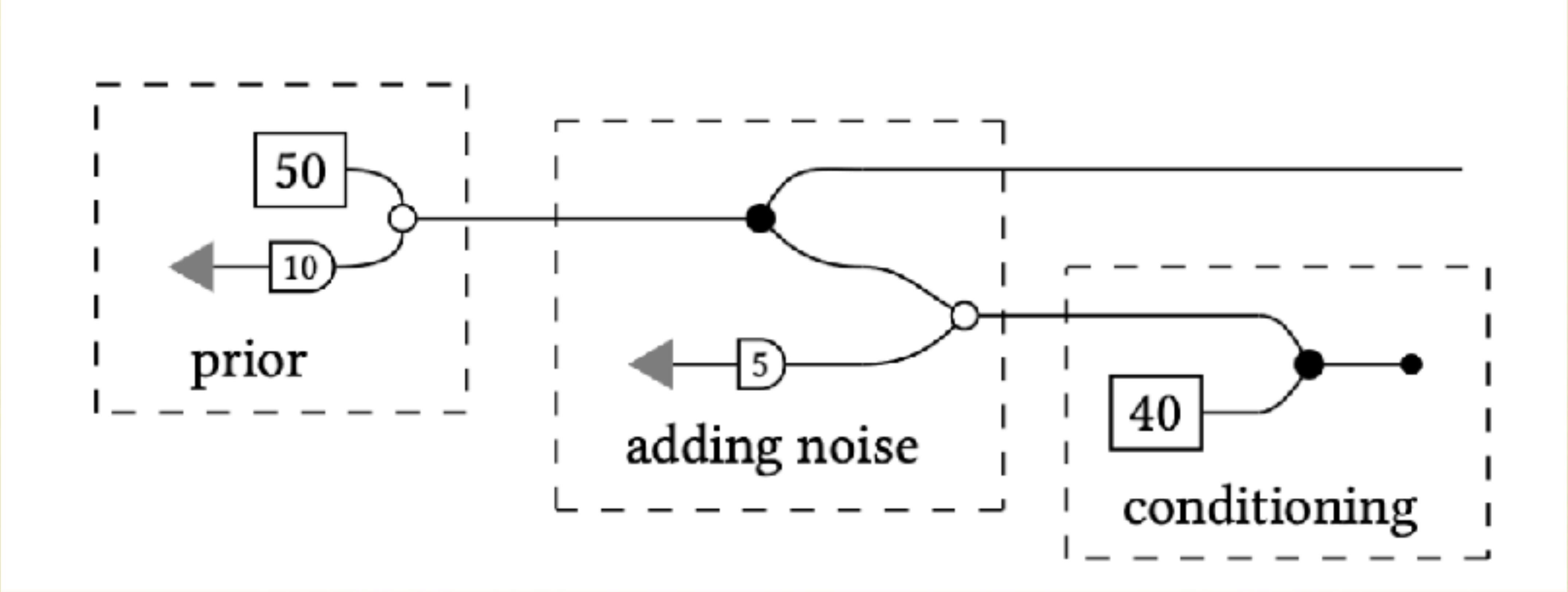
What we have proven:
x has mean $A^+ b$.

The stochastic method
leads to the same result as OLS.



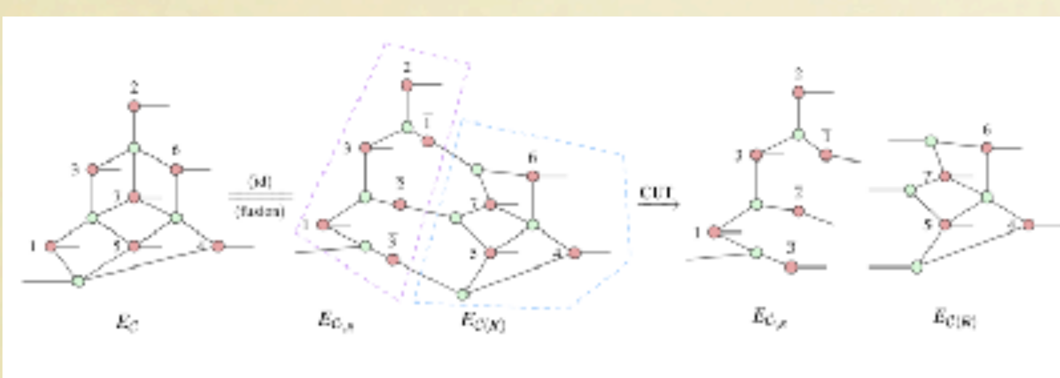
Probabilistic Programming in GQA

```
let value = 50 + 10 * normal() in
let measurement = value + 5 * normal() in
measurement =:= 40;
return value
```

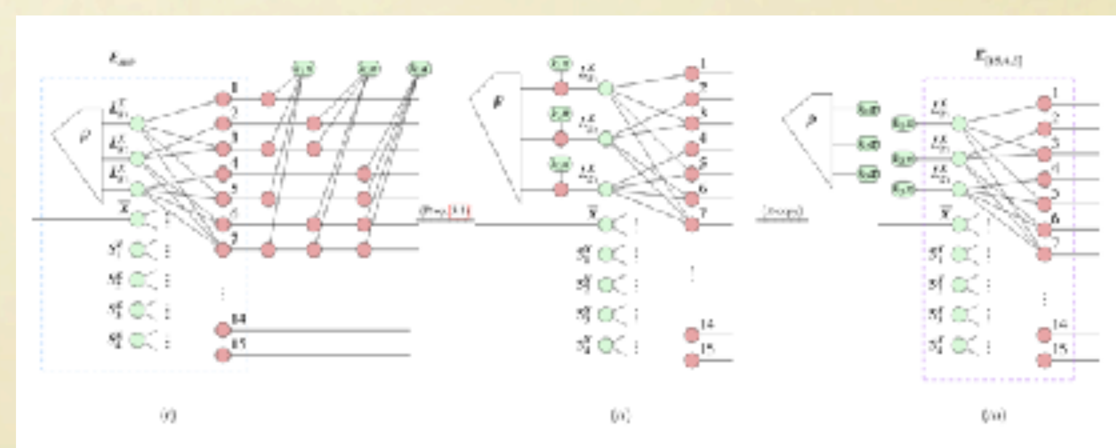


Other Notable Extensions: the ZX-Calculus

Quantum Error Correction



Quantum Circuit Compilation



Conclusions

Graphical Linear Algebra is an alternative foundation for linear algebraic formal reasoning.

It is **abstract**, **visual**, **compositional**.

It reveals symmetries and dualities underpinning linear algebra (monoids/comonoids, Hopf algebras, Frobenius algebras,….).

It provides a unifying framework for many different applications: control theory, concurrency, quantum, probabilistic computation, optimisation, and more.

To Know More

- P. Sobocinski et al. - graphicallinearalgebra.net.
- P. Sobocinski - *Linear algebra…with diagrams*, Chalkdust Magazine, 2017.
- F. Bonchi, P. Sobocinski, F. Zanasi - *Interacting Hopf Algebras*, Journal of Pure and Applied Algebra, 2017.
- F. Zanasi - *Interacting Hopf Algebras: the Theory of Linear Systems*, PhD Thesis, ENS Lyon, 2015.
- R. Piedeleu and F. Zanasi. - *Introduction to String Diagrams for Computer Scientists*, CUP, 2025.